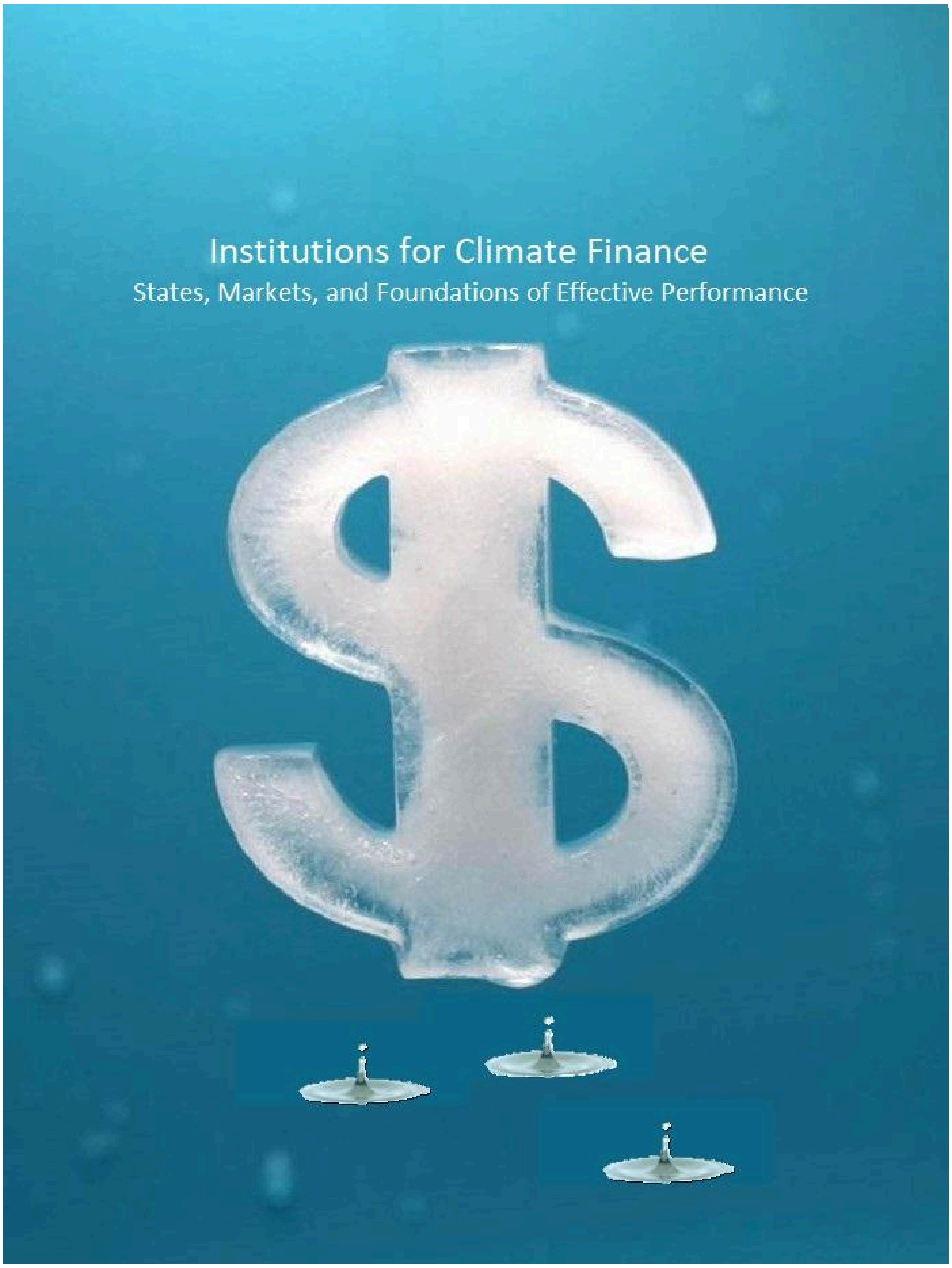


Liam Clegg & Julio Galindo Gutierrez

Acknowledgements

Colleagues across the University of York have provided a fantastically supportive institutional environment for the development, from inception to completion, of this book. We have in particular benefited from three aspects of life in and around the Department of Politics and International Relations.

First, and perhaps most fundamentally, the Department is an extremely collegiate place to work, in which we are interested in and enthused by our peers' work, and (we hope!) they by ours. Social bonds ward off the potentially atomistic nature of writing a book, and have embedded us in a sense of wider collective endeavour. Second, over the last decade or so, the Department has become home to internationally-leading quantitative political analysis. When we started in the Department (in 2010 and 2021 respectively), we entered as qualitatively-oriented researchers. We have both benefited hugely from the willingness of quantitatively-oriented colleagues to share both their knowledge, and their technical skills. This support has come from colleagues whose doors have always been open to answer our questions and resolve our frustrations, and from members of the Comparative Politics and Public Policy Research Cluster and the Political Economy Research Cluster for feedback on earlier versions of the empirical analysis presented within this book. Third, we have been extremely fortunate that the long-standing and enthusiastic commitment to interdisciplinarity between colleagues in Political Science and Economics at York has been maintained. The School of Politics, Philosophy, and Economics supplies intellectual foundations for this interdisciplinarity, and the Economic History and Political Economy Research Cluster and Centre for Applied Macro-Finance (CAMF) practical mechanisms that bring together colleagues from both sides of University Road. Without the willingness of colleagues within CAMF to de-mystify the finer points of bond market performance and analysis, this book would not have emerged in anything approaching its present shape.

We extend a heartfelt 'Thank You' to the colleagues who have helped us along the way, and whose input has helped clarify and strengthen the final work. Any shortcomings, of course, are owned by us.

To Mum, Dad, Rachel, and Eleanor

Para Angie

Demystifying Climate Finance: A Note on Bond Market Terminology

We were motivated to write this book because of the obvious contemporary importance of climate finance. Without adequate financing, efforts to meet the challenge of climate change cannot succeed. We were driven by a desire to identify and understand factors that might accelerate the flow of financing to support climate transition, and lessen overall greenhouse gas emissions and resultant global warming. Through the book, we put forward analyses of domestic government spending on climate transition, the provision of climate aid, and the operation of green bond markets. We have written this book to be accessible to an interested reader, with little prior knowledge of these topics. When we focus on the operation of green bond markets, a financial product of huge quantitative importance to overall climate finance flows, we range into a world that will be less familiar to many readers. We have written these green bond market-focused chapters with the less familiar reader in mind. While detailed knowledge of bond market terminology is not required, we provide this glossary for readers wishing to expand their familiarity and refresh their understanding.

ask-price, bid-price, mid-price. The ask price refers to a baseline price that a **bond** holder communicates to potential investors who may want to purchase their bond holding. When publishing an ask price, the holder makes a commitment to sell their holding if this price is met. The bid price refers to the price that an interested investor publicly commits to buying an asset at, should a holder be willing to sell at that price. The mid-price refers to the mid-point between these ask and bid values. So, with an ask-price of US$1.06m and a bid price of US$1.04m, we would have a mid-price of US$1.05m. These prices are listed in dynamic trading platforms such as the London Stock Exchange International Order Book. Once their positions have been listed in an Order Book, potential sellers and potential buyers can make contact to complete a transaction. Market data platforms such as

Bloomberg or Refinitiv provide analysts with access to a wide range of Order Book market data, including daily ask-, bid-, and mid-price data.

basis points, bps. In ordinary use, when talking about percentages we tend to use whole numbers (e.g. 5 percent). But financial market participants, because it can be useful for them to talk precisely about small differences in percentage values, commonly use a language of 'basis points' to, in effect, talk about percentage values down to two decimal places. So, to describe the difference between an interest rate of 4.31 percent and an interest rate of 4.39 percent, a market participant may refer to a difference of 8 basis points (commonly abbreviated, in written form, to 8 bps).

Bond. A financial instrument through which a borrower seeking investment funds is matched with a syndicate of investors seeking an investment opportunity. A bond is, in essence, a form of long-term loan; a promise made by the issuer to repay a borrowed sum of money with supplementary interest payments, over an agreed period of time. To launch a bond, the **issuer** (working via an **underwriter**) announces the quantum of cash they wish to borrow, and an annual rate of interest they are willing and able to pay. Typically, a syndicate of investors will each supply a portion of the total investment funds, and receive back an equivalent share of the principal repayment and interest payments. Investors are able, if they so wish, to sell-on their portion of the bond, with the subsequent owner then receiving the share of principal repayments and interest payments.

Coupon. The annual rate of interest that an issuer of the bond agrees to pay to the purchaser of the bond periodically across the lifetime of the bond, according to an agreed schedule of payments. The coupon rate can be fixed by the issuer at the point of **issuance** across the bond's lifespan, can be varied over time according to a schedule that was agreed at the point of issuance, or can be varied over time in response to market conditions.

Green bond. A bond where the issuer has made a commitment to use the cash raised to invest in an activity that will generate some form of positive environmental impact. A green bond may be labeled as such by virtue of an issuer's self-declaration within their bond issuance paperwork, or by a process of external verification through which an external third-party to the bond issue evaluates and assures the environmental impact credentials.

Greenium. A reduced cost of borrowing achieved by the issuer of a green bond that accrues by virtue of the bond's 'green' designation. There is a vibrant body of scholarship across Financial Economics exploring factors that support or inhibit the achievement of a greenium, with no clear consensus on whether green bonds systematically receive improved borrowing terms off the back of their designation.

Issuance. The completion of the process of launching a bond, at which point in time cash from investors is transferred to the **issuer** in return for the issuer's promise to meet a specified repayment schedule.

Issuer. The corporation or government who issues a given bond to raise investment funds. The characteristics of an issuer can shape the language used to describe a bond. Where the issuer is a national government the bond is commonly referred to as a 'gilt', reflecting the historic practice from some governments of writing down their bond characteristics and repayment schedules on fancy gilt-edged paper. Where the issuer is perceived to have such weak finances that their capacity to meet bond repayments is in question a bond may be referred to as a 'junk' bond, a term ultimately derived from a Middle English term for human detritus.

Lifespan. The period of time between bond issuance, and the final scheduled repayment by the issuer of the principal and interest.

Mid-yield. A measure of a bond's value within the **secondary market**, on a given day. Detailed understanding of mid-yield is not necessary to understand the general insights of the book - we only analyse mid-yield values in Chapter Five, and you will be able to understand the flow of Chapter Five without understanding the detail of mid-yield calculation. But for those who would like to develop a deeper understanding, the mid-yield is a value that captures the **mid-price**, expressed as the return on investment that would be enjoyed by a bond holder if they were to purchase at the mid-price. A worked example can be used to clarify this mid-yield measure. Imagine that, in the **primary market**, Investor A initially paid US$1m for a bond with a one-year repayment schedule and a fixed rate of interest at 10 percent. A repayment flow of US$1.1m would ultimately flow to Investor A, and the bond would be said to have a **yield** of 10 percent (the US$1.1 total return in the pipeline to Investor A is 10 percent higher than her US$1m initial investment). Should Investor A wish to quickly sell her asset before any repayments have been received, she might ask other market participants for US$1.06m (wishing to turn a quick profit on her US$1m outlay). Not wanting to feel like he was being fleeced, Investor B might be willing to bid only US$1.04m to take the bond off A's hands. These contrasting valuations from Investor A and Investor B give us a **mid-price** of US$1.05m. The mid-yield, in this case, is 4.76 percent, because the US$1.1m repayment flow would generate a 4.76 percent return on a hypothecated **mid-price** of US$1.05m. The utility of the mid-yield is that it provides a measure of the real investment value of a given financial asset within the market conditions that prevail at a given point in time, which is readily comparable with other assets and across other points in time.

Plain vanilla bond. A bond with a coupon rate that is fixed across the bond's lifetime. The 'plain vanilla' designation is used to reflect the mainstream conventionality of this fixed coupon structure.

Primary market. Interactions between the bond issuer and investors who are potentially interested in purchasing a portion of the bond or who do purchase a portion of a given bond, leading up to the bond's **issuance**.

Secondary market. Interactions pertaining to the re-sale of a bond holding between investors who at a given point in time own a portion of a given bond, and investors who at that point in time are potentially interested in purchasing a portion of that bond or who do at that point in time purchase a portion of that bond. The average value of a trade in the secondary market is relatively high, typically in the ballpark of US$2m.[1]

Underwriter. A large financial institution contracted by the bond issuer to manage a given bond issue. The underwriter will suggest the rate of interest that may need to be offered by the bond issuer to secure the required quantum of investment funds, and liaise across their networks to find potential investors in the bond. The underwriter often will purchase a portion of the bond, either because they were unable to secure insufficient demand across the primary market and so had to mop-up an unsold portion of the bond, or because of a desire to hold the asset for a shorter portion of time prior to trying to generate a profit through re-sale, or because of a desire to hold the asset for a longer period of time to accrue return on investment through the repayment of principal and interest from the **issuer**.

Yield. The rate of return accrued by a bond holder on their investment in the bond, expressed as the proportional difference between total return and total outlay. Where, for example, an investor paid US$1m and eventually received US$1.05m back, she would have received a yield of 5 percent.

[1] See 'So why do bonds trade OTC?', International Capital Market Association website, available at https://www.icmagroup.org/market-practice-and-regulatory-policy/secondary-markets/Bond-Market-Transparency-Wholesale-Retail/So-why-do-bonds-trade-OTC-/#:~:text=Second%2C%20the%20average%20size%20of,as%20%E2%82%AC7500%20or%20less. Accessed 8th May, 2025.

1. The Political Economy of Climate Finance

1.0. Introduction

Across the long sweep of human history, finance has played a central role in shaping and re-shaping social organisation. John Maynard Keynes (1930: 11-12) suggests that human experimentation with finance may well have emerged with the thawing of the last glacial period, some 12,000 years ago. As this earlier global warming rendered previously inhospitable climes more amenable to human existence, conditions for a revolution in agricultural productivity emerged. In this context, institutions for organising and expanding the use of finance became an integral component of increasingly complex social systems. For Keynes, the detail of these origins of finance sit within pre-history, 'lost in the mists when the ice was melting'.[2] Historic global warming catalysed the emergence of finance; now, in response to the pace of contemporary human-induced climate change, finance is being repurposed as a tool to push-back against global warming. As glacial ice continues to melt away, in this book we explore the institutions that support the unlocking of accelerated flows of climate finance.[3]

There is a tendency within significant bodies of social science scholarship to view finance, at a foundational level, to be inimical to the public good. Karl Polanyi (2001: 48-66) suggests to us that the emergence of money, and the attendant rise of more market-based societies, brought with it moves towards more atomised social systems. Previously, social bonds and principles of diffuse reciprocity

[2] As covered by Davies (2002: 34-65), the earliest known written records of accounting come from Mesopotamia and date to around 3,100 BCE. Cuneiform tablature dating from around 1,800 BCE show that, in Babylon, banking operations by temples and large landowners were sufficiently institutionalised that political authorities provided standard overarching rules. The Code of Hammurabi, named after the Babylonian ruler from 1792-50 BCE and taking the form of carved diorite standing seven feet tall (currently in the Paris Louvre), arguably constitutes our earliest known example of a banking regulatory framework.

[3] Throughout the book, we use the terms climate finance, climate transition finance, and green finance interchangeably. While the precise meaning is contextually determined, we in general take the terms to refer to financial flows that support movement toward reduced greenhouse gas emissions and more sustainable modes of living and working.

may have structured patterns of consumption, such that surplus was communally shared and individual want forestalled. With financialisation, there is increased capacity on the part of an individual with surplus to monetise this surplus, and effectively self-insurance against their own future want. In this context, however, scope may be increased for the needs of others to be left unmet. At a system level, finance has been said to value profit over social purpose. More heavily financialised social systems may prioritise the needs of those with higher incomes, and fail to deliver adequately for those on lower incomes (Montgomery 2019, Clegg 2017, Federici 2014, Soederberg 2014). While a countervailing view holds finance to provide a key to broadened efficiency and opportunity (Ahmad and Malik 2009, Fama 1970), critical interpretations continue to sit at the mainstream across the social sciences (Huber et al 2022, Pellandini-Simányi 2021).

Mirroring these critical reflections on the problematic relationship between finance and the public good, we have interrogations that more specifically explore tensions between finance and the environment. For Bruce Rich (1994), there is a definitive tendency on the part of institutionalised finance toward 'mortgaging the earth'; in the drive to achieve the desired returns on investment, ecological security is sacrificed at the altar of enhanced productive capacity. In addition to a range of academic voices warning about the danger of financial institutions' tendency toward 'greenwashing' (Galletta et al 2024, de Freitas Netto et al 2020, Lyon and Maxwell 2011), there is awareness within the sector itself that public relations discourse on sustainability may not always match up to the reality of environmental performance.[4] The WhatsApp messages sent by a climate finance practitioner to Aneil Tripathy (2022: 33) during ethnographic research into climate finance points to significant internal scepticism: 'Sustainable finance, as you know, is mostly self-serving BS', wrote a research subject to the ethnographer. Climate finance, clearly, is a topic surrounded by controversy.

As a starting point for this book, we acknowledge the inherent tensions that exist between current systems of government and market-based financing on the one hand, and ecological and planetary boundaries on the other. These existing systems for allocating financing are perpetuating

[4] See 'Greenwashing in Financial Services: Tips for Avoiding Common Pitfalls', Deloitte LLP website, available at https://www2.deloitte.com/nl/nl/pages/financial-services/articles/greenwashing-in-financial-services.html. Accessed 20th February, 2024.

cycles of over-intensive production and consumption, and driving our current trajectory towards emissions levels that will push global warming well above the agreed target of 1.5-2°C. And it is precisely because of the brokenness of the current system that we need to place enhanced attention on unlocking accelerated flows of climate finance, flows that constitute an absolutely necessary foundation for efforts to move from our current carbon-intensive societies towards more sustainable modes of working and living.

Through this book, we take a broad approach to climate finance. Our breadth of scope is shaped by shifts in thinking about the key sources of climate financing. In the earlier years of the 21st century, conventional wisdom held that governments would be central actors for financing climate transition. By the Paris Agreement 2015, consensus had shifted such that government financing would need to be supplemented by government ability to catalyse market financing. And as the sluggish nature of the long-term recovery from the Global Financial Crisis in the late 2000s and the fiscal constraints from the global Coronavirus response in the early 2020s become clear, there is a wider acceptance that market-based financing will be of preponderant importance.[5] We know that government spending, interaction between states and markets, and market-driven dynamics all matter in shaping the future of climate finance, and we explore all three dimensions across this book.

The analytic focus of our study is climate finance. The United Nations Framework Convention on Climate Change (UNFCCC 2014: 5) offers the following definition:

> Climate finance aims at reducing emissions, and enhancing sinks of greenhouse gases and aims at reducing vulnerability of, and maintaining and increasing the resilience of, human and ecological systems to negative climate change impacts.

We remain guided by this definition throughout our study. With this holistic conceptualisation, we are able to offer a review of the breadth of flows from multiple sources that can contribute to supporting climate transition. We know that, ultimately, the effectiveness of our collective response to the challenge of climate change will be contingent on financing coming from a combination of public and

[5] For an exploration of the relative roles of, and interaction between, public and private climate financing, see CPI (2023).

private sources, and through the book as a whole we probe foundations for effectiveness across each of these modes of climate finance.

Overall, through this study we identify institutional factors that support effective climate finance performance across these multiple levels and in these different forms. Our baseline view of effective performance here is one of scale and efficiency; larger overall climate financing flows, or enhanced demand for green bonds and reduced borrowing costs for green bond issuers, are taken as indicators of effective performance. Consequently, where an institution is shown to be contributing to larger flows of climate financing or to enhanced green bond demand and reduced green bond borrowing cost, we hold that institution to be contributing to more effective climate finance performance. The efficacy with which resources are translated into on-the-ground interventions and results is, of course, of huge real-world importance. Our approach is to explore factors that determine the overall size of the climate financing envelope, and we very much acknowledge the vital work being done elsewhere to push for equitably and efficacious translation of money into action.[6]

The purpose of this introductory chapter is to provide a contextualisation of our study, and to give an overview of the headline message that we distil from across the subsequent empirically-focused chapters. Towards this end, we develop this chapter through the following structure. In the first section below, we provide an introduction to the regime governing climate change and climate financing. We then in the second section lay conceptual foundations for our study by presenting our engagement with scholarship on institutions and political economy, and outline the approach to institutions we deploy through the book as a whole. In the third section, we review key scholarship on the political economy of climate transition and financing, to contextualise our overall contributions. We conclude with an overview of the remaining chapters of the book. By identifying institutions that impact on climate finance effectiveness, we lay foundations for what we term 'evidence-based optimism'. By understanding the ways in which party politics, election cycles, and cultural values shape government climate spending and aid, and the ways in which international organisation operations, central bank interventions, and underwriter characteristics shape green bond

[6] While these efforts are innumerable and internationally dispersed, at the global level initiatives such as Climate Action Tracker and Climate Action Network deserve particular mention in this regard.

market dynamics, we can seek to extend institutional practices that work, and challenge institutional practices that don't. Optimism can come from the enhanced capacity for action that comes from a strengthened evidence base regarding institutional foundations of effective climate finance.

1.1. Climate change and climate finance

The focus on climate transition and climate financing has risen in tandem with broadening understanding of the scale of global warming, and the climactic change being induced by human activity. In the paragraphs below, to contextualise the subject-matter of our study, we provide a brief overview of the emergence of scientific consensus over the role of human activity in accelerating climate change, and explore contemporary developments in the governance of climate change and climate financing.

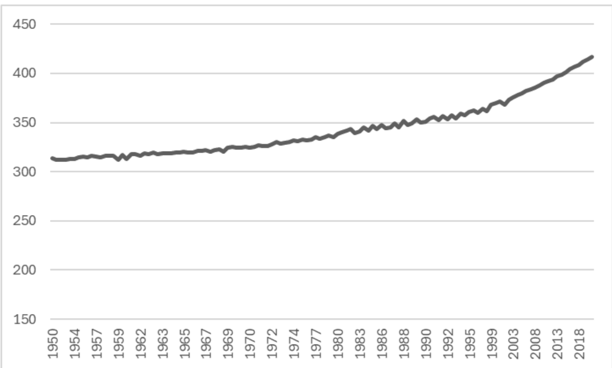
The intellectual foundations upon which understanding of global climate change came to be built run deep. It was around two centuries ago that, in 1827, Jean-Baptiste Joseph Fourier published his 'Remarks on global and planetary temperatures'. Fourier's work described, in embryonic form, the phenomenon that later came to be known as the 'greenhouse effect'. With insights drawn from a combination of practical experiment and abstracted reason, Fourier suggested that the earth's atmosphere functioned to capture heat being radiated from the sun (Peirrehumbert 2004).[7] A significant refinement to understanding of this atmospheric effect was introduced some three decades hence, through Eunice Foote's experimental observations of gaseous responses to thermal radiation. Specifically, in experiments conducted in 1856 and reported the same year in the *American Journal of Science and Arts*, Foote demonstrated the elevated heat absorption capacity of CO_2 relative to other atmospheric gasses. Moreover, by hypothesising that increases in the atmospheric concentration of

[7] In his review of Fourier's work, Pierrehumbert provides a translation of 'On the Temperatures of the Terrestrial Sphere and Interplanetary Space' as a supplementary file.

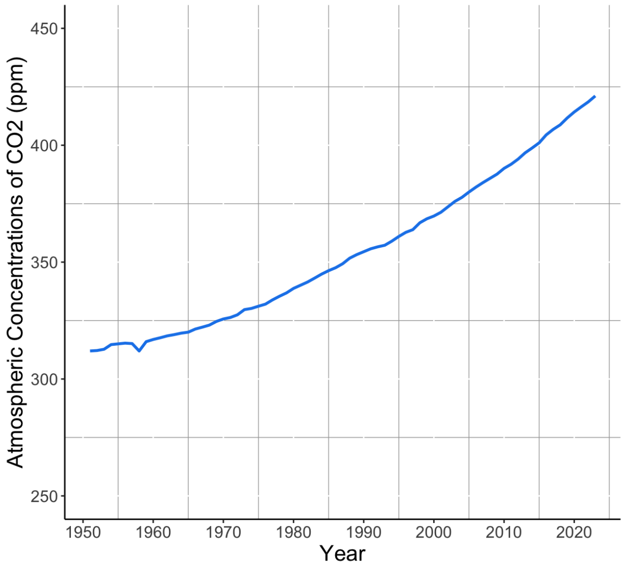
CO_2 could be associated with increases in atmospheric temperature, Foote pointed early attention to climatological interactions that are of foundational importance to climate change (Jackson 2020).[8]

From these intellectual foundations, it was necessary to demonstrate two phenomena to establish a robust understanding of human-induced climate change. First, it was necessary to show that human activity through the modern era had led to meaningfully elevated concentrations of CO_2 and other greenhouse gases within the earth's atmosphere. And second, it was necessary to show that global temperatures were, in aggregate, increasing. By the 1960s, technical capacities for establishing both of these dynamics were well on the way to being laid (Alley 2015, Jouzel 2013, Menne 2012, Robinson and Henderson-Sellers 1999), and by the turn of the 21st century it was clear that both CO_2 concentration and global temperatures were rising at pace (see Figures 1.1 and 1.2).

Figure 1.1: Atmospheric CO_2 concentration (ppm), 1950-present

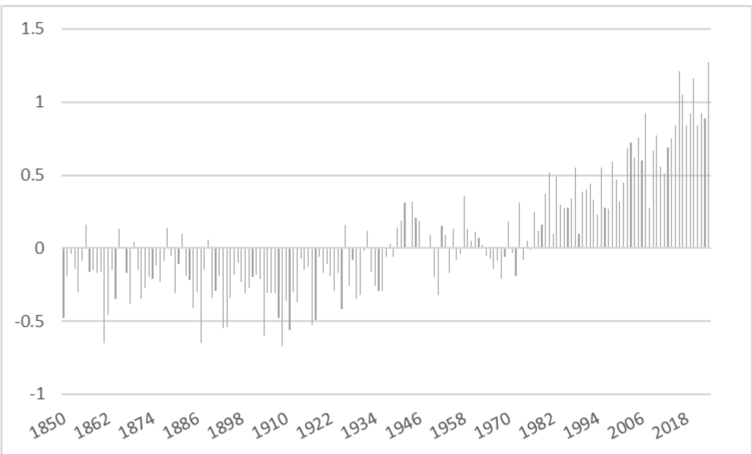


[8] In his intellectual history of this aspect of climatological science, Jackson (2020) notes that - as all too often with female contributions - Foote's work was largely ignored or forgotten, with John Tyndall's study some three years later generally taken as delivering proof of the heat-absorbing capacity of CO_2 and its role in delivering a greenhouse effect.

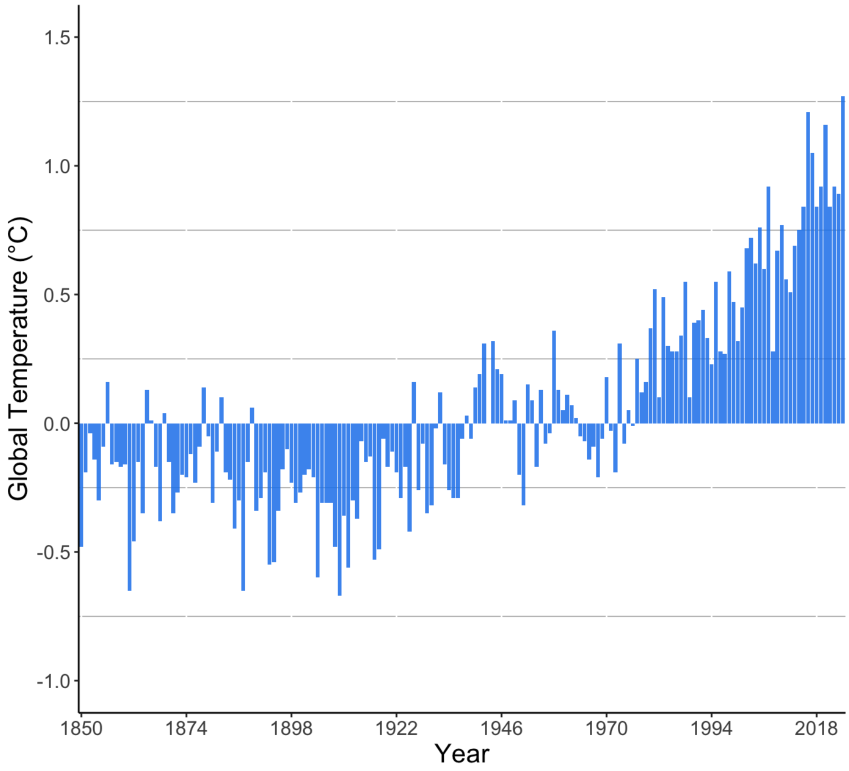


Source: US Environmental Protection Agency.[9]

Figure 1.2: Global average surface temperature (ºC)



[9] See US EPA website, available at https://www.epa.gov/climate-indicators/climate-change-indicators-atmospheric-concentrations-greenhouse-gases#:~:text=This%20figure%20shows%20concentrations%20of,monitoring%20sites%20around%20the%20world. Data collected 1st March, 2024.



Source: US National Centers for Environmental Information.[10]

The translation into policymaking arenas of scientific consensus that increasing atmospheric concentration of human-sourced greenhouse gases was raising global temperatures took place incrementally. Over time, epistemic communities formed that sought to pool data and insights relating to human-induced climate change, with transnational collaboration often serving to catalyse these processes of network-building (Gough and Sharkey 2001, Haas 2015, Maliniak et al 2021). The World Climate Conference constituted a signal event in this regard. Held in 1979 under the auspices of the World Meteorological Organization (WMO) and the United Nations Environment Programme (UNEP), the World Climate Conference brought together scientists primarily from across North America and Europe. Over a subsequent series of workshops, sufficient agreement was reached amongst participants over the nature of the climate change challenge that, in 1985, a collective call to action was issued. The Villach Declaration declared that:

[10] See NCEI website, available at https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/global/time-series. Data collected 4th March, 2024.

> '[I]n the first half of the next century a rise in mean temperature will occur that is greater than any in man's history… [S]cientists and policymakers should begin active collaboration to explore the effectiveness of alternative policies and adjustments' (cited in Agrawal 1998a: 608).

The long road toward responding to the challenge of climate change was begun. Coming in the wake of the Villach Declaration, the formation of the Intergovernmental Panel on Climate Change (IPCC) constituted an important next step in this journey.

Following the Villach Declaration, the WMO and UNEP sought to embed the linkage between scientific and policymaking communities with the formation of the IPCC. The founding purpose of the IPCC was to explore and report back to UNEP member governments on the scientific understanding of climate change, the impacts of climate change, and responses to climate change. Three parallel Working Groups were established to address these three areas of concern. A protocol developed that saw the IPCC bureau identifying, with member government input, the teams to lead the report writing and contribute to the review process. Working Groups would follow an assessment cycle schedule, beginning with a plenary session approving upcoming focus and workflow, running through a process of drafting, extensive peer review, and redrafting, and concluding with a plenary session line-by-line review of the 'policymaker summaries'. The IPCC procedures represent an attempt to balance an expert-led approach with an acknowledgement that buy-in from national political and governmental actors was needed for progress towards required policy adjustment to be realised (Agrawal 1998a, Agrawal 1998b). As had been hoped for by its initial backers, IPCC outputs paved the way for international agreement to be reached on measures to combat climate change.

The Panel's first Assessment Report communicated certainty that human activity was increasing the concentration of greenhouse gases in the atmosphere, and predicted that these rising concentrations would increase global temperatures by on average 0.3°C each decade through the 21st Century (IPCC 1990: xii). Driven by this message, through 1992 a UN International Negotiating Committee laid foundations for what was to become the UN Framework Convention on Climate Change (UNFCCC). The UNFCCC was mandated to stabilise greenhouse gas concentrations 'at a level that would prevent dangerous anthropogenic interference with the climate system', with its annual Conference of the Parties (COP) bringing signatories together to assess progress and establish commitments. The 1997 Kyoto Protocol and 2015 Paris Agreement functioned as extensions to the UNFCCC. The former committed a grouping of 37 industrialised countries and the European Union to, by 2012, reduce greenhouse gas emissions by an average of 5 percent relative to 1990 levels. While these targets were largely met, annual global emissions in fact rose by some 40 percent by the end of the Kyoto census period, owing to the non-participation of leading emitters including the US and China in the Protocol. In marked contrast, the Paris Agreement was designed to be universal in scope, and to periodically ratchet-up ambitions to support greenhouse gas emissions that would lead to 1.5-2°C average global increases through to 2100.[11]

Under the Paris Agreement, signatory countries set their emissions reduction targets through Nationally Determined Contributions (NDCs). At present, 165 countries have lodged NDC plans and targets with the UNFCCC secretariat. Unfortunately, the UNFCCC assessment suggests that current NDC plans are not, in aggregate, compatible with limiting global warming to 2°C, let alone the headline ambition of 1.5°C: 'This information', notes the UNFCCC, 'implies an urgent need for a significant increase in the level of ambition of NDCs' (2021: 6). NDC plans are to be renewed every five years, with each iteration providing more exacting targets. Given the collective overshoot in greenhouse gas emissions identified by the UNFCCC, it is clearly vital that this ratchet has dramatic, and rapid, effects.

Our understanding has grown substantially of the real-world impacts that will come from human-induced climate change. Drought, extreme weather events, famine, flood, rising sea levels, and wildfire will continue to be increasingly common, and increasingly deadly, occurrences. Those with the fewest resources, who have contributed the least to creating the climate emergency, will often be left to face the heaviest impact. The effects of climate change will remain extraordinarily variegated; some of the world's currently colder climes may become more temperate and enjoy a boost to agricultural productivity, while many of the world's warmer and more arid regions may move toward absolute uninhabitability.[12] Rigaud et al (2018) estimate that, across Latin America, South Asia, and Sub-Saharan Africa, some 147 million people may by 2050 be forced to leave their homes as a result

[11] Technically, the 1.5-2°C target refers to the average global temperature over a 20 year period.
[12] For a useful review of scholarship on the impacts from climate change, see Wallace-Wells (2019).

of climate change. Climate change may, over coming decades, generate a large proportion of overall migration flows.[13]

To respond effectively to the challenge posed by climate change, rapid decarbonisation of human activities is required. While countries, regions, municipalities, and individual households face different opportunities and challenges, some required areas for change are clear across the world's currently highest-consuming societies. The carbon intensity of electricity grids needs to be drastically reduced, through greater use of renewables, and more effective demand management to smooth-out inefficiencies generated by unpredictable peaks and troughs. Transportation systems need to be remade, so that the internal combustion engine is replaced as the mode of choice by active travel, public transportation, and electric vehicles. Buildings need to become more efficient, requiring less energy for heat during cool periods and for cooling during times of excess heat. Diets need to move toward more plant-based options, with attention paid to efficiency of growing and distribution networks. Approaches to land management need to maximise the scale of carbon sinks that capture and store greenhouse gas and so attenuate their contribution to atmospheric heating.

Looking across the contemporary climate change regime, there is clear and significant variation in policymakers' levels of engagement and ambition. Focusing on national-level governments, the inconstant commitment from the US has attracted particular attention. Climate change in the US remains a highly partisan issue domestically, with political divisions contributing to the US' non-ratification of the Kyoto Protocol in the late 1990s. More recently, partisan cleavages saw President Donald Trump's withdrawal from the Paris Agreement in January 2025, after his initial 2020 withdrawal had been reversed under the administration of President Joe Biden. Germanwatch, a leading environmental non-government organisation, remains sufficiently underwhelmed by the international response to climate change that it continues to leave the top of its Climate Change Performance Index (CCPI) unfilled. In the most recent iteration of the CCPI Denmark is the highest-ranked country, with a score of 75.59 on account of an assessment of its NDC emissions

[13] As noted by Kaczan and Orgill-Meyer (2020), a wide range of predictions have been made regarding current and future migration driven by climate change.

targets, renewable energy, and climate policy. However, to emphasise the lack of appropriate leadership from any quarters, Denmark is ranked 4th.[14]

Climate change transition requires finance. And while it is not possible to put a rough-and-ready figure on the climate finance required to support transition towards pathways that fit with the Paris Agreement commitment to limiting global warming to 1.5-2°C, we can gain a sense of the quantum of resources needed by reviewing some key contributions. McCollum et al (2018) offer an assessment of the financing needed for a global energy system transformation consistent with Paris Agreement goals. To decarbonise energy systems sufficiently to meet with Paris Agreement commitments, total investment of around US$2.3tn would be needed globally each year through to 2050, representing increased investment of around US$0.8tn per year above a 'business as usual' baseline. Turning specifically to the needs of developing countries, the UNFCCC (2021b: 7) estimates a gap of some US$600bn per year in climate financing when analysing NDC documentation. While there is little precision over the scale of investment required from private sector actors, there is a clear assumption that this source will supply a large portion of additional climate finance required.[15]

When reviewing current climate finance flows, the UNFCCC (2022:1549) notes highly divergent patterns across countries and regions, and a slowing rate of growth. We know that climate aid, domestic government climate spending, and market-based financing will continue to constitute key pillars of climate finance. Through the book's empirical chapters, we survey climate finance flows across these sources. While observed volumes are below the necessary scale, we see some headline evidence of progress. On aid, we see that the much-vaunted US$100bn per year benchmark has recently been reached (Gabbatiss 2024). On domestic government climate finance, we have seen major commitments under the US Inflation Reduction Act and the EU New Green Deal, albeit with both being subjected to varying degrees of political contestation. Green Bond issues, which represent

[14] See Climate Change Performance Index website, available at https://ccpi.org/wp-content/uploads/CCPI-2024-Results.pdf. Accessed 4th March, 2024.
[15] In the case of the UK, for example, the government has estimated that most of the additional £46bn per year required to meet climate change targets will come from the private sector. See Committee of Public Accounts (2023: 3).

the major mechanism for catalysing private climate finance, stood at around US$500bn in 2024,[16] a scale that is particularly noteworthy given the relatively recent movement, since the 2010s, of this asset class toward the mainstream.

Scientific consensus on the reality of human-induced climate change is clear. Policy-maker commitment to introduce changes needed to limit planetary warming has been demonstrated, although there is a pressing need for more rapid decarbonisation to be locked-in and delivered. Private sector action is accepted as a necessary component of successful transition to sustainable patterns of working and living. And we see that, while not yet adequate to the task of ensuring that Paris Agreement commitments can be met, significant sources of climate finance have begun to flow. The task we set ourselves through this book is to identify and explain variations in these flows of climate finance, so that we can more fully understand the factors that are pushing the climate finance envelope and enabling more effective response to the climate emergency.

1.2. Understanding institutions - our approach

A central insight that we develop through the book as a whole is that, for flows of climate finance, institutions matter. Institutions provide the political economic topography from which climate finance emerges, and systematic study can identify particular features of the landscape that lead to higher flows and greater capacity for action. In the paragraphs below, to provide a thematic contextualisation of our study, we review approaches to studying institutions that have been developed across core politics and political economy literature. We close the section by introducing the approach to understanding institutions that informs our study of climate finance.

Historically, the study of politics was to a very large extent constituted by the study of institutions. With the consolidation of politics as an identified field of inquiry within higher education,

[16] This figure is reported by the Climate Bonds Initiative, who apply a relatively strict definition of 'green bond' that eliminates a significant volume of the assets reported by other analysts of sustainable finance. See CBI (2025).

interest in cross-national institutional variation gained particular prominence. Scholarly focus was placed on the causes and consequences of differences in civil service organisation, electoral system, judicial system, legislative structure, party system, and state-economy relations (Rothstein 1996: 135). There has been a strong classificatory and typological drive within this institutional scholarship. We see, for example, Hall and Soskice's (2001) differentiation between 'liberal market economies' and 'coordinated market economies', with the former characterised by union density and job-specific skill levels that are are relatively low, inter-firm relationships that are more competitive rather than collaborative, and employer-employee relationships more adversarial. We see in Esping-Andersen (1990) the division of capitalist political economies into 'corporatist', 'liberal', and 'social democratic' worlds of welfare, based on the level of decommodification, social stigma, and stratification associated with social safety net provision.

Institutional architectures are, across the discipline, taken to provide defining characteristics of substantively different political systems and policy constellations. They are also taken to play vital roles in conditioning processes of change. Through the 1990s, we saw a surge of interest in the relationships between institutions and democratic transition and consolidation, motivated by Latin American moves from military dictatorship to democracy through the 1980s, and the shifts away from one-party rule across Eastern and Central Europe (Linz and Stepan 1996). Equally vibrant debates took place over the role played by institutions in catalysing economic development. Chalmers Johnson (1982) pointed to the planning capabilities of the Ministry of International Trade and Industry and Trade, and its convening power across core sectors of economic activity, as helping explain 'the economic miracle' of Japan's post-war transformation. Discussion of the institutional foundations of the 'developmental state' remains ongoing (e.g. Chang 1999, Wade 2018, Woo-Cummings 2019, Haagh 2019).

A repeated point of observation from scholarship on institutions in political economy is of the potential for institutions to facilitate the emergence and embedding of solutions to otherwise intractable problems. The value of Ostrom's (1990) work on institutions was to show that, in practice, groups were able to create institutional structures that supported effective common ownership. By serving to open space for individuals to make binding contracts that constrain individual behaviour in accordance with long-term visions of the public good, institutions for Ostrom played a vital role in structuring capacity to effectively govern scarce shared resources.

The depth of consideration given to institutions within the study of politics has been such that, periodically, attempts have been made to rationalise and bring order to an increasingly diverse body of scholarship. A particularly useful intervention in this regard is provided by Kato (1996: 553), who reminds us that a key dividing line in institutionalist scholarship relates to conceptualisations of the relationship between institutions and individuals. On the one hand, rational choice institutionalism lodged that the individual level provided the ultimate source of agency; while institutions shape the nature of interactions that occur, it is individuals who carry preferences into these spaces in which they seek to shape outcomes. On the other hand, more sociologically-orientated institutionalism viewed agency as being constituted by the complex interplay between individuals and institutions. Here, individual preferences, their ideas about favoured resolutions to the particular socio-political challenges and contexts they find themselves confronted by, are shaped in part by institutional context. Across a range of fields, we see acknowledgement of the complementarity of these more rationalist and sociological variants. Whether in the realm of electoral studies (Birch 2008, Sudulich and Trumm 2019), feminist scholarship (Waylen 2009, Krook and Mackay 2011), or international politics (Jupille et al 2003, Nielsen et al 2006, Weaver 2007, Clegg 2013), pragmatic acceptance has emerged that both individual preferences and institutions matter. Institutions provide the terrain on which interactions occur that seek to translate individual preferences into outcomes, and two-way relationships can exist between the individual and the institution. Institutions can condition individual ideas and behaviour, and individual ideas and behaviour can recondition institutions.

Sitting alongside these nuanced views on the ontological properties of institutions are a range of competing definitions of the phenomenon. To help us navigate the contested meaning of institutions, it is useful to consider the marker laid down by North (1990: 3-4). North offers us a broad-based account, when framing his study of institutions and economic performance:

> Institutions are the rules of the game in a society, the humanly devised constraints that shape human interaction… Institutions reduce uncertainty by providing a structure to everyday life. They are a guide to human interaction, so that when we wish to greet friends on the street, drive an automobile, buy oranges, borrow money, form a business, bury our dead, or whatever, we know (or can learn easily) how to perform these tasks… [I]nstitutions define and limit the set of choices of individuals.

For North, institutions run the gamut of social forms. They can be highly formalised; codified 'rules of the game' that permit or prohibit certain behaviours under certain conditions. And they can be entirely informal; the conventions and codes of behaviour that provide the subconscious scripts and rules of thumb that pilot us through everyday life.

North's approach has been highly influential, across the social sciences in general. However, the analytic value of such an all-encompassing vision has been subjected to considerable debate. In the words of Rothstein (1996: 145), if such a maximalist approach to institutions is adopted, 'the drawback is that the concept risks becoming too diluted… [I]f it means everything, then it means nothing'. If, as most observers would accept, the function of and impact from a national constitution is categorically and foundationally different to the functioning of and impact from culture, habits, and routines, our capacity to clearly identify and explore these points of difference will be enhanced through the development of a more fine-grained conceptualisation. The way in which concepts are defined can either dial up, or dial down, the precision of the analytic insights we are able to generate.

The alternative conceptualisation is provided by March and Olsen's notably more restrictive approach to institutions. March and Olsen (1989: 16) foreground what they term 'administrative organisations' in their study of political processes and outcomes. Anchoring their work in the earlier 'institutional economics' tradition of John Commons, Thorstein Veblen, and William Willoughby, March and Olsen (1-2) take as their point of entry an observation that 'many of the major actors in modern political and economic systems are formal organisations'. By deploying exacting criteria relating to coherence and autonomy, March and Olsen suggest that we can consider institutions to be entities that exert independent influence across economic, political, and social outcomes. Administrative organisations function as relatively stable features of political economic landscapes, which can identify system-level characteristics and points of difference, and can shape variation in performance and outcome. As elaborated by March and Simon (1993: 22-23), the relative structural stability of these formalised organisational entities facilitates their capacity to interact in an orderly manner with their wider environments. It can be useful, following Scott (2014: 106-7), to conceptualise these structured institutions as existing within 'organisational fields', entities constituted by 'a collection of diverse, interdependent organisations participating in a common meaning system'.

Our approach to understanding institutions builds on these contributions from March and Olsen, March and Simon, and Scott. When examining flows of climate financing, we are interested in the influence from the type of institutions pointed to by these contributions. We build on these sources in the conceptualisation of institution that we deploy across our study. For the purpose of our study, we define institutions as *organisational structures or entities that display a sufficiently high level of coherence and autonomy to exert meaningful influence over significant political economic outcomes*. In the context of our study, the political economic outcomes of interest are state- and market-based flows of climate financing.

Given our broad-based focus on state- and market-based modes of climate financing, we encounter a range of institutions in our analyses. Across our empirical work, we identify institutions including political parties, elections and election outcomes, national values, international organisations, central banks, and financial market underwriters as exerting significant independent influence over climate finance performance. In many cases, we find a positive contribution being made by the identified institution to the volume of efficiency of climate finance flows. Where we consider the World Bank influence on green bond market development and the European Central Bank on green bond market performance, our findings point to potentially constraining effects from these institutions' activities and suggest a need for further interrogation and operational refinement.

Overall, in advancing these insights, we orient our study by taking direction from the pragmatic conceptualisation of institutions offered by Olsen, March, Simon, and Scott, and by drawing inspiration from Olstrom's insights into the power of institutions to allow for more effective governance of shared resources. We know from existing scholarship that institutions can have a critical influence over trajectories of political economic transformation. We here build from this body of work to show how institutions are providing foundations for effective climate financing, and to show where there is space for improved institutional performance.

1.3. Situating our contribution

Commensurate with the importance of the issue, there is a large and vibrant body of scholarship that seeks to interrogate responses to climate change, and evaluate the adequacy of our current and future adjustments toward more sustainable forms of living and working. Our focus through this book on the political economy of climate financing is relatively unusual, as demonstrated by the scarcity of scholarship explicitly focusing on this topic. When situating our study, we here consider contributions that specifically focus on climate financing, and also works that focus more broadly on climate change commitment and performance that are of wider thematic relevance.

The wide bloc of scholarship on the political economy of climate change is, of course, divided by differences in empirical focus, methodological approach, and theoretical commitment. However, to generate a degree of order when contextualising our contribution, we organise this literature with reference to the headline message being presented by particular contributions. We specifically disaggregate between three general strands of scholarship: work that emphasises systemic constraints to meaningful climate transition, work that identifies institutional weaknesses that exert drag on climate transition, and work that foregrounds the existence of 'green shoots' and emergent progressive action. Throughout the paragraphs below, we explain the relevance of our study to each of these three strands of work. We go against the grain of much of the systemic constraint literature, by emphasising institutional foundations that currently exist in support of effective climate financing, and identifying specific areas of institutional performance requiring attention and reform. We engage more directly with scholarship on institutional weaknesses inhibiting effective climate transition and the green shoots studies of emerging good practice through our fine-grained identification of the independent influence from a range of institutional factors and foundations. To give a sense of these extensions to existing scholarship, in the paragraphs below we review each of the three blocs of literature in turn.

Scholarship emphasising systemic constraint suggests that, without significant change in system-level organisation or properties, the dysfunction and suboptimality that is present in prevailing political economic regimes will be reproduced through efforts to finance and operationalise sustainable transformation. Some contributions to this bloc of scholarship are explicitly anchored within Marxist and other critical political economy traditions, and others deploy alternative terminologies and framings when elaborating their critique.

Through *Climate Capitalism*, Newell and Paterson (2010) argue that national and international responses to climate change are introducing significant shifts in prevailing approaches to economic regulation and production. In a work that engages closely with scholarship on neoliberalism and neoliberalisation, Newell and Paterson emphasise the need to acknowledge the limits to action that exist from the world as it is. From this standpoint, Newell and Paterson deliver a forensic critique of prevailing systemic constraint. In identifying market-based mechanisms for addressing climate change as reproducing rather than challenging the essentially unsustainable quest toward capital accumulation and perpetual economic growth, Newell and Paterson suggest that climate capitalism is doomed to fail to deliver required climate transition. Overall, Newell and Paterson (10) remain 'highly sceptical of the idea that capitalism can deliver either a socially just or sustainable future'.

The theme of the relationship between capitalism and nature is expanded upon by Bohm et al (2012), in their interrogation of the functioning of carbon markets. Drawing on historic and contemporary Marxist theory, Bohm et al (1619) develop the contention that 'the dynamics of capitalism tend to propel economic processes beyond the limits of controllable growth'. While carbon markets have been established to price environmental externalities in to production costs, Bohm et al (1630) suggest that 'the likelihood that they will succeed' in decarbonising economies 'seems bleak'. Broadly parallel insights are offered through explorations of 'eco-socialism', which emphasise the necessity to transition away from capitalism to effectively respond to climate change (Pepper 2002, Albritton 2019, Satgar 2022).

Perhaps the most prominent recent call for system change to facilitate an adequate response to climate change has come from Raworth's work on *Doughnut Economics*. In arguing against policy-makers' and economists' fixation with Gross Domestic Product as *the* yardstick of national performance, Raworth builds on an established line of critique from feminist scholarship on the devaluation of womens' work that is imminent to this economic measure (e.g. Gustafsson 1995, Waring 1999). By offering a conceptual framework around which to operationalise a sustainable measure of economic activity, one that is sensitive both to the need for baseline needs to be met and for the limits posed by an ecological ceiling to be respected, Raworth (37-45) presents a tool for a dramatic reorientation of how we approach economic fundamentals.

With a similar focus on re-framing how we think about sustainability, Helm (2023) presents to us an inter-generational logic of calculus. We should, for Helm, be judging current human activity on the basis of its impact on the assets we are able to pass on to the next generation, defined as the stock of human capital, physical capital, social capital, and natural capital. If we want truly to move toward sustainable working and living, we must reframe how we define and measure sustainability at the systemic level. There is much in common between Helm's approach and the 'ecological modernisation' associated with the Free University, Berlin. As noted succinctly by Mol and Spaargaren (2000: 19), adherents of ecological modernisation accepted 'the need for fundamental transformations within the modernisation project to restore some of its structural design faults that had caused severe environmental destruction'. For Warner (2010: 539), ecological transition required that strategies be deployed to 'subvert modern forms of states, markets and industrial institutions'.

In *The Value of a Whale*, Buller (2022) takes a broader look at the relationship between capitalism and the environment. Buller argues that contemporary market-based assumptions skew our approach to climate change, pushing us towards forms of adjustment that will have a beneficial impact on 'the economy'. From within this worldview, green capitalism is seen by Buller to offer, at best, incomplete solutions to socio-ecological crises, prioritising technocratic solutions that ignore the inherently politically contested nature of the ecological challenge we face. Looking at several case studies of climate finance, Buller argues, for instance, that carbon pricing remains mostly insignificant, and that carbon markets become a tool for distraction and a prioritisation of offsetting schemes involving the expulsion of local communities and land seizure without effectively generating additional carbon emission reduction. Buller also explores how, through the increased role of international financial institutions and the growing influence of asset managers, the logic of climate finance has been transformed to a strategy of leveraging the private financial system's considerable resources to combat the environmental breakdown. In these efforts, Buller criticises the excessive focus on shifting risk from the private sector towards public institutions and funds.

In a broadly similar vein, Christophers' (2024) *The Price is Wrong* assesses the interplay between asset managers, financialisation, and response to climate emergency. Cristophers argues that electricity provides an example of what Polanyi dubbed a 'fictitious commodity', a resource that is unsuitable to be traded in the market. Christophers observes a negative pricing loop in the renewable energy market, which ultimately drives away investors in renewables. Government support is deployed to increase the supply of renewable energy, but expanded supply drives down price which has the effect, in the absence of wider regulatory fixes, of diminishing investors' expected returns and therefore necessitates additional government support. Rather than perpetuating this negative cycle, Christophers instead calls for direct government intervention in each step of the energy chain, from financing, to generation, to distribution, offering this a prescription for accelerated decarbonisation.

In part, our study of climate financing pushes against scholarship emphasising systemic constraint to sustainable transition. While we very much acknowledge the pressing need for the effectiveness and scale of climate financing to be ramped-up, we prioritise the identification of institutional foundations that are aligned with strengthened performance. From across this scholarship, our approach coheres closely with that of Newell and Paterson reviewed above; systemic constraints undoubtedly exist, and within these constraints we seek to identify institutional mechanisms and structures capable to impacting meaningfully on the pace of change.

The second strand of scholarship on the political economy of climate change that we situate our study alongside is constituted by work that prioritises the role of institutional constraint in inhibiting commitment to climate transition and climate financing. In contrast to the works focusing on macro-level systemic issues, this scholarship is constituted by a meso-level focus on organisational structures and organisational fields.

UNFCCC institutional processes are one site of organisational interaction that have attracted significant attention. While some analyses offer guarded optimism about outcomes reached and processes commenced around the Paris Agreement (Dimitrov 2010, Falkner 2016), it is common to find more critical commentary. Hovi et al (2013), for example, identify the lack of meaningful commitment devices as providing a critical limitation to the effectiveness of intergovernmental climate negotiations, with limited institutional capacity to incentivise compliance with national commitments and targets or to attenuate the possibility of future withdrawals. Inadequate progress within these institutional fora on 'loss and damage' funds from the carbon-intensive global North to compensate for climate impacts wrought across the global South has commonly been identified as a specific point of failure. The Green Climate Fund, established in 2009 and tasked with raising climate aid of US$100bn per year by 2020, has been criticised for falling far short of its resource mobilisation targets (Cui et al 2020), and for permitting donors to re-announce and re-cycle previously committed climate aid flows and to divert funds away from other aid initiatives rather than supply additional aid resources (Shalatek et al 2010, Roberts et al 2021).

Beyond the UNFCCC, a range of additional international institutions have been foregrounded for their shortcomings on climate change and climate financing. In this regard, responses from the centrepiece organisations of global economic governance have attracted notable critical commentary. Both Zaman (2013) and Cima (2017) argue that World Trade Organisation rules are poorly aligned with the exegeses of the climate challenge, the former suggesting that intellectual property protections inhibit green technology transfer and the latter suggesting that rules on subsidies fail to adequately account for environmental benefits being generated by government support for clean technology. In relation to the International Monetary Fund we see criticism of an institution that has adjusted its core activities relatively marginally, with a limited focus on the adequacy of climate financing available to climate vulnerable states, and little movement toward sustainable transformation in its wider lending and surveillance operations (Kentikelenis et al 2022, Ramos et al 2022). Across the road from the IMF at the World Bank, we see a longer-standing and more deeply embedded engagement with climate change and climate finance. The World Bank has, for example, moved to integrate analysis of the coherence between borrowing countries' development plans on the one hand, and Paris Agreement commitments on the other. The World Bank also played an important role in establishing green bonds as a new asset class in the late-2000s (Clegg 2024). Nonetheless, the World Bank has been labelled a 'climate change profiteer', criticised for the lack of transparency in its support for climate financing through the Clean Development Mechanism, and for shortcomings in its mainstreaming of climate change goals (Galindo-Gutierrez 2024, Redman 2008). More recently, the Bank's promotion of 'business friendly' climate change policy frameworks has been criticised for 'reinscribing both causes and effects of uneven development' (Bigger and Webber 2021: 36).

The lack of commitment from political leaders, operating within both government and party political institutional structures, has commonly been identified as a road-block to sufficient progress on climate change and climate financing. Discursive strategies are identified by Kurz et al (2010) that contest and rebut appeals for rapid climate action, replacing these with calls instead to prioritise 'the national interest' and 'our way of life'. Backsliding and limited ambition on climate commitments from national political elites have been similarly documented in cases including Chile (Parker et al 2013), France (Forchtner 2019), the UK (Carter and Clements 2015), and the US (Tesler 2018).

Beyond this discursive reframing, organisational linkages between political institutions and a collection of market-based institutions dubbed the 'pro-economy lobby' have been identified as a source of national-level foot dragging (Vesa et al 2020). And corporate organisations have been identified as climate transition constraining institutions in their own right. Yang et al (2020) identify a tendency for multinational corporations to engage in 'greenwashing' - that is, to exaggerate the scale of their environmental credentials and commitment - upon entering an emerging country host market with limited regulatory capacity and significant potential for capture of high market shares. More broadly, Coen et al (2022) are critical of the 'symbolic rather than substantive' climate change commitments enunciated through corporations' public reports and statements. Likewise, the Science

Based Targeting Initiative suggests that, globally, only around 800 companies have performance targets that are accredited as being aligned with Paris Agreement goals (SBTI 2022).

Whereas the institutional constraints scholarship emphasises mid-level road-blocks, the 'green shoots'-oriented literature points attention to organisational successes. The green shoots literature is constituted by works that highlight emerging innovative practice across institutional constellations that has shown the potential to lock-in enhanced climate ambition and financing, albeit often in nascent form.

Within the field of comparative politics and comparative political economy, over recent years we have seen the emergence of notable studies of factors associated with climate change leadership. We see, for example, that Tobin (2017) and Farstad (2018) find associations between left-wing party orientation and national-level climate commitment, while Liefferink et al (2009) and Oberthur and Dupont (2021) point towards the positive impact on this front from membership of the European Union. Beyond these national-level institutional catalysts of climate commitment, influence from sub-national organisational structures have also been highlighted. From Eckersley (2018) and Clegg (2023) we see the influence of local party orientation on climate performance at this sub-national level, and from Papin (2020) and Heikkenen et al (2020) we see evidence that transnational municipal networks have accelerated municipal authorities' climate responses.

Specifically on the subject of climate finance, we have in recent years received studies that provide insight into contemporary developments and trends. Bryant and Webb (2024) provide us with a rich review of market approaches to creating and valuing 'climate capital', through mechanisms including green bonds. Cash and Swatuk (2022) adopt an innovative approach to identifying effective climate financing. With a particular focus on climate aid, Cash and Swatuk's study aims to transpose insights and good practice from pre-existing development practice, emphasising for example the importance of donor coordination and recipient ownership as foundations for success. Michaelowa's (2022) compendium of contributions, too, provides nuanced reflection on a range of specific interventions and general dynamics in the realm of climate finance. We have, for example, chapters on the history and legitimacy of the Green Climate Fund (Basak and Karlsson-Vinkhuyzen 2022), on the relationship between multilateral development banks and Paris Agreement goals (Gebel et al 2022), and on national-level climate funds (Gomez-Echeverri 2022). Through our study, we extend these valuable existing contributions to the study of climate finance. Our 'value-added' to this most proximate body of work comes from our highlighting, through quantitative comparative analysis, of specific institutional foundations associated with increased flows of financing. As detailed in later empirical chapters, our insights complement and build from these existing scholarly foundations.

Considered together, scholarship on the institutional constraints to climate change responses and financing, and on the 'green shoots' of emerging effective performance in these areas, present intensive exploration of the histories behind contemporary practice, and reflection on defining features of particular climate policy fields. We see significant attention being placed on the intervening influence from institutional factors, whether institutions are seen as offering constraint and sources of footdragging, or a catalytic effect toward improved performance. Overall, our extension to these two blocs of scholarship comes from the systematic comparative focus we place on the multiple forms and levels of climate financing. Our approach to analysis allows for specific political and economic institutions to be identified that exert a significant and systematic influence over flows of climate financing. Accross our empirical chapters, we develop fine-grained contributions to scholarship on state-based climate finance, on the interaction between states and market-based climate finance, and on market-driven dynamics. As a starting point we acknowledge the presence of system-level constraint and the need for improved performance. And by demonstrating the important role being played by a range of political, economic, and social institutions in supporting strengthened climate financing performance and identifying some cases where reform may be needed to more effectively drive progress, we in particular take insights from and contribute to the institutional constraint and the green-shoots-focused scholarship.

1.4. Overview of book

Through this opening chapter, we have laid the foundations on which the later elements of the book stand. We specifically have introduced our approach to studying institutions, and situated the contribution being made to scholarship on the political economy of climate transition and climate financing. Our conceptualisation of institutions builds from March and Olsen (1989), March and Simon (1993), and Scott (2014), prioritising institutional structures that are sufficiently robust and stable to be exerting an identifiable impact on specified climate financing outcomes. Our contribution to scholarship emphasising systemic constraint to climate transition comes from the focus we adopt on the foundations of within-system progressive change, and we extend scholarship on institutional constraints to climate transition and on the 'green shoots' of improved performance through our identification of specific institutional foundations associated with higher financing flows.

In Chapter Two, we turn our attention to the first of our empirical chapters. Drawing primarily on OECD data on governments' climate change spending patterns across 2001-19, we probe in particular the influence of party politics and elections in influencing overall commitments.[17] Our main contribution comes from the insights generated into the pathways through which Green party electoral performance impacts on climate spending. We show that, across this cohort and timeframe, Green party coalition presence is associated with elevated climate change spending. We also show Green influence through inter-party competition, with ruling parties typically responding to Green electoral success by increasing their climate spending in order to try and capture the newly-demonstrated pool of environmentally-motivated voters. We here show how party political and electoral institutions have underpinned elevated state-based climate financing flows.

Through Chapter Three, we explore organisational structures associated with larger bilateral climate financing flows. Building from scholarship on the political economy of bilateral aid, we explore in particular influence from political structures and societal values. Focusing on measures of climate aid from OECD member states, we show that countries exhibiting post-materialist values and

[17] An earlier version of the analysis in this chapter appeared in Clegg and Galindo-Gutierrez (2025).

with higher trust in foreign nationals provide larger climate financing flows. We also show that where governments are in a relatively safe electoral position with a firm governing majority, and where governments are more left-leaning, climate aid generosity increases. Social institutions, in the form of national values, and political institutions, in terms of electoral outcomes and party politics, provide foundations for larger flows of climate aid, which for many countries across the global South constitutes a core component of transition financing.

Chapter Four progresses to consider the intersection of states and markets. When exploring institutions and international climate financing, we specifically focus on the role played by the World Bank in developing green bond markets across emerging markets. Although the World Bank was central in establishing green bonds as a new asset class from the latter years of the 2000s, through the chapter our core findings suggest that World Bank operations may be exerting a constraining effect on green bond market development. We specifically find that World Bank green bond issuance into a domestic market serves to crowd out subsequent green bond issuance within that market. This dynamic raises important questions over the efficacy of the World Bank's existing approach, suggesting that rather than directly issuing green bonds into development markets, the Bank's impact may be better supported through other forms of support for green bond market development. We further interrogate this relationship between state-based governance structures and market-based climate financing through Chapter Five. Here, our analysis of the ECB market interventions suggest a systematic negative impact on green bond performance from the institution's operations.[18] The suggestion again is that existing institutional structures and practices may require revision and reform in order to more effectively catalyse market-based climate financing.

In Chapter Six, we close the quantitatively-based empirical component of the book by exploring the influence of private institutions on market-based climate financing. We explore the impact of underwriting on green bond performance, using a study of European green bond performance to demonstrate the positive impact that follows from bond issuers' use of home-market underwriters. Where an underwriter is organising and supporting a green bond issue into their own

[18] An earlier version of the analysis within this chapter appeared in Galindo-Gutierrez (2025).

home market, we find that a 'home-turf effect' will reduce the cost of borrowing significantly. In this case, it seems that stronger networks and information flows between an underwriter and market participants in their home market may constitute institutional foundations for strengthened performance, with on average a bond issuer saving over US$27m by using a home-market underwriter.

We close the book through Chapter Seven, which offers forward-looking reflection on current trends in climate financing. We continue to follow the thematic organisation of the book as a whole, exploring in turn dynamics around state-based climate financing, influence from states over private climate financing flows, and the impact from private actors and initiatives on market-based climate financing. We outline the volatility in US and EU that is being driven by contemporary politicisation of climate commitment and financing, contrasted to the medium-term stability from the Chinese government with its depoliticised insulation from sources of climate backlash. We explore the ways in which regulation of green bonds and sustainable finance can provide tools for states to bolster the performance of private climate financing, and the emergent tendency of Sovereign Wealth Funds to use their investment power to catalyse a greening of financial flows and practices. And we assess the potential for the Carbon Disclosure Project and the Partnership for Climate Accounting Financials to provide enhanced information flows and institutional networks for an acceleration of market-based climate finance. We close the book with a call for what we term 'evidence-based optimism'. By extending our understanding of institutional structures that support effective climate financing, a robust evidence base can be used to inform advocacy for such institutions. By extending our understanding of institutional structures that inhibit or constrain effectiveness, evidence can underpin advocacy for enhanced accountability and reform. It is very clear that improved performance from climate financing systems is needed, to move to emissions trajectories that are more aligned with Paris Agreement goals. But the future of climate financing has yet to be written; given this indeterminacy, it is vital that organisational and individual agency is deployed to drive forward our collective transition toward sustainable modes of working and living.

2. Institutions for Domestic Climate Finance: Parties, Elections, and Government Climate Spending

2.0. Introduction

As the core site for organising collective action, the state represents a central arena for marshalling climate finance to support sustainable transformation. We can gain an immediate sense of the importance of states to national economic structures by considering the sheer size of government spending flows with, across advanced-industrialised states,[19] government expenditure typically equating to between 35 and 50 percent of GDP. And we know from history that state resources can prompt rapid and deep reorganisation. Periods of 'total war', during which the power of the state was used to orient national production around the maximisation of military capacity, represent one such example of this transformative potential (Black 2010). The construction of expansive social safety nets provides a peaceable manifestation of this transformative agency, which saw state finances being channelled to support pension provision, disability support, assistance through periods of illness and unemployment, and the supply of affordable housing (Esping-Andersen 1990: 30-32, Scanlon et al 2014: 1-3).

Through this chapter, we explore variation in the extent to which domestic climate financing is being delivered by national governments. Effective action on this front is a central pillar of efforts to meet Paris Agreement targets. We particularly probe the relationship between party politics, elections,

[19] Through this and subsequent chapters, we use the term advanced-industrialised states as a placeholder for member states of the Organisation for Economic Cooperation and Development. For a list of the 38 OECD member states see OECD website, available at https://www.oecd.org/about/members-and-partners/. Accessed 12th March, 2024.

and flows of national government climate financing. From our focus on these political institutions, we demonstrate the existence of direct and indirect mechanisms of influence from Green party electoral performance on national climate spending. The direct pathway occurs through Green coalition presence, with this condition being associated with an increased flow of public climate finance. The indirect pathway functions through inter-party competition. In the aftermath of strong Green electoral showing, incumbent governments move to try and attract support from across this newly-revealed pool of environmentally-motivated voters by boosting their climate spending. Much scholarship uses composite measures of environmental policy performance as the outcome of interest against which to gauge the influence of political institutions. Part of the value we add comes from our incorporation of a precise measure of governments' national-level climate change expenditure into our analysis, which we achieve by taking advantage of an OECD climate spending database that was released in 2021.

In presenting our analysis of political institutions and national-level climate financing flows, we develop the chapter through the following structure. In the first section below, we review contemporary trends and dynamics in domestic climate spending by national-level governments. We then progress, in the second section, to specifically derive expectations about likely institutional drivers of variation in this form of governmental climate change commitment. We explain decisions taken over data sources and analytic approach in the third section of the chapter, before in the fourth discussing results of our analysis. To conclude, we recap the headline insights, and reflect on implications that follow from our empirical work.

2.1. National-level climate finance - key dynamics

On September 25th, 2023, French President Emmanuel Macron unveiled a major package of climate change commitments from his government. The interventions were designed to more-than double the pace of decarbonisation across the country; in comparison to a baseline annual reduction in greenhouse gas emissions of around 2 percent, Macron declared that the climate package would allow a transition pathway to be shifted to with emissions reductions of 5 percent each year through to 2050. Climate financing commitments were laid out to support clean-ups of the country's 50 largest-emitting industrial sites, increased rail commuter capacity across the largest urban centres, a leasing scheme that would allow those on lower incomes to lease a new electric vehicle for €100 per month, and investments in energy system transformation. In the short term, these enhanced commitments would see domestic spending on climate change increase by €7bn per year.[20]

These extended commitments from the French national government came in the wake of accelerated action from across the Atlantic in the US, under the administration of President Joe Biden. Late in the summer of 2022, after the conclusion of tortuous negotiations and votes on Capitol Hill, the Inflation Reduction Act was signed into law. Provisions of the Act were designed to curb inflation included mechanisms for reducing the federal government budget deficit, reducing the prices paid for regulated prescription drugs, and pushing down utilities charges by catalysing investment in new energy production. It is the detail of legislative provisions pertaining to clean energy production and electrification of transportation, however, that led the Environmental Protection Agency to declare the Inflation Reduction Act 'the most significant climate legislation in US history'.[21] According to Bistline et al (2023), the Inflation Reduction Act included planned spending of up-to around US$1.1tn in support of climate transition in the decade to 2031, through a combination of direct spending and tax credits. As explored in Chapter Seven, across the US and EU we now see volatility in climate commitment and financing, with party political and electoral factors contributing to contemporary constraints and some row-back on climate commitments.

Moving beyond these individual prominent cases that point to interplay between political institutions and variation in government climate spending, we can turn to systematic comparative study of national climate change and financing to gain insight into more general trends in the strength

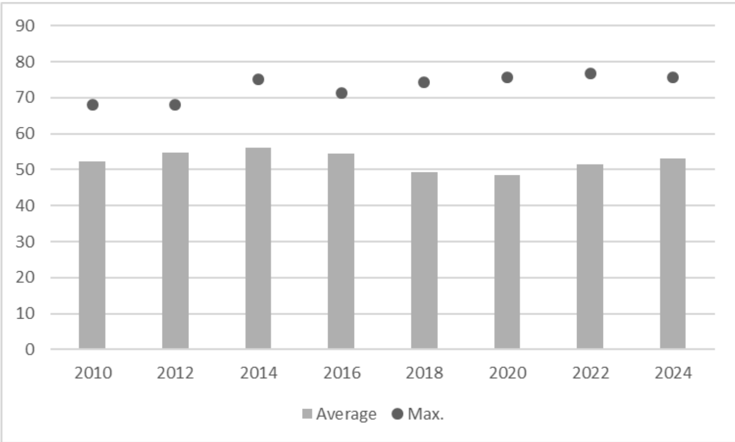
[20] Responses to the climate package were mixed. The French government referred to these measures as "ambitious, coordinated, and far-reaching", while opposition leaders were less effusive in labelling the interventions "just pathetic, a pittance". See Willsher (2023).

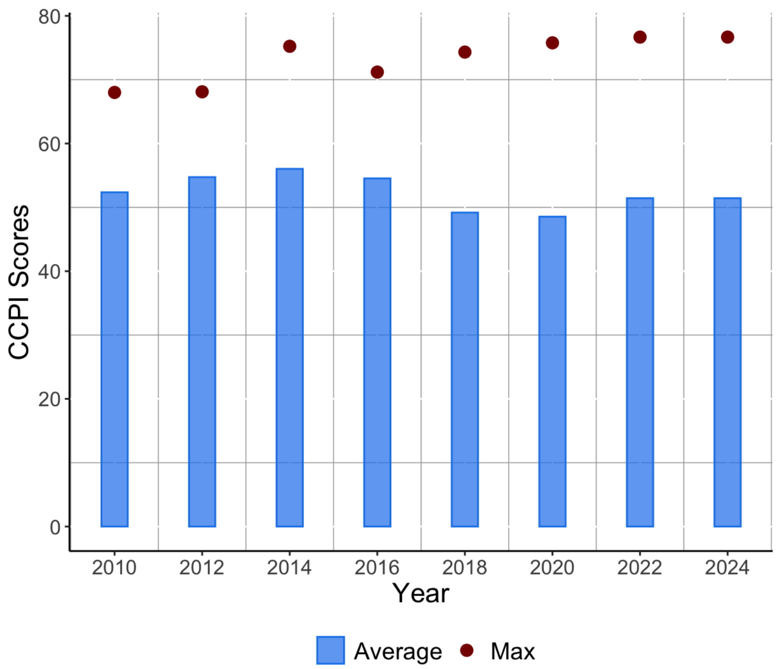
[21] See Environmental Protection Agency website, available at https://www.epa.gov/green-power-markets/summary-inflation-reduction-act-provisions-related-renewable-energy#:~:text=The%20Inflation%20Reduction%20Act%20of,of%20new%20clean%20electricity%20resources.. Accessed 12th March, 2024.

of national commitments. We see valuable nuance from Mills-Nova and Liverman's (2019) comparative analysis of the Nationally-Determined Contributions documentation submitted by governments to the UNFCCC. Mills-Nova and Liverman identify a cleavage between the content and framing of UNFCCC documentation across different groupings of states, based on climate vulnerability and emissions characteristics. On the other hand, the most climate vulnerable states tend to have more precisely-costed transition plans, which often note the inadequacy of climate aid flows as a major barrier to effective implementation. On the other, from across the world's ten largest greenhouse gas emitters, we see a lower focus on climate financing. While the existential challenge being posed to the most climate vulnerable countries provides a potent motivation for climate transition planning and costing, this level of engagement seems to be less present amongst the states who are most responsible for cumulative emissions, and whose decarbonisation is most necessary for collective achievement of Paris Agreement aims.

The multidimensional nature of national governments' climate change responses, and the spread of transition pathway planning across the decades from now through to 2050 and 2100, makes the issue a challenge to gain analytic traction. One established tool that provides some insight in this regard is Germanwatch's Climate Change Performance Index. Germanwatch has compiled the Index since the late 2000s, benchmarking national governments' climate commitments and outputs. The Climate Change Performance Index rolls together expert evaluations of government climate policy and financing, climate targets, and the prevalence of renewable and sustainable sources within national energy mixes, to generate an aggregate score with a maximum of 100. Looking at government performance since 2010 (Figure 2.1), we see that when viewed in aggregate performance has remained relatively stable. The average national grade of 52.36 in 2010 closely matches 2024's figure of 53.05. However, belying this aggregate stability, there is notable change in country-specific performance. The strongest performer has moved from Brazil with a score of 68 in 2010, via Sweden and Denmark, through most recently to Denmark with its score of 75.59 in 2024. Denmark's 20-point increase between 2010-18 represents a dramatic acceleration, while Russia's 18-point decline over the same time is a notable application of foot-dragging.

Figure 2.1. Trends in Climate Change Performance Index gradings





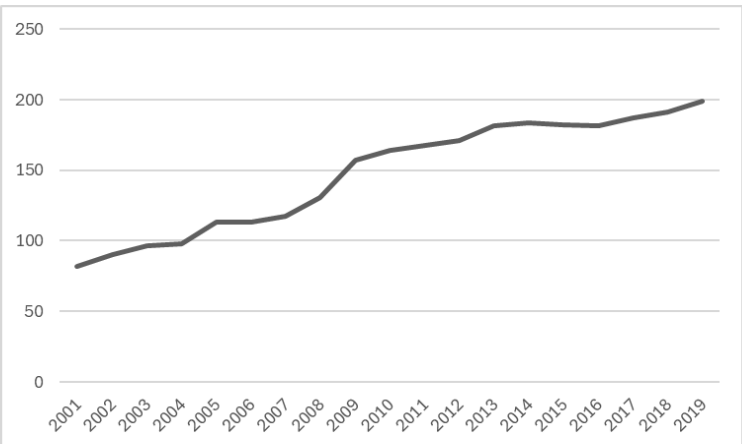
Source: Authors' analysis of Climate Change Performance Index Annual Reports

In recent years, an initiative from the Organisation for Economic Cooperation and Development has allowed us to gain a more direct understanding of government climate change spending across advanced-industrialised economies. Through 2019, an OECD team worked to map the

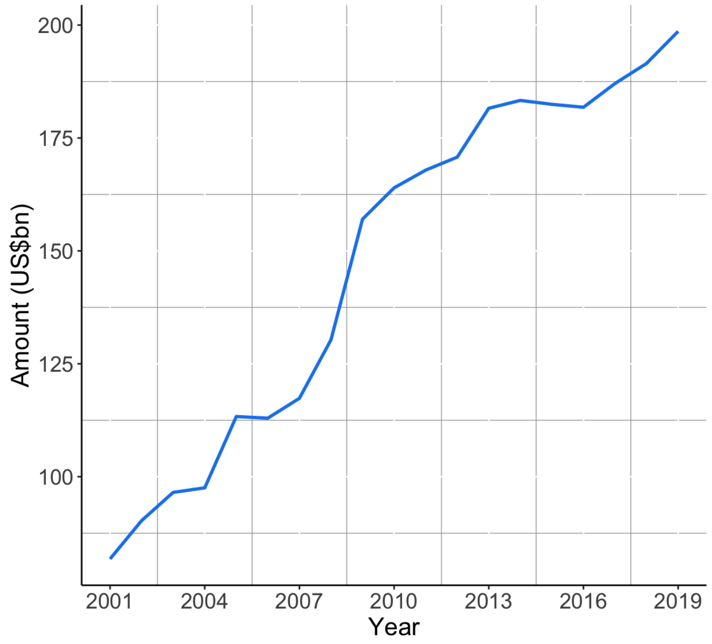
EU Taxonomy of Sustainable Activities onto the lines of government expenditure that the OECD routinely collates from its member countries. In parallel with the passage of EU regulations that codified and rolled-out the Taxonomy, the OECD presented its new database on members' climate expenditure, covering the period 2001-19. The OECD aimed to present a holistic measure of climate-relevant spending, that brought together lines of government spending from across agriculture, energy, forestry, transport, and beyond (OECD 2022).[22]

When we look at aggregated data on OECD member states' national-level climate spending, we see an upward trajectory through the 2001-19 period (see Figure 2.2).[23] From a starting point of total national-level climate spending across OECD member states sitting at US$81bn in 2001, we see a steady increase through to the figure of US$199bn in 2019. Expressed as a percentage of GDP, this represents a rise from an average climate spend equivalent to .74 percent of GDP, to a figure of .89. There is a high level of variability in national climate spending commitments across the OECD's members. At the laggard end of the scale we have Japan's 2006 climate spending at .16 percent of GDP. Leading the pack, we have Slovenia's 2015 climate spending at 2.89 percent of GDP.

Figure 2.2.: OECD national climate expenditure (US$bn)



[22] While the database provides extremely valuable insight through its deployment of a direct and comprehensive measure of climate change-relevant expenditure, the non-participation from the US in the initiative constitutes a limitation.
[23] The OECD database disaggregates between climate spending and climate investment. In this overview, we focus on the former.



Source: OECD.[24]

From this review of contemporary dynamics in national-level flows of climate financing, it is clear that there are notable patterns of variation across both time and place. We now turn our attention to exploring the role played by political institutions in shaping these outcomes.

2.2. Explaining performance variation - existing literature

In the paragraphs below, we explore existing scholarship from which we can derive insights into the plausible drivers of variation in national governments' commitment to climate financing. Given the restricted range of scholarship focusing specifically on comparative analysis of national climate financing, we consider works that more broadly shed light into the political economy of national climate change performance and commitment. We organise our review so as to firstly cover scholarship that directs attention to political institutional structures, before then progressing to

[24] See OECD website, available at https://stats.oecd.org/Index.aspx?datasetcode=SGCFD. Data collected 1st March, 2024. Values are presented in real terms, to support comparability over time. To generate the national-level figure, we subtract the subnational spending figure from the total climate spending figure.

consider additional factors of interest that are highlighted. The purpose of this section is to establish points of interest and parameters for the empirical analysis that we present later in the chapter.

Throughout this study, we hold institutions to be organisational entities that display sufficient levels of coherence and autonomy to plausibly exert meaningful influence over flows of climate financing. In the context of governments' national-level climate financing, we find that political parties, and interactions between political parties and elections, constitute likely institutions of interest. The role of Green parties in shaping national climate commitment has been subjected to considerable focus, and debates are ongoing as to the wider influence from the partisan orientation of governing parties. We see also from existing scholarship that, as a supplementary political institutional factor, membership of the EU may be associated with enhanced national-level climate commitment and performance.

Through the 1990s and 2000s, in many contexts Greens for the first time emerged with significant vote shares in national elections and representation in national legislatures. The Finnish Green League became the first example of greens in national government, with the party entering a national rainbow coalition through Pekka Haavisto's role as Minister for Environment (Kontinnen 2000). Motivated by a concern that such changes in Green party fortunes had taken place below the radar of academic analysis, Rihoux and Rudig (2006) worked to extend scholarship on the topic. Overall, Rihoux and Rudig point to individual cases in which Greens have plausibly influenced outcomes, including in relation to food safety and agricultural standards. However, there was an overall suggestion from Rihoux and Rudig (12-20) that, by the early-2000s, Greens had displayed a fairly limited ability to shape outcomes. The lack of experience in government, and the need to cut against the grain of government bureaucracies' conventional wisdom and established practice to achieve their core policy transformations, was seen as constraining Greens' effectiveness in this regard.

Over more recent years, scholarship has emerged that points to evidence of systematic Green party influence over environmental policy and outcomes. Studies have pointed to the existence of direct mechanisms of influence, through which Greens have used positions within coalition government, or more broadly positions within legislative chambers, to shape environmental and climate outcomes. We also see findings of indirect influence through an inter-party competition mechanism. With this indirect mechanism, incumbent governments respond to Green electoral success by adopting greener positions, to try and attract support from amongst the newly-demonstrated pool of environmentally-motivated voters.

We have some case study-based insights into Greens' influence through direct coalition and legislative presence, with rich detail drawn from across Europe (Muller-Rommel and Poguntke 2002, Rootes 2002, Rudig 2002, Spoon 2009), and beyond (Bale and Dann 2002). We receive useful insights into systematic Green legislative influence from quantitatively oriented studies. Through a study of air quality indicators across OECD member states from 1980-99, Neumayer (2003) identifies a likely positive Green legislative presence effect. Specifically, Neumayer's study finds the combined parliamentary presence of Green and left-libertarian parties to be associated with cleaner air. Whereas Neumayer identifies a joint Green and left-libertarian effect, from Debus and Tosun (2021) we receive insights specifically pertaining to Greens in isolation. Debus and Tosun's work presents a discursive analysis of parliamentary debates, overall suggesting that Green members play a key role in advancing what they term a 'green agenda'. From their study, Debus and Tosun (933) find that 'it is the presence of Green parties in parliaments that determines to what degree… the green agenda is placed on the legislative agenda'.

Overall, then, it seems that parliamentary presence can provide a pathway for Green influence over aspects of national-level climate engagement and performance. Transposing these insights to our area of study, we derive a first hypothesis to be tested through our study:

> *H_1: A stronger Green national parliamentary presence will be associated with a higher level of national government climate financing.*

The intuition here is that elected Greens may be able to use, for example, their capacity to shape parliamentary agendas, committee processes, and legislative amendments to push for enhanced climate commitments.

In the aftermath of an election that fails to return a party with a governing majority, negotiations will commence over the formation of a coalition. Coalition negotiations provide potential governing partners to reconcile their individual policy preferences into an agenda for government that is acceptable to all. There can be great variation in the intensity and length of coalition negotiations; Hellstrom et al (2023) identify the 589 days of discussions between political leaders in Belgium through 2010-11 as the longest coalition formation process, while many cases exist of processes involving near-immediate resolution.

We have mixed evidence of Green coalition influence in existing scholarship. Exploring social investment policies of OECD member states from the 1970s to 2015, Roth and Schwander (2020) find that Green coalition presence is associated with stronger policies in this area. While not focusing on the environment, Roth and Schwander nonetheless suggest that Greens in coalition are likely to favour more extensive welfare provision. Knill et al (2010) do incorporate a focus on environmental outcomes when testing for a Green coalition effect. However, when exploring the relationship between Green coalition presence and environmental outcomes as measured by a composite index of policy outcomes achieved across 1970-2000, Knill et al find no evidence of a significant systematic effect. While there is limited supporting evidence, these studies deploy outcome measures that involve complex causal pathways. Given that it is intuitively easier for Greens to bargain-in elevated expenditures than to bargain-in and then successfully deliver policy outcomes and outputs, we on balance derive a second hypothesis to be tested through our empirical analysis:

> *H_2: A Green presence in a national governing coalition will be associated with a higher level of national government climate financing.*

The underlying expectation is that Greens will be able to use their bargaining power during coalition negotiations to secure increased commitment to climate change, and to use threats of exiting from the coalition to secure their delivery.

Beyond these direct legislative presence and coalition pathways for Green influence over national government flows of climate financing, we also have the indirect party-competition pathway for influence. Insight into Green electoral performance and inter-party competition on the environment is provided by Spoon et al (2014). Through a study of responses to Green electoral performance, Spoon et al suggest that parties who view the Greens as an electoral threat are more likely to dial-up environmental commitment, as articulated through their subsequent election manifesto. A more benign economic context is found by Spoon et al to catalyse this form of environmental inter-party competition, the suggestion being that in better times there is higher potential for mainstream parties to foreground areas beyond economic policy and fiscal aims. Through case study analysis focusing on Ireland, Norway, and the UK respectively, Carter and Little (2020), Farstad and Aasen (2022), and Carter and Farstad (2017) provide supplementary accounts of mainstream adjustments in response to Green electoral successes. Drawing on these findings, we present a third hypothesis to be tested in our empirical work:

> *H_3: A higher Green vote share will be associated with a higher level of national government climate financing.*

Inter-party competition, we expect, will bid-up national climate commitment.

Moving from Green party performance, to identify our next political institutional focus of analysis we turn to scholarship on the European Union. The EU has for some time been considered a climate change leader, catalysing improved performance from its member states by virtue of regulatory interventions and socialisation. Following the Single European Act of 1987, the EU gained the environment as a core policy competency. The EU was held to have exerted a positive impact on the consolidation of UNFCCC climate governance, having supported and cajoled member states to set and deliver emissions reductions through the Kyoto Protocol (Vogler and Bretherton 2006). In their exploration of EU environmental performance through the 2000s, Scott and Rajamani (2012) note the 'ambition' of the European Commission and Parliament, citing particularly the approach taken to aviation regulation and emissions trading. From Tobin's (2017) study of national climate ambition as measured by the Climate Change Performance Index, we see EU membership being identified as exerting a significant and positive effect. Broadly parallel findings come from Liefferink et al (2009) and Janicke (2005). With the former, EU states were shown to display systematically stronger commitment to environmental policy, as operationalised by an assessment of positions on 40 issue areas including clean air regulation, climate change, land management, sustainable energy, and wildlife conservation. With the latter, we see the demonstration between EU membership and environmental trend-setting, defined as the initial development of a policy solution that is subsequently adopted by peers.

With the consistency of these findings of the positive links between EU membership and environmental performance, we put forward the fourth hypothesis to be tested empirically:

H_4: Membership of the EU will be associated with a higher level of national government climate financing.

The intuition here is that EU members may be exposed to regulatory and socialisation effects, which have the effect of systematically raising their government domestic climate commitments.

The final political institution that we touch on for its potential relationship to national domestic climate financing flows pertains to governing parties' partisan orientation. There are longstanding debates as to whether we should consider climate change to be a partisan issue characterised by leftwing issue ownership, or a valence issue over which parties of the left and right compete to demonstrate effectiveness (Carter and Clements 2015). However, we have had a series of contributions that suggest there may be systematic differences between parties of the left and right on climate change. Tobin's (2018) study of variation in states' performance in relation to the Climate Change Performance Index and Farstad's (2017) study of variation in national climate commitment as revealed through manifesto content point in this direction. More recently, we have seen interrogations of rightwing populism and climate change, which again suggest systematic rightwing populist disengagement from the issue (Fiorino 2022). As a consequence, the final hypothesis we present regarding political institutions and domestic climate finance is:

H_5: Leftwing national government will be associated with a higher level of national government climate financing.

The intuition behind this expectation, as set out in this scholarship on partisan orientation and environmental policy commitment, is that leftwing parties may be more inclined to deploy the type of policy interventions associated with accelerated environmental protection. These policy tools may include measures to regulate market processes, and to enhance state capacity to directly support processes of economic transformation.

Turning now to the broader control variables we incorporate into our models. Explorations of the ‘environmental Kuznets curve’ represent perhaps the most prominent line of study of the determinants of governments’ green performance. Stern et al’s (1996) work constitutes a key intervention in this field, formalising the expectation of an inverted u-shaped relationship between income and environmental degradation. Under this environmental Kuznets curve model, economic growth is predicted to initially exacerbates degradation before, as more affluent citizens begin to prioritise non-material outcomes, plateauing and then entering a down phase during which subsequent growth supports improved outcomes. Cole et al’s (1997) empirical study of OECD countries in the early-to-mid-1990s, generated nuanced findings that presented evidence overall for the existence of a Kuznets-type relationship for local airborne pollutants that generate direct local-level harms, but not for CO_2 with its less direct and more global-level harms. Empirical work on the environmental Kuznets curve has often generated corroborative findings (Dasgupta et al 2002, Sarkodie and Strezov 2019, Ansari 2022), although climate leadership from lower-income states has been taken as evidence of a need for some recalibrations and some concerns expressed over the robustness of generalisations from observations across a narrow range of states and limited timeframes (Stern 2004). On balance, though, this strand of scholarship provides a plausible expectation that, amongst high-income OECD member states in particular, economic expansion be associated with stronger government commitment to climate change. As such, we control for this factor in our empirical work.

It is in addition plausible that a government’s climate change-related spending is likely to be positively associated with the country’s vulnerability to the impact of climate change. Over the past decade, there has been a growing policy focus on identifying and addressing ‘vulnerability gaps’;

situations in which infrastructure and institutional capacity falls short of increasingly likely extreme weather occurrences (Mastrandrea et al 2010). The United Nations Environment Programme provides annual 'national adaptation gap' updates that aim to both catalyse governments' domestic action and the provision by advanced-industrialised states of 'loss and damage' funds to support adaptation across climate-vulnerable states of the global South (UNEP 2022), with a parallel initiative from the OECD Task Force on Climate Change Adaptation.[25] While we know that decision-makers tend to under-prepare for future crises (Pelling 2012), it nonetheless remains plausible that a higher adaptation gap will be associated with a stronger drive to catch-up, as reflected by higher levels of current expenditure.

2.3. Data and method

To operationalise national government climate financing flows, we use the OECD's Government Climate Finance database.[26] With this database, the OECD has identified lines of government spending that fit with the European Union Taxonomy of Sustainable Activities. These spending lines are aggregated to form measures of 'climate-significant' expenditure and investment. The OECD has collated this measure of domestic government climate financing flows to cover the period 2001-19. Coverage extends to 27 of the OECD's member states.[27] As such, the OECD database provides a powerful insight into advanced-industrialised states' domestic governing climate financing flows, in a field where typically studies have had to rely on indirect proxy measures of governments' climate commitment or performance.

[25] See 'Strengthening Resilience for a Changing Climate', OECD website, available at https://www.oecd.org/climate-change/theme/resilience/. Accessed 6th December, 2022.
[26]See 'Government Climate Finance Database', OECD website, available at https://stats.oecd.org/Index.aspx?datasetcode=SGCFD. Accessed 18th March, 2024.
[27] The cases included in the database are Australia, Austria, Belgium, Czeck Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Latvia, Lithuania, Luxembourg, Netherlands, Norway, Poland, Portugal, Slovenia, Spain, Sweden, Switzerland, and United Kingdom.

In our analysis, we run separate analyses of the OECD's measures of governments' national climate financing for the expenditure- and the investment-based measures. The rationale for running these separate analyses comes from the expectation that expenditure and investment flows may respond with systematic difference to variation in institutional conditions. It seems plausible that climate-significant investment flows, which involve acquiring or creating assets that deliver a climate impact over the medium- or longer-term, may involve notable lead times prior to the inclusion of a project in a government budget, with these lead times making the relationship between institutional conditions and investment response more complex and contingent. In contrast, climate-significant expenditure, which tends to be heavily shaped by staff costs, may respond more rapidly to institutional shifts.[28]

Through this analysis our focus is on the impact on governments' national-level climate financing that comes from variation in political institutional features, with our primary interest being on the strength of particular party families within national political constellations. Given uncertainties over the temporal relationship between political institutional shifts and changes to climate financing, we develop a twin-track empirical strategy.

First, to test for the existence of a somewhat dispersed impact from Green electoral performance on climate spending, we aggregate the outcome variable across individual 'election cycles'. An election cycle begins in a year that contains a national-level election, and concludes in the year preceding the next national-level election. We exclude financial data from the election year itself, given the likelihood that budgets have been set the previous year under the previous administration. Under this election cycle approach, we calculate the mean value of climate outlays across all other years of the given cycle. So, for example, in a country where there were national elections in 2001 and 2005, a first election cycle would run 2001-04 and the climate financing outcome variable would be constituted by the mean figure for 2002-04. Under this first approach, we generate a time series cross-sectional (TSCS) database, where across each political cycle the outcome variable is fixed.

[28] For useful reviews of the lengthened time horizons involved in public sector investment planning and delivery, see Tarschys (2003) and Srithongrung et al (2019).

The second analytic approach we adopt tests for a more immediate link between electoral performance and climate outlays, and focuses on financing flows in the year following an election. Again, the intuition here is that the budget for the election year itself will largely have been set in the previous cycle, and that it is in the subsequent year that responses to political institutional shifts are likely to be seen. With this second approach, we generate a cross-sectional database that includes years in which an election occurred and associated outlays from the following year.

To test for H_1, which expects a positive relationship between climate financing and Green party presence within the legislature, we turn to Armingeon et al's (2023) Comparative Political Dataset (CPDS). We initially seek to explore the impact on government climate financing from the existence of a Green party institutional presence in a national legislature. To do so, we created a 'Green seat (any)' dummy variable that was coded 1 to capture observations where one or more legislative members represented a Green party, and given a value of 0 where there was no Green representation. We also created an alternative specification to explore the impact from the strength of the Green party institutional presence by creating a 'Green seat (share)' variable. Where, for example, Greens held 7 percent of all available seats in the legislature, a value of 7 is taken by this variable. In constructing these variables, we aggregated together data relating to all members of the Green party family within the given national parliament, as listed in the CPDS.

To test for H_2, which expects a positive relationship between Green presence in a national coalition and the scale of government climate financing, we turn to Roth and Schwander's (2020) database on Greens in national coalitions across OECD states. Using this source, we created a 'Green coalition' dummy variable with a value of 1 for observations Greens were within a governing coalition, and a value of 0 elsewhere. Roth and Schwander's database coverage runs only to 2015; to extend the 'Green coalition' variable through to our time series end date of 2021, we undertook additional analysis of scholarship and reporting on the composition of national governments across these recent years. Roth and Schwander's narrative information on post-2015 government composition provided a valuable foundation for this data extension, and when making extensions we followed the protocol of coding 1 where there was a Green ministerial presence or formal coalition agreement involving the Greens.

H_3 directs our attention towards Green influence through inter-party competition. To operationalise this measure we returned to the CPDS, incorporating the combined vote share from members of the Green party family from this source into our analysis. The expectation here is that a higher Green vote share will nudge incumbent governments to ramp-up their climate financing commitments, to try and attract support from this newly-revealed pool of environmentally-motivated voters. To operationalise government partisan orientation, which features in the expectation from *H_4* of a positive impact on climate financing from leftwing leadership, we turn to the CPDS Schmidt Index score. With the Schmidt Index offering a five-point scale with a value of 1 for leftwing hegemony, 2 for leftwing dominance, 3 for a centrist government, 4 for rightwing dominance, and 5 for rightwing hegemony, we would expect the lower values to be associated with higher flows of climate financing. From *H_5*, we expect EU membership to be associated with higher climate financing commitment. We draw on the CPDS to construct an 'EU member' dummy variable, to test for the expected influence from the EU institutional presence.

Beyond these political institutional factors, we control for the influence of level of economic development, the scale of climate vulnerability gaps, and decentralisation in our analyses. Work on the 'environmental Kuznets curve' suggests that the relationship between national economic performance and environmental protection will follow an inverted u-shape. After an initial phase in which economic expansion brings increased environmental harm, countries are expected to transition into a phase where post-material values prioritise environmental improvement (Cole et al 1997, Dasgupta et al 2002, Sarkodie and Strezov 2019, Ansari 2022). To control for this effect, we incorporate data from CPDS on the real GDP annual growth rate. Over the last decade or so, the focus on identifying national climate vulnerability gaps has increased markedly. A vulnerability gap is said to exist where infrastructure and bureaucratic response capacity falls short of the scale of response required by extreme weather occurrences (Mastrandrea et al 2010, UNEP 2022). While we know that decision-makers tend to under-prepare for future crises (Pelling 2012), it nonetheless remains plausible that a higher adaptation gap will be associated with a stronger drive to catch-up, as reflected by higher levels of current expenditure. To control for climate vulnerability, we incorporate the University of Notre Dame's Global Adaptation Index score.[29] The benefit from this index is the incorporation of measures of vulnerability to climate change-related events and of existing capacity to respond to these events. A higher score represents a higher vulnerability gap, which may be associated with upward pressure on climate-relevant spending. Finally, we control for decentralisation to capture the likelihood that countries with greater subnational policy and spending autonomy will see climate financing transposed to the subnational from the national level. For this, we turn to Shair-Rosenfield et al's (2021) Regional Autonomy Index (RAI). With the RAI, a higher score represents greater decentralisation.

Drawing on the above sources, our resulting dataset covers 27 OECD states across 2001-19. Tests focusing on the political cycle as the unit of analysis were run on a time series cross-sectional dataset. To ensure consistency across the TSCS dataset, we removed observations that were part of an election cycle that carried over from a pre-2001 election. In the case of Belgium, for example, the 2001 and 2002 observations were dropped as they represented the final years of a 1999-2002 cycle. Our analyses of the TSCS database deployed a panel-corrected standard errors (PCSE) estimator. The PCSE approach is appropriate for analysing collections in which temporally and spatially correlated errors and heteroscedasticity may be present, as is the case here. To capture temporal and spatial fixed effects, we incorporated dummy variables capturing individual years and countries across the TSCS dataset. This analytic approach fits with Beck and Katz (2011: 342), and allows for a higher level of confidence in the robustness of our findings relative to the deployment of a PCSE estimator without incorporating measures to control for unobserved temporal and spatial effects. For tests focusing on the post election year climate spending outcomes we created a cross-sectional database, and deployed a Generalised Least Squares estimator. Again, to capture unobserved temporal and spatial effects and therefore strengthen confidence in results generated, we incorporated dummies for individual years and countries.

[29] See 'Country Index', University of Notre Dame Global Adaptation Index website, available at https://gain.nd.edu/our-work/country-index/. Accessed 18th March, 2024.

2.4. Results and discussion

The OECD data on national government climate financing covers both climate-relevant spending, and climate-relevant investment. When presenting and discussing our results, we first cover our analysis of the interaction between institutions and governments' climate spending, before progressing to climate investment. Across both of these analyses, we run two specifications of the outcome variable; one that deploys the 'political cycle' measures of aggregated climate financing from the year following an election through to the subsequent election year, and another that focuses on the single year following an election. Table 2.1 provides descriptive statistics relating to the analyses of climate spending over political cycles, while Table 2.2 relates to climate spending over the post-election year.

Table 2.1: Descriptive statistics, political cycle climate spending

Variable	Obs	Mean	Std dev.	Min.	Max.	Frequency
Outcome						
Climate spending (cycle)	432	.837	.484	.182	2.205	~
Independents						
Green seats (share)	432	3.520	4.718	0	22.2	~
Green seats (any)	432	.525	.500	0	1	0 (205), 1(227)
Green coalition	432	.088	.284	0	1	0 (394), 1 (38)
Green vote share	432	4.398	4.759	0	21.7	~
EU membership	432	.803	.398	0	1	0 (85), 1 (347)
Partisan orientation	432	2.826	1.267	1	5	1 (78), 2 (99), 3 (132), 4 (66), 5 (57)
Controls						
Economic growth (cycle)	432	7.892	8.990	-19.929	36.183	~
Climate vulnerability	432	32.384	3.586	24.909	40.503	~

Decentralisation	432	13.541	10.797	0	35.478	~

Table 2.2: Descriptive statistics, post-election year climate spending

Variable	Obs	Mean	Std dev.	Min.	Max.	Frequency
Outcome						
Climate spending (year)	109	.819	.475	.187	2.231	~
Independents						
Green seats (share)	109	3.984	5.425	0	22.2	~
Green seats (any)	109	.541	.501	0	1	0 (50), 1 (59)
Green coalition	109	.101	.303	0	1	0 (98), 1 (11)
Green votes share	109	4.963	5.485	0	21.7	~
EU membership	109	.807	.396	0	1	0 (21), 1 (88)
Partisan orientation	109	2.697	1.266	1	5	1 (24), 2 (23), 3 (37), 4 (12), 5 (13)
Controls						
Economic growth (year)	109	2.070	2.993	-7.664	12.264	~
Climate vulnerability	109	32.459	3.544	24.904	39.794	~
Decentralisation	109	13.841	10.922	0	35.478	~

In headline terms, there is significant similarity in the climate change-relevant spending outcome variable across the two collections. Mean annual climate expenditure is, overall, around 0.8 percent of GDP. The minimum and maximum values for the cycle-wide and single year observations are the same across both collections, standing respectively at 0.2 percent (Japan, 2005-08 and Germany, 2009) and 2.2 percent (Hungary, 2010-13 and Luxembourg, 2018).

In relation to the political institutional variables of interest, we have the measures pertaining to Green electoral performance, legislative representation, and coalition presence. In 50 of the 109 election events across the OECD, Greens gained a seat share of zero. The Greens' strongest electoral performance came with the 2009 Icelandic election, where Greens gained 22 percent of the seats share and votes. There was a Green presence within a governing coalition in around 9 percent of observations across the political cycle-focused database (38 from 432 cases), and a similar pattern in the single year-focused database (10 percent, or 11 from 109 cases). Government partisanship was on average centrist but with a higher representation of right- over left-leaning executive balance, and around 80 percent of the country/year observations related to EU members.

Table 2.3 provides an overview of our analysis of the political cycle-focused database, on which (as explained above) we deployed a PCSE estimator with dummy variables to capture temporal and spatial fixed effects. Models 1 and 3 operationalise direct legislative Green influence through the seat share measure, and Models 2 and 4 use the measure of any legislative presence measure.

Table 2.3: Explaining variation in climate spending across political cycles

	Model 1		Model 2		Model 3		Model 4	
Variable	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error
Independents								
Green seats (share)	-.029	.011**	~	~	-.025	.015	~	~
Green seats (any)	~	~	.070	.037	~	~	.019	.050
Green coalition	.070	.028*	.058	.029*	.085	.032**	.084	.032**
Green vote share	.034	.010***	.005	.005	.037	.014**	.015	.008*
EU membership	.053	.065	.057	.073	~	~	~	~
Partisan orientation	-.007	.008	-.005	.007	-.004	.009	-.003	.009
Controls								
Economic growth	-.000	.001	-.000	.001	-.000	.001	-.000	.001
Climate vulnerability	.003	.001	-.003	.001	-.016	.024	.010	.024
Decentralisation	-.000	.028	.015	.029	.046	.031	.053	.031
n	432		432		347		347	

X^2	.664***	.662***	.677***	.677***

Note: *p≤0.05, **p≤0.01, ***p≤0.001

The results from Model 1 show that, across the whole OECD cohort, there is significant indirect Green influence on climate expenditure through inter-party competition, and significant direct influence through participation in coalition government. When we deploy the alternative specification of legislative influence in Model 2, we see that the inter-party competition effect loses its significance, which questions the robustness of the initial positive finding on this front. On the basis of Models 1 and 2, we conclude (in line with the expectation of H_2) that, across the OECD cohort, Green participation in coalition government is associated with higher climate expenditure across the subsequent political cycle. Having a Green coalition presence boosts climate spending by between .07 percent of GDP (Model 1) and .06 percent of GDP (Model 2). In contrast, across the OECD cohort the expectations from H_1 (that Green legislative presence should generate increased climate spending) and H_3 (that we should see inter-party competition translating a higher Green vote share into increased climate spending) were confounded. Models 1 and 2 are significant ($p<.001$), and explain around 66 percent of observed variation in the outcome variable ($\chi2$=.664 and .662 respectively).

Through Models 3 and 4, we turn our attention to the sub-grouping of EU member states. While we found no independent influence from EU membership on climate spending through Models 1 and 2, Models 3 and 4 seem to confirm the expectation from existing scholarship of systematic difference in EU member states' climate change performance. Across Models 3 and 4, we see again that Green participation in coalition government is associated with higher levels of climate expenditure. Across EU members the Green coalition effect is slightly higher, raising climate spending by around .08 percent of GDP. We also see that, amongst EU member states, larger Green vote shares are associated with higher climate spending. As such, we see that amongst EU member states inter-party competition seems to function as a pathway that drives-up climate spending. A rise of one point in Green vote share translates into an increase in climate spending of between .04 percent of GDP (Model 3) and .02 percent of GDP (Model 4). Overall, then, through this analysis of the EU sub-grouping, we see that H_1 (Green legislative presence effect) continues to be confounded, and that

H_2 (Green coalition effect) and H_3 (inter-party competition) are confirmed. Models 3 and 4 are significant ($p<.001$), and explain around 68 percent of observed variation in climate spending (in both cases, $\chi2=.677$). When comparing the full OECD cohort and the EU sub-group, we see that EU membership exerts a conditional effect on the impact of Green electoral performance on climate spending. Contextual factors associated with EU membership seem to facilitate inter-party competition on climate change.

When we turn to the post election year measure of climate spending, we find no systematic difference from the EU sub-grouping. While in both cases a Green coalition presence effect remains, the inter-party competition effect is no longer observed. Table 2.4 shows the results from these analyses of climate-relevant expenditure in post election years. Models 5 and 6 include the full cohort of OECD members, and toggle between the 'Green seats share' and 'any Green seats' measures of legislative presence respectively. Models 7 and 8 include only EU members in order to further probe systematic variation from this sub-grouping, again toggling between alternative legislative presence specification.

From across these analyses, intriguingly we see evidence of Green coalition presence having a significant and negative impact on climate expenditure in the post-election year. A possible explanation for this dynamic is that Greens' coalition partners want to signal in the short term that they remain firmly in control of the governing agenda, and so ensure that any additional climate spending becomes back-loaded toward the subsequent years of an administration. It also may be the case that Greens in coalition experience an initial transition phase of 'learning the ropes', before subsequently being able to more successfully release financing for climate change-related spending. It is perhaps less surprising to see no evidence of the inter-party competition effect carrying over from spending across a political cycle in the EU to spending in a post election year in the EU; it may, for example, be the case that governments wish to strengthen their 'green' credentials only as a the next election becomes closer, rather in the first year of the election cycle. We discount the finding from Model 6 of the legislative presence effect given its dependence on the use of the 'any Green seats' specification. The finding in Models 7 and 8 of a significant and positive effect from decentralisation on post election year climate spending is puzzling and difficult to interpret, given that a higher level of decentralisation should intuitively be associated with lower national-level spending.

Table 2.4: Explaining variation in climate spending across post election years

	Model 5		Model 6		Model 7		Model 8	
Variable	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error
Independents								
Green seats (share)	.008	.018	~	~	.017	.023	~	~
Green seats (any)	~	~	.188	.083*	~	~	.167	.093
Green coalition	-.188	.087*	-.182	.084*	-.333	.113**	-.320	.112**
Green vote share	.012	.017	.005	.010	.005	.022	.005	.013
EU membership	.153	.452	.176	.375	~	~	~	~
Partisan orientation	-.021	.019	-.021	.019	.016	.026	.007	.026
Controls								
Economic growth	.006	.010	.002	.001	.017	.012	.012	.012
Climate vulnerability	-.041	.054	-.013	.017	-.045	.059	-.022	.057
Decentralisation	-.019	.018	-.020	.017	.048	.015**	.056	.015***
n	109		109		88		88	
p	***		***		***		***	

Note: *p≤0.05, **p≤0.01, ***p≤0.001

To complete our discussion of results, we turn briefly to the analysis of climate-relevant investment financing from central governments. As was expected on account of the likely longer and more variable time lags surrounding national government climate investment flows, our analysis establishes less clear insights into this area. As reported in Annex I (Tables A2.1 and A2.2), Models 9 to 12 focus on climate investment across political cycles, and provide preliminary evidence of an inter-party competition effect. Models 13 to 16 focus on climate investment in the post-election year, where it seems that there is systematic difference between the impact of political institutional features within EU member states. Amongst EU member states, but not across the wider OECD cohort, there seems to be a positive Green coalition effect on these short-term investment flows. The explanatory power of the models focusing on climate investment are lower than their spending-focused counterparts, with an $\chi 2$ range .486-.514.

Overall, then, our findings provide significant support for H_2, which led us to expect that the presence of a Green coalition partner would be associated with stronger climate change commitment. Results also suggest that EU membership functions to 'switch on' the inter-party competition pathway for raising climate spending. As such, through our study we provide conditional support for H_1, which led us to expect that inter-party competition would work to translate a higher Green vote share into stronger climate change commitment. The evidence also shows a sensitivity to outcome variable operationalisation, with the above outcomes holding true for climate change spending across political cycles, but not when we deployed the post election year spending measure. It takes time, it would seem, for Green coalition partners to translate their spending preferences into reality, and for EU governing parties to start competing for Green voters by ramping-up their green credentials. Across our analyses, we find no evidence of an independent effect from EU membership and no evidence that leftwing orientation was associated with increased government climate financing, confounding the expectations of H_4 and H_5 respectively.

2.5. Conclusion

Through this chapter, we have probed relationships between political institutional factors on the one hand, and national government climate financing flows on the other. Focusing on OECD data on advanced-industrialised countries' climate-relevant spending from 2001-19, we probed three expected pathways for translating Green electoral success into increased flows of climate financing, as well as expected positive effects from EU membership and leftwing government orientation.

Following our definition of institutions as organisational entities that display a sufficiently high level of coherence and autonomy to plausibly exert meaningful influence over climate financing, we here confirm in particular that party political structures matter. Specifically, Green coalition presence appears to be associated with stronger climate financing performance, with Green coalition presence raising national climate spending by between .06 and .09 percent of GDP. We also see that, within the EU, incumbent governments respond to Green party electoral success by increasing their climate expenditure. The impact from these coalition and inter-party competition effects appear to be back-loaded toward the later phases of an election cycle. For the coalition effect, this dynamic may reflect Green coalition partners' growing influence over time. For the inter-party competition effect, this dynamic may reflect the incentive for the incumbent government to enhance their climate credentials closer to the subsequent election, at which they aim to capture increased support from the pool of environmentally-motivated voters whose existence was demonstrated by earlier Green success.

In addition to demonstrating these pathways through which Green electoral success can drive forward national performance on climate financing, our null finding on partisan orientation suggests that parties of the political left and political right do not exhibit systematic variation in this area. While it does appear that there is clear differentiation in the way that parties of the left and right talk about climate change (e.g. Carter and Clements 2015, Farstad 2018), there is an underlying similarity in their delivery of climate financing across the time period we have considered. Given the possibility that the prevalence of rightwing party control of national governments we have witnessed over recent decades may continue into the future,[30] evidence that rightwing parties may not exert a systematic downward drag on domestic climate financing may carry significant real-world importance.

[30] From the OECD sample used in our analysis, the mean cabinet political orientation as captured by the CPDS Schmidt Index is 2.8, showing that the average cabinet is marginally to the right-of-centre. We also see that 177 cabinets are right-of-centre, as compared to 123 left-of-centre (and 132 centrist).

3. Institutions for Bilateral Climate Finance - National Values, Governing Majorities, and External Transfers

3.0. Introduction

When exploring government motivations for supplying foreign aid, Dudley and Montmarquette (1976: 132) offered a pithy summary. Looking at trends in evidence through the 1960s, and considering the dominance of France, the United Kingdom, and United States as originators of development assistance through this period, the proffered diagnosis was one of external transfers being motivated by ‘cold war and colonialism’. Over the years, bilateral aid flows have become notably more complex, as major new donors have entered the field, and transfers have spread to an increasingly wide array of states across the global South. Our understanding of the institutional foundations that shape these aid flows has expanded considerably, with studies suggesting that factors including economic openness, elections, and government orientation may influence the volume of aid flowing from OECD member states (Fuchs et al 2014, Annen and Strickland 2017).

Through this chapter, we extend scholarship on the political economy of aid by considering specifically the determinants of bilateral climate change financing. In 2009, the OECD was tasked with collating data on climate change-related aid flows, which are disaggregated by bilateral donor. We draw on this resource to explore institutional features associated with increased bilateral climate aid effort. We show that national values constitute an important social foundation for climate aid generosity, with high levels of trust in people from other nationalities and postmaterial commitments supporting stronger performance. And we show that political institutions matter, with government partisan alignment and electoral vulnerability exerting a significant influence. More electorally safe governments, and non-right-wing governments, display enhanced effort on climate aid.

In establishing these insights, we develop the chapter through the following structure. In the first section below, we provide a brief history of climate change-related aid. We then in the second section introduce expectations regarding potential institutional drivers of variation in propensities to give bilateral climate finance, which highlight in particular potentially significant roles for social and political institutional foundations. In the third section of the chapter, we explain our operationalisation of variables and the analytic approach adopted, before then providing a discussion of results in the fourth section. To conclude, we review core insights from the chapter, and present thoughts on the implications that follow from our analysis.

3.1. Bilateral climate finance - key dynamics

On 19th November, 2000, an international network of environmental activists convened for a two-day summit on Climate Justice at The Hague, Netherlands. Timed to coincide with COP 6, participants used the forum to raise global awareness of the environmental causes that they were fighting for. Owens Wiwa spoke of the struggle to evict Shell from Nigeria's Ogoniland, and of the government's execution of his brother Ken Saro-Wiwa for supporting this cause. Margie Richards spoke of the efforts of the local community in Norco, Louisiana, to use 'citizen science' to establish and seek redress against harms to health from a nearby refinery and chemical works. In addition to this important awareness-raising, one of the lasting impacts achieved by participants was, by virtue of the event's name and core focus, to embed the term 'climate justice' within the lexicon of academics, climate change activists, and, increasingly, policymakers (Whitehead 2014).[31]

When considering climate justice, we can usefully disaggregate between its theorisation, and its practice. At heart, theorisations of climate justice direct our attention towards iniquities in the

[31] See also Third World Network website, Yin Shao Loong: 'An account of The Hague Climate Justice Summit', available at https://www.twn.my/title/twr125j.htm. Accessed 24th April, 2024.

distribution on the one hand of the harm caused by climate change, and on the other hand of the material gains from deployment of the carbon-intensive practices that have accelerated climate change. Elaborations of the concept push us to explore the obligations held by today's 'winners' to today's 'losers' in the complex intergenerational game of climate change distributional politics. Prominent calls continue to be made for acknowledgement of, and compensation for, the gendered and racialised patterning of climate change harms (Terry 2009, Schlosberg and Collins 2014, Williams 2021). Turning from these prominent theorisations to mainstream practice, we see some discursive attachment to the interplay between climate change and gendered and racialised inequalities.[32] However, when exploring UNFCCC negotiations and agreements, we see that actual progress towards operationalising new mechanisms for providing financing to address these iniquitous impacts from climate change has remained extremely slow-paced.

Under the prevailing UNFCCC practice, which has gradually evolved since the early-2000s, obligations are increasingly seen to be held by populations within advanced-industrialised states, to those in lower-income countries with particularly high climate vulnerabilities. Countries across the global North, through their historic and contemporary carbon-intensive modes of living and working, have generated more rapid economic development while also priming the earth's atmosphere with heat-retaining particulates. Within the global South, we have countries with far smaller footprints of accumulated emissions, with greater exposure to the impact of weather-system shifts and less capacity to adapt to these risks.

It was with the 2002 Delhi Declaration from COP 8 that a need to establish mechanisms to support more vulnerable states was initially signalled. At Delhi, Parties specifically committed to 'capacity-building for the integration of adaptation concerns into sustainable development strategies', with financial support to be targeted at developing countries and small island developing states.[33] Discussion over the nature of these obligations to developing countries continued through subsequent COPs. The creation of new mechanisms for delivering on climate obligations has proved

[32] For an overview of UNFCCC work on its gender action plan, see 'Review of the enhanced Lima work programme on gender', UNFCCC website, available at https://unfccc.int/gender/final-review. Accessed 7th May, 2024.
[33] Relevant content from the 2002 Delhi Declaration is quoted in Roberts and Huq (2015: 148).

to be challenging; the launch in 2013 of the Warsaw International Mechanism for Loss and Damage was, for example, followed by a five-year work programme at the end of which 'the overall question of fund mobilisation remained unresolved' (Pill 2022: 2). While there was willingness to agree that the global North had a responsibility to provide additional financing to the global South to attenuate climate vulnerabilities, consensus on the details of who should pay what remained elusive. An important staging post was reached at the Dubai COP 28 in 2023 where, a decade after the Warsaw commitment, the Loss and Damage Fund was established. While its launch was accompanied by announcements on contributions from a range of donor states, the eventual operational scale of the Loss and Damage Fund remained unclear for some time.[34]

In the face of these delays with the creation of new mechanisms for delivering on bilateral climate obligations, countries have primarily turned to existing aid delivery structures for deploying climate aid. Back in 2009 at the Copenhagen COP, developed countries committed to mobilise US$100bn per year in climate financing for developing countries, with the ambition that this level would be reached by 2020 and retained through to 2025. The pledge referred to the aggregated flows from multilateral, bilateral, and private sources, with no clear indication on relative contribution shares. The OECD was charged with monitoring progress towards this US$100bn goal, with shared commitment to meeting the US$100bn target being reiterated at subsequent COPs. OECD monitoring identified notable progress through the decade from 2010, with significant increases to climate aid flowing in particular through bilateral and existing multilateral channels. However, as 2020 came and went, OECD reporting showed that the US$100bn target had been missed. By 2021, total climate aid stood at some US$90bn, with around 40 percent coming from bilateral and multilateral sources, and the remaining 20 percent from private finance that had been mobilised by government backing or co-financing (OECD 2023: 8). The OECD's preliminary analysis of climate aid flows in 2023

[34] For an overview of this launch, see Emma Shumway's 'Observations from COP 28 on the Loss and Damage Fund', Columbia University Law School Climate Law blog, available at https://blogs.law.columbia.edu/climatechange/2023/12/20/observations-from-cop28-on-the-loss-and-damage-fund/#:~:text=A%20decade%20after%20its%20first,recommendations%20of%20the%20Transitional%20Committee. Accessed 25th April, 2024.

suggested that the US$100bn goal may have been met, with projections suggesting that it would be exceeded in 2024.[35]

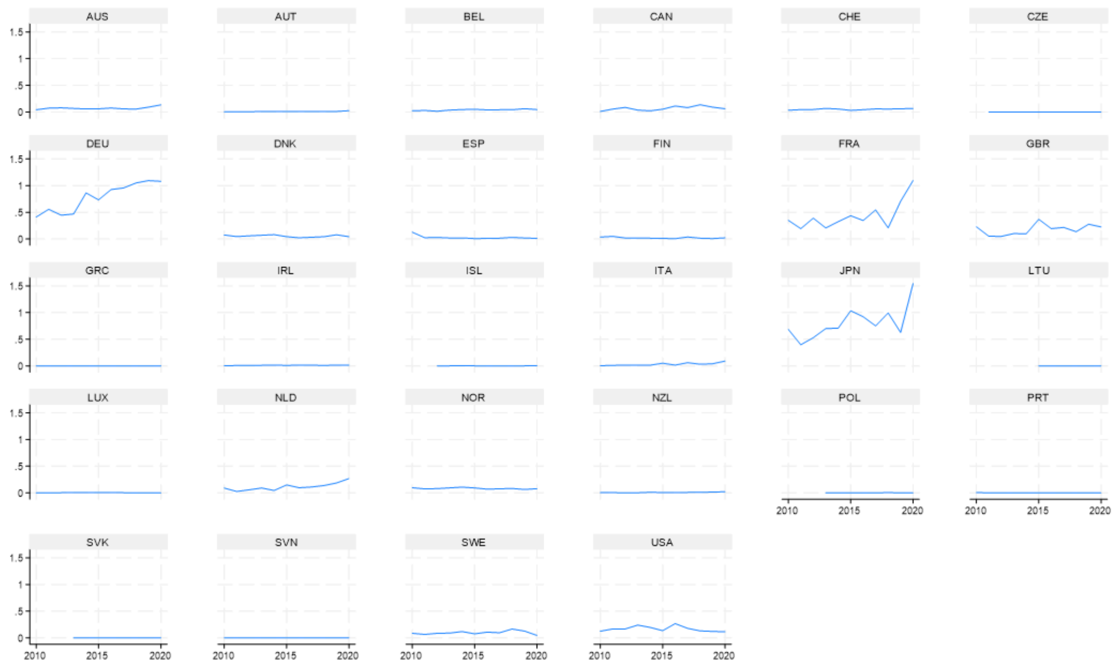
While individual donor countries have sought to draw attention to the scale of their own delivery,[36] controversy continues to surround bilateral climate financing. Climate advocacy groups have shamed donor governments who are seen to be falling short on meeting their 'fair share' of climate aid commitments. Based on an assessment of historic greenhouse gas emissions, national income, and population size, the Overseas Development Institute has, for example, highlighted underperformance from countries including the UK with a suggested 33 percent shortfall in climate aid (when assessing 2021 disbursals against a hypothecated 'fair share' measure), Spain with a 64 percent shortfall, and the US with a 79 percent shortfall (Pettinotti et al 2023). Donor governments have been criticised for a lack of transparency over climate finance commitments, with governments' repeated announcements of the same financing commitments exacerbating the difficulties faced by actors working to hold developed countries to account for their under-delivery of climate aid (Horton 2024). Non-governmental organisations including ActionAid, Christian Aid, and Save the Children have criticised donors for 'robbing Peter to pay Paul', by shifting bilateral aid from vital pre-existing poverty reduction projects and programmes towards climate change, rather than ensuring that additional financing was being supplied to meet climate pledges.[37]

Bilateral climate financing, then, remains a heavily contested issue. And when we look a little more closely at the OECD data on bilateral climate aid, we see evidence of divergent behaviour between donor states. Focusing on climate aid flows expressed in constant US$ values (Figure 3.1), we see that a small number of donors have significantly increased the magnitude of these flows between 2010-20. Japan, France, and Germany are particularly noteworthy in this regard, with their respective total climate change aid flows having risen to US$1.5bn, US$1.1bn, and US$1bn by the

[35] See 'Statement by the OECD Secretary-General on future levels of climate finance', OECD website, available at https://www.oecd.org/newsroom/statement-by-the-oecd-secretary-general-on-future-levels-of-climate-finance.htm. Accessed 26th April, 2024.
[36] See, for example, 'Japan's climate-related support', Japanese Ministry of Foreign Affairs website, available at https://www.mofa.go.jp/ic/ch/page23e_000446.html. Accessed 26th April, 2024.
[37] See 'The UK is reneging on climate finance commitments', ActionAid website, available at https://www.actionaid.org.uk/latest-news/uk-reneging-climate. Accessed 26th April, 2024.

end of this period. When we take the more fine-grained measure of bilateral climate aid expressed in constant US$ per capita values (Figure 3.2), we see further evidence of country-level variation. Viewed through this prism, by 2020 we find the OECD leaders to be France, the Netherlands, and Norway, with transfers of US$161, US$154, and US$148 per head of population. Through the main body of this chapter, we work to shed light on the drivers of variation in these countries' bilateral climate change aid generosity, to allow us to understand more fully the institutional foundations that support improved performance on this mode of climate financing.

Figure 3.1. OECD bilateral climate change aid (US$bn), 2010-20



Source: Authors' analysis of OECD Development Finance for Climate database.[38]

Figure 3.2. OECD bilateral climate change aid (US$ per capita), 2010-20

[38] See OECD website, available at https://www.oecd.org/dac/financing-sustainable-development/development-finance-topics/climate-change.htm. Accessed 26th April, 2024.

Source: Authors' analysis of OECD Development Finance for Climate database.[39]

3.2. Explaining performance variation - existing literature

With a small number of exceptions, bilateral climate aid has yet to be the subject of sustained political economy analysis.[40] Consequently, in deriving expectations over potential drivers of variation in countries' levels of bilateral climate aid effort, we turn primarily to two strands of existing scholarship. On the one hand we have a developed body of work that explores the drivers of variation in donors' bilateral aid generosity, and on the other we have studies exploring the politics of climate change that illuminate dynamics that are of potential salience. In the paragraphs below, when elaborating the hypothesised institutional foundations of bilateral climate aid generosity, we review and bring together these bodies of work.

[39] See OECD website, available at https://www.oecd.org/dac/financing-sustainable-development/development-finance-topics/climate-change.htm. Accessed 26th April, 2024.

[40] Klock et al's (2018) study of bilateral climate aid 2011-15 provides the work that most closely matches our own focus, and we build from these foundations in our empirical work. As detailed below, we extend this earlier work by considering a range of political institutional factors that are excluded from Klock et al's analysis.

We begin our exploration of the institutional factors that lie behind variation in climate aid flows by turning to scholarship on the interplay between public opinion and aid. Informed by Tingley and Milner (2013: 390), who suggest that public attitudes toward development aid are ‘consistent and structured’, in this context we hold public opinion to constitute an institutional structure capable of exerting meaningful impact over outcomes of interest.

The intersection of political outcomes on the one hand, and public opinion and national values on the other, has a well-established scholarly lineage. The challenges of probing these relationships are laid out clearly by Page (1994: 26), who highlights the necessity of ‘sorting out causality’. In many issue areas, it can be extremely difficult to meaningfully disentangle the direction of causality; where there is correspondence between popular opinion and policy, do we know whether the policy has been influenced by prevailing national sentiment, the policy itself has lead to a shift in national sentiment, or whether there is there some interplay between the two factors through positive feedback cycles. Within this complexity, however, prominent contributions on popular opinion and aid have suggested an important role from the former in shaping the latter. In an early study of the topic, Mosley (1985: 373) suggests that we can view aid as ‘a public good for which there is a market’, with electoral demand serving to induce higher flows. Collier (2007: 183-4), drawing on rich insights into the theory and practice of development assistance, directs our attention to the beneficial impact that can come from public engagement on bilateral aid: with greater support, aid agencies can be empowered to develop a more entrepreneurial approach that goes beyond the sense of risk aversion that can prevail when agencies are more isolated.

These initial insights into potential links between popular sentiment and bilateral climate aid effort are strengthened by wider comparative analyses of national environmental performance. Through Dolsak (2001, 2009) and Harrison and Sundstrom (2007), we see that governments’ climate mitigation efforts can be driven forward in contexts where there is stronger public support. The suggestion that bilateral climate change aid flows are accelerated by more supportive national sentiment is confirmed by Klock et al (2018), through their earlier study of OECD climate flows.

From this body of work, then, we derive our first expectation on the drivers of variation in bilateral climate aid effort:

H_1: Where national values are more supportive of climate financing, bilateral climate aid will be increased.

The underlying intuition here is that governments are systematically responsive to national sentiments in this area, and that their own preferences may be influenced by prevailing national values.

For our second expectation on the drivers of variation in bilateral climate aid effort, we continue to focus on popular opinion, and move toward their intersection with political institutional factors. Specifically, it seems likely that the scale of governing majority may interact with national sentiments to generate influence on climate aid flows.

The scale of a governing majority can be considered an institutional variable under the parameters laid out by our working definition of an institution, with governing majority constituting a feature of competitive party politics that has a sufficiently stable influence to exert a predictable and observable influence over political economic outcomes. From scholarship on the relationship between electoral competition and policy positions, we know that where individual candidates face a tight election, they will align their policy positions more closely with that of prevailing public opinion (Griffin 2006).[41] Moving the analytic focus from the individual candidate to the national party level, Franchino et al (2022) show that governing parties facing a tighter upcoming election will be more responsive to median voter positions and concerns. From these general expectations of governing party behaviour, we derive our second hypothesis:

H_2: Governments with small governing majorities will be more responsive to national values relating to climate aid.

Facing a tougher test in upcoming elections, we expect more electorally-vulnerable governments to try to align their behaviour more closely with prevailing popular opinion.

[41] Median voter approaches have been particularly influential in economic modelling of political performance, building from Harald Hotelling's (1929) theory that, given a competitive environment and a one-dimensional preference continuum, politicians would gravitate towards the 'centre-ground' of the median voter's position.

For the final of our expectations regarding institutional drivers of variation in bilateral cliamte aid effort, we remain focused on formal political factors. Specifically, it is well established in existing literature that the partisan orientation of a national government has an impact on the scale of development assistance. Krasner's (1978) study of US external economic management in the initial post-World War II decades provides a foundation stone for scholarship on the interaction between ideology and foreign policy, to which aid can be considered an important constituent component. When probing the determinants of US members of Congress' voting on aid approvals, Milner and Tingley (2010) establish that left ideology is indeed associated with stronger support for aid. The pathway identified by Milner and Tingley stems from voter preferences, with members of Congress responding to constituents' ideological positioning when voting. Studies including Brech and Potrafke (2014) and Pickering and Mitchell (2017) identify a direct partisan effect from government orientation on aid intensity. Given the identification of left-wing ideology with a preference for deploying state financing capacity to deliver redistributive outcomes (Laver and Garry 2000), this elevation of aid from left-leaning governments can be expected to be reproduced widely.

Importantly for our purposes, from Greene and Licht (2018) we receive additional reasons for expecting left-wing government to be associated with increased bilateral climate aid effort. Greene and Licht move beyond exploring partisan orientation effects on overall aid flows by interrogating the types of foreign aid favoured by parties of the left and right. From this analytic focus, we see a tendency from left parties to favour aid flows that are more humanitarian in nature, and right parties to favour flows that are more directly linked to economic performance and that may be materially beneficial to donor-country exporters. The operational alignments that exist particularly between climate adaptation financing and humanitarian concerns, with adaptation financing targeting the reduction of vulnerabilities that may have been revealed through previous humanitarian disaster events,[42] supports an expectation of enhanced climate aid effort from left-wing government. In the

[42] For an overview of humanitarian agencies' engagement with climate change see 'Humanitarian aid donors' declaration on climate and environment', European Union Civil Protection and Humanitarian Operations website, available at https://civil-protection-humanitarian-aid.ec.europa.eu/what/humanitarian-aid/climate-change-and-environment/humanitarian-aid-donors-declaration-climate-and-environment_en. Accessed 29th April, 2024.

light of this scholarship on partisan orientation and aid flows, we present the first hypothesis to be tested through our empirical work:

H_3: Left-wing governments will provide larger flows of climate aid.

The underlying intuition here is that climate assistance from left-leaning governments will reflect the stronger ideological commitment to redistributive flows in general, and humanitarian assistance specifically.

Beyond these three expected institutional foundations for greater climate aid generosity, in our empirical analysis we control for a range of additional political, economic, and geographic factors.

In terms of political factors, we control for membership of the European Union. It is well-established across existing scholarship that EU structures have served to catalyse improvements in member states' climate performance (Liefferink et al 2009), and specifically that EU positions on climate change and approaches to climate change transition feature as an important component of its overall external relations (Oberthur and Pallemaerts 2010, Allwood 2021). It is, therefore, likely that EU membership, in and of itself, may be associated with increased bilateral climate aid effort.

Moving to economic factors, we follow the lead of Fuchs et al (2014) in controlling for the overall level of economic development, and for the rate of economic growth. The expectation here is that, at higher levels of national income and with stronger economic growth performance, citizens and governments are more willing to increase the volume of aid flows. The expectation is informed by both the Environmental Kuznets theory, which suggests that as incomes increase individuals become more willing to prioritise environmental policies, and empirical observation of a positive relationship between income levels and the provision of development assistance. We also control for economic openness, given that countries with higher trade-to-GDP ratios have been shown to display higher willingness to supply bilateral aid. We also control for overall aid flows, so that our models effectively isolate variation in climate aid specifically. The final factor we control for in our models of variation in bilateral climate aid flows comes from climate change vulnerability. Climate change vulnerability refers to the quantum of impact on existing patterns of living and working from the types of shift in climactic conditions that are likely within that national context. The underpinning assumption is that higher vulnerability will be associated with larger bilateral climate aid flows, as vulnerable donors may be more attuned to climate transition and may carry this focus into their external economic policies (Klock et al 2018).

3.3. Data and method

Through our analysis, we are interested in explaining variation in national levels of bilateral climate assistance generosity. To operationalise this outcome variable, we turn to the Organisation for Economic Cooperation and Development's creditor reporting system (CRS) database. Specifically, within the CRS, the OECD collates a series of aid activities targeting global environmental objectives. At the Copenhagen COP of 2009, the OECD was specifically tasked with monitoring aid flows that were in support of climate change objectives, and to gain traction over this outcome we use OECD data on bilateral flows for climate mitigation and adaptation.[43] The OECD disaggregates between flows that have been earmarked by donors as 'primarily' targeting climate mitigation and adaptation or as making a 'significant contribution' to these areas, and residual flows with a less clear-cut alignment. To support confidence in our findings, we include only flows that are marked as primarily or significantly targeted at climate mitigation and adaptation. For comparability over time, we use the measure of bilateral climate aid expressed in constant US dollar values, and we denominate against national population to create a per capita measure.[44] The OECD climate aid presents the new commitments that have been made by a government in a given year, rather than actual expenditure that may take place over a longer timeframe given the nature of projects and programmes being supported. In terms of timeframe, we draw on the OECD measures of climate aid from 2010 through to 2021, the most recent year for which data is available at the time of writing.

[43] See 'OECD Data Explorer: Aid Targeting Global Environmental Objectives', OECD website, available at https://data-explorer.oecd.org/. Accessed 7th May, 2024.
[44] We take national population figures from the Comparative Politics Dataset (Armingeon et al 2023).

To test for the H_1 expectation of a significant relationship between national values and bilateral climate flows, we build from the approach of Bayram (2017) on trust and development aid, Klock et al (2018) on trust and climate aid, and Mostafa (2016) on postmaterialism and climate commitment. All three studies draw on the World Values Survey to explore the role played by national-level trust in foreigners or postmaterial commitments in shaping positioning on development assistance and climate change.[45] Bayram's study of the 'moral basis of public support' for development aid posited that 'generalised trust' was likely to constitute 'a crucial component of the moral calculus of publics in donor countries' (134). Where an individual has a higher level of trust in other people, they are seen by Bayram to be more likely to support the provision of financial assistance to others. Bayram demonstrates an association between this indicator of trust on the one hand, and support for poverty reduction as revealed through the WVS questionnaire on the other. Klock et al (2018) also draw on the WVS measure of trust, finding that where trust is higher flows of climate aid are increased. From Mostafa, we see that postmaterial values are associated with climate change commitment, with respondents who display a stronger commitment to postmaterial goals also professing higher engagement with climate change.

Guided by these studies, we created 'National climate aid values' measure, constructed from WVS responses on trust and postmaterialism. The seventh wave of the World Values Survey ran from 2017-22, and so most closely aligns with our own timeframe for analysis. With WVS 7, questions relating to trust in foreigners and to postmaterial commitments are ordered on scales ranging from one to four. To generate the 'National climate aid support' variable, we calculated the mean national

[45] See 'World Values Survey Wave 7', World Values Survey website, available at https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp. Accessed 7th May, 2024.

response to each individual question, and then combined the scores.[46] As noted below, we re-ran our models to test for independent effects from national values on trust and on postmaterialism.

With H_2, we expect to find an interaction between governing majority and climate aid generosity. We develop two specifications to probe this relationship. As a first specification, we start from the assumption that climate aid will be unpopular to the median voter. We therefore develop a specification to test whether governments with a vulnerable majority systematically wind-down their commitments in this area. The suggestion from Tobin et al (2022) of public aversion to certain types of development assistance, and from Abbott and Jones (2021) of voter resistance to development assistance during times of economic constraint as characterised much of the 2010s, inform our adoption of this founding assumption. To operationalise this initial specification, we created a dummy variable to identify governments with a vulnerable majority. Building from Clegg and Davies (2024), we identify a government as having a vulnerable position if it holds a parliamentary majority of under 10 percent, as captured within the CPDS. As a second specification, we interacted the dummy indicator of government vulnerability (where a value of 1 denoted a vulnerable majority, and a value of 0 captured safer governments) with the 'National climate aid values' measure. This second specification probes whether governments might respond in a nuanced and contextually-determined manner to electoral vulnerability.

The third of our institutional foundations of interest relates to government orientation, and the expectation from H_3 that left-wing control will be associated with higher levels of bilateral climate aid effort. To operationalise this government orientation variable, we remain with the Comparative Political Dataset. The CPDS includes a Schmidt Index measure of partisan orientation, with values

[46] In relation to postmaterial values, respondents were asked to select the statement that most accords to their view of the most important national priority over the next decade. The index was designed to test commitments to postmaterial values, with a score of 1 attached to the 'A high level of economic growth' response, through to a score of 4 for the 'Trying to make our cities and countryside more beautiful' response. In relation to trust in individuals from other nationalities, respondents were invited to position themselves on a scale where 1 indicated 'Trust completely' and 4 indicated 'Do not trust at all'. In their original form, the postmaterial and trust-related measures were mis-aligned; whereas a higher numerical score on the former captured a more intense presence of postmaterialism, a higher score on the latter captured a more intense lack of trust. To ensure a meaningful combination, we inverted the trust scores prior to combining the two measures.

running from 1 where there is a right-wing hegemony, through 3 for centrist governments, and up to 5 to capture the presence of a left-wing governing hegemony.

Turning to our control variables, we constructed a dummy variable for EU membership by drawing directly on the CPDS. To control for economic development, we created a GDP per capita variable that brought together World Bank data on GDP expressed in constant US$ values,[47] with data from the CPDS on the size of national populations. For rate of economic growth, we drew on the CPDS measure of the annual change in GDP, expressed as a proportion of the previous year's value. We remain with the CPDS to operationalise economic openness, drawing on its measure of volume as trade expressed as a proportion of GDP. To control for overall bilateral aid generosity, we combined OECD data on development assistance flows with the CPDS data on national populations to create an aid per capita measure.[48] Finally, to control for climate vulnerability, we turn to the University of Notre Dame Global Adaptation Index (ND-GAIN).[49] We specifically draw on the ND-GAIN country exposure score for this purpose. Through the exposure score, ND-GAIN combines measures including projected impact from climate change on food systems by analysing changes to future crop yield, health systems through consideration of changed disease vectors and death rates, and challenges to infrastructure through increased flood hazard and sea level rise. A country's ND-GAIN exposure score is a continuous value between 0 and 1, with the higher values representing greater exposure.

By combining the above data sources, we are able to construct a database that covers OECD member state climate aid flows from 2010-21. Given that the outcome variable represents budgeted commitments, we run all independent and control variables contemporaneously. The assumption here is that the political and economic factors of interest in a given year will have influenced the budget-setting process in that same year. There is some loss of observations owing to incomplete data. Most notably, a lack of WVS coverage means that we are unable to include the Belgium, Ireland, or

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[47] See 'GDP (current US$)', World Bank website, available at https://data.worldbank.org/indicator/NY.GDP.MKTP.CD. Accessed 8th May, 2024.
[48] See 'Development finance data', OECD website, available at https://www.oecd.org/dac/financing-sustainable-development/development-finance-data/. Accessed 8th May, 2024.
[49] See 'Country Index', University of Notre Dame Global Adaptation Index website, available at https://gain.nd.edu/our-work/country-index/. Accessed 8th May, 2024.

Luxembourg cases, and a lack of CPDS coverage means that we are unable to include the South Korean case. Our dataset contains 234 country/year observations. Given the unbalanced time series characteristics of the dataset, we run our models using a panel corrected standard errors estimator. In line with Beck and Katz (2011), to control for potential year- and case-based fixed effects, we include year and case dummy variables in our models.

3.4. Results and discussion

Our dataset contains 234 country/year observations, covering 25 of the OECD's member states.[50] There is significant variation in countries' revealed levels of climate aid effort across. The Slovak Republic is the overall climate aid laggard, with its 2016 climate aid commitment of just US$0.1m translating into a per capita value of just US$0.02 (Table 3.1). In contrast, Norway is a clear climate aid leader. The country was the most generous per capita donor for eight of the 11 years covered by the dataset, and its 2014 total of over US$211 in climate aid per person represents the highest commitment overall. While the downturn across 2015 and 2016 bucked the trend, there is an overall tendency for climate aid generosity to increase over time (Figure 3.3).

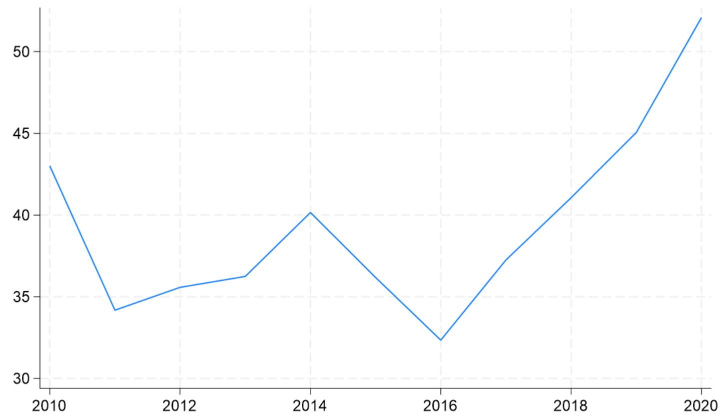
Table 3.1. Descriptive statistics

Variable	Obs	Mean	Std dev.	Min.	Max.	Frequency
Outcome						
Bilateral climate aid per capita (US$)	234	38.14	45.17	.02	211.47	~
Independents						
National values	234	-.243	.464	-1.364	.501	~

[50] The following OECD members were included in the analysis: Australia, Austria, Canada, Czechia, Denmark, Finland, France, Germany, Greece, Iceland, Italy, Japan, Lithuania, Netherlands, New Zealand, Norway, Poland, Portugal, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, United Kingdom, and United States.

Vulnerable gov't	234	.825	.381	0	1	0 (41), 1(193)
Vulnerable gov't*National values	234	-.227	.415	-1.364	.501	~
Partisan orientation	234	2.278	1.363	0	5	1 (103), 2 (33), 3 (48), 4 (30), 5 (20)
Controls						
EU membership	234	.662	.474	0	1	0 (79), 1 (155)
GDP per capita (US$)	234	40,726.57	17,691.69	11,432.82	87,123.45	~
GDP growth	234	1.847	1.790	-7.087	6.304	~
Aid per capita (US$)	234	138.86	159.09	2.14	849.57	~
Climate exposure	234	.400	.059	.273	.520	~

Figure 3.3. OECD average annual climate aid commitment (constant US$ per capita)



Source: OECD climate aid database.

Turning to the first of our independent variables, our measure of national values in support of climate change runs on a potential scale from -3 to 3, with a higher value capturing national values that are more propitious for enhanced climate aid efforts.[51] The average country score, at -.243, sits toward the centre of this scale. The Greek score of -1.364 represents the national sentiment that is least

[51] A score of -3 would result from a mean trust in people from other nationalities of -3 (indicating the lowest trust score) and a mean postmaterialist value of 1 (indicating the lowest postmaterialist score). In contrast, a score of 3 would result from a mean trust in people from other nationalities of -1 (indicating the highest trust score) and a mean postmaterialist value of 4 (indicating the highest postmaterialist score).

supportive of climate aid, and the Icelandic score of 0.501 represents the national sentiment that is most supportive. For the second of our independent variables, we see that across the cohort there was a prevalence of governments with vulnerable majorities. Overall, 193 of the 234 observations were coded as featuring vulnerable governments, whose parliamentary majorities were under the 10 percent benchmark. Under our alternative specification for government vulnerability, we interacted this dummy with the national values score. With this alternative measure, we see that where there were vulnerable governments, the national climate aid sentiment was largely aligned with results from across the cohort as a whole. Our final independent variable incorporates the Schmidt Index measure of partisan orientation. With this measure, we see evidence of a notable right-wing bias in government control. Overall, 136 case observations include governments with a rightwing hegemony or rightwing dominance, with just 50 cases of a leftwing hegemony or leftwing dominance.

The results of our models are displayed in Table 3.2. Through Model 1, we deploy the original specification of the government vulnerability independent variable, which captures cases where governments' parliamentary majorities are under 10 percent. Variables included in Model 1 exert a significant effect on per capita climate aid commitments (p=.000), with the model explaining around 87 percent of observed variation (x^2=.867).

Table 3.2. Explaining variation in climate aid effort

	Model 1		Model 2		Model 3	
Variable	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error
Independents						
National values	24.385	6.492***	28.728	8.399**	26.002	6.143***
Vulnerable gov't	-14.8	5.233**	~	~	~	~
Vulnerable gov't*National values	~	~	-5.845	6.519	~	~
Vulnerable*Left	~	~	~	~	2.082	5.146

Vulnerable*Right	~	~	~	~	-11.204	4.599*
Partisan orientation	2.309	1.117*	1.659	1.146	~	~
Controls						
EU membership	-22.107	20.866	-25.699	21.551	-26.631	20.311
GDP per capita	-.000	.000	-.000	.000	-.000	.000
GDP growth	-.475	.781	-.824	.78	-.724	.765
Aid per capita	.146	.050**	.155	.051**	.155	.050**
Climate exposure	-10.128	45.279	-1.315	45.329	-4.362	45.319
Year FE	Yes		Yes		Yes	
Country FE	Yes		Yes		Yes	
n		234		234		234
x^2		.867***		.851***		.862***

Under Model 1, in line with the expectations of H_1, we see a positive association between national values and bilateral climate aid. Where national values are more supportive of climate aid, as gauged by the composite score of trust in people from other nationalities and postmaterial commitments, we see higher bilateral commitments. A one-unit increase in this national values indicator is associated with an additional bilateral climate aid commitment of around US$24 per capita. With Model 1, we also see a confirmation of the H_2 expectation of a negative relationship between government vulnerability and climate aid generosity. Governments with a parliamentary majority of under 10 percent of seats, relative to their more secure peers, display reduced climate aid flows of around US$15 per head of population. This finding gives support for the intuition that climate aid might be considered to run counter to median voter preferences, which underpins this variable specification. We probe this negative effect from government vulnerability in subsequent models. Turning to H_3, we see from Model 1 that left-wing governments do indeed seem to display higher levels of climate aid commitment. For a one-unit move in the Schmidt Index score towards the left-wing pole, we see that governments increase climate aid commitments by around US$2 per capita.

With Model 1 we see limited effects from control variables, with only aid per capita displaying the expected positive relationship to climate aid flows.

To explore further the mechanisms around the government vulnerability effect, through Model 2 we deploy the alternative specification that captures the potential interaction between vulnerability and national values. The intuition here is that vulnerable governments may feel pressure to align their behaviour more closely with national sentiment. The results from Model 2 find no significant association between this alternative specification and bilateral climate aid generosity. Under Model 2 the expected national values effect on climate aid continues to be shown, and the expected partisan orientation effect is no longer displayed. This shift illustrates the sensitivity of the partisan orientation effect to variable specification.

Model 3 presents findings from our final attempt to probe in more detail the mechanisms that lie behind the variation in climate aid effort associated with governing party vulnerability. Drawing on insights from Isotalo et al (2020) that parties of the left and right may hold competing views on what constitutes a median position on a given policy issue, we test whether there is an interaction between partisan orientation and government vulnerability. Based on the intuition that vulnerable right-wing governments may want to shore up a base that is more sceptical of climate change and more hostile development assistance, we would expect to see vulnerable right-wing governments display significantly reduced climate aid efforts. We integrated 'Left vulnerable' and 'Right vulnerable' variables into Model 3, created by interacting the government vulnerability dummy with the Schmidt Index score.[52] Results from Model 3 confirm a systematic difference in the way that governments of the left and right respond to electoral vulnerability. On average, governments of the right who were electorally vulnerable wound-down climate spending by around US$11 per capita, relative to all other governments. No significant difference was found between climate aid effort from governments who were left-wing and electorally vulnerable and the generosity of other governments. It seems likely,

[52] Specifically, where governments were vulnerable and had a Schidt Index score of 4 or 5 they were designated as left vulnerable, and where they had a score of 1 or 2 they were designated as right vulnerable. The capacity of Model 3 to probe overarching partisan orientation effects was limited by the fact that all of the 50 governments of the left also met the vulnerability specification. The partisan orientation variable was excluded from the parameters adopted for Model 3, although its inclusion did not significantly affect results.

then, that in times of electoral constraint right-wing governments row-back on climate aid commitments to appeal to their base.

Overall, taking account of the results from Models 1-3, we can be confident that the institutional foundations provided by national values, electoral vulnerability, and government partisan orientation have significant influence on bilateral climate aid generosity. We see from all models that national values of trust in people from other nationalities and prioritisation of postmaterial goals support effective performance in this area, with effective performance being defined here as higher per capita flows of climate aid. We see from Models 1 and 2 that governments' partisan orientation matters, with Model 3 introducing the possibility that this effect may be moderated by electoral vulnerability. And we see from Models 1 and 3 that government vulnerability exerts a dampening influence on climate aid generosity, with right-wing governments in particular seeming to respond to vulnerability with reduced climate aid commitments. These social and political institutional structures matter, when it comes to national effort on bilateral climate change assistance.

3.5. Conclusion

The nature of the obligations held by advanced industrialised states to their peers across the global South have have long been debated. Advanced industrialised states are widely held to have a moral requirement to acknowledge and compensate for historic acts of exploitation that brought material advantage to their populations, with principles of distributive justice also being seen to provide compulsion for significant transfer payments (Opeskin 1996). Over the last decade or so a new dimension has emerged to these debates, with the increasing focus on the requirements of climate justice. There is now broad-based consensus across the international community that, for principled reasons and from the pragmatic judgement that strengthened climate transition capacity across the global South will bring collective gains, bilateral climate assistance must be embedded and extended.

Through this chapter, we have probed variation in the scale of these bilateral climate flows emanating from OECD member states.

Focusing on a measure of climate aid expressed in constant US$ per capita values to support comparability across time and between cases, we found high levels of variation. We also found clear evidence of individual climate aid leaders, with the Norwegian government leading the pack by featuring as the most generous donor in eight of the 11 years covered by the dataset. When seeking to explain the institutional foundations of this variation, we drew on scholarship on the political economy of development assistance and on the politics of climate change to derive core expectations. Our analyses presented evidence in support of the expected positive role from social institutional foundations, with nation values that were more strongly trusting of foreigners and committed to postmaterial priorities being associated with higher climate aid effort. We also found support for the existence of political institutional foundations pertaining to governing majorities and partisan orientation. Where governments have more vulnerable majorities they appear to wind-down their supply of climate aid, with the behaviour of vulnerable right-wing government in particular seeming to explain this dynamic. Given our focus on institutions for stronger bilateral climate aid performance, national values and secure governing majorities constitute important foundations.

Back in 2009, advanced industrialised states committed to providing US$100bn per year in climate assistance by 2020. With the deadline missed, it was several years subsequent that the target was belatedly met. Given the difficulty in meeting this commitment, and given the clear need to surpass this level of resource flow in coming years, it is vitally-important that our understanding is improved of the institutional factors that are supportive of accelerated climate aid effort. We present our findings on the social and political institutional foundations of climate aid generosity as a contribution towards this overarching aim.

4. Institutions for Catalysing Private Climate Finance: The World Bank and the Challenge of ‘Market-Making’

4.0. Introduction

This chapter constitutes a point of transition in the book, as we begin to turn our attention from state-based mechanisms of climate financing and toward market-based mechanisms. Multilateral development banks have come to occupy a prominent position at the intersection of state- and market-based climate financing, with World Bank staff having since the late 2000s worked to support for the emergence and mainstreaming of green bonds. By exploring the relationship between the World Bank and the mobilisation of private climate finance, we here probe the interplay between state- and market-based institutions.

On the surface, the relationship between the World Bank and market-based climate financing seems to be a positive one. The World Bank was central to the establishment of green bonds as a new asset class from the late 2000s, and over the years has itself issued green bonds to the tune of some US$36bn. The World Bank has, as such, played a direct role in channelling large volumes of private finance toward sustainable transformation, and has been praised as laying foundations for more effective climate change performance.[53] In this chapter, we go beneath the surface by presenting an interrogation of the ‘market-making’ impact of World Bank climate finance operations. In this context, we use the term market-making to refer to an action that leads to an increased scale or efficiency of private climate financing flows. Consequently, effective market-making can be said to have occurred

[53] See, for example, ‘Green Bond of the Year’, Environmental Finance website, available at https://www.environmental-finance.com/content/awards/environmental-finances-bond-awards-2022/winners/green-bond-of-the-year-supranational-world-bank.html. Accessed 18th June, 2024.

where we see evidence of an increased scale or efficiency of sustainable financing flows after a particular World Bank intervention. To achieve this goal of evaluating the market-making impact of World Bank climate finance operations, we build on analysis undertaken by the World Bank Independent Evaluation Group (IEG 2020). By developing and deploying a more sensitive operationalisation of World Bank green bond market presence, we allow for a more fine-grained analysis of World Bank impact. We also pay particular attention to the World Bank impact on green bond markets across emerging and developing economies, a focus that seems apposite given the World Bank's operational prioritisation of financial market expansion across this cohort.

The findings from our analysis present a nuanced picture regarding the impact on climate financing effectiveness from World Bank operations, and overall it is a picture that questions the positive market making impact as reported by the IEG. When initially following core elements of the IEG operationalisation, we reproduce the headline IEG finding of a positive association between World Bank green bond issuance in a given market and the average size of a corporate green bond issuance in that market. However, questions over the nature of the World Bank's impact are raised when we deploy our alternative specification, which more effectively differentiates between corporate green bonds that were (not) preceded by a World Bank issue. Using this alternative specification, we find a negative association between World Bank green bond issuance and the size of subsequent corporate green bond issuance in that market. The suggestion of a negative association between World Bank issuance and the size of subsequent corporate green bonds is amplified as we hone-in specifically on emerging and developing markets. Across this subsection of countries, under our preferred alternative specification of World Bank market presence, we again find that where there has been a World Bank green bond placement prior to a given corporate green bond issue, the size of the corporate green bond is on average reduced. It seems possible that, rather than making markets, the World Bank may be functioning as an unintentional market breaker, reducing the vibrancy of national green bond markets potentially by crowding-out subsequent corporate green bond issues.

To develop our core line of analysis, the chapter proceeds through the following structure. In the first section below, we contextualise the World Bank's approach to market-making. We review the increased operational focus that has been placed by the World Bank on financial sector expansion over recent decades, as well as the more recent turn towards market-based sustainable finance and green bonds. In the second section we lay foundations for our empirical model, by identifying likely determinants of the size of a bond issue. We then in the third section of the chapter explain the decisions taken when operationalising our study, and outline our method of analysis. Through the fourth section of the chapter we present and discuss our results, which overall suggest that the World Bank may have unintentionally functioned as a market breaker across green bond markets. We conclude by providing a recap of the core insights put forward, and reflecting on our findings.

4.1. The World Bank and market-making - key dynamics

The World Bank is very much a state-based institution. When established back in 1945, the World Bank was empowered to lend to its member states for the purpose of post-war reconstruction and development. The World Bank's core capital came directly from its member states, who acquired voting power in line with the volume of resources that were paid in. However, from the outset, the World Bank was also an organisation with a close relationship to financial markets. Below, we review the evolution of the World Bank relationship to market-based financing, through to its synthesis of market financing and climate transition with its support for green bonds.

Structurally, the World Bank was designed to directly plug-in to flows of private finance. The institution's resourcing plan was for the organisation to use bond issuance to leverage the lending power of its paid-in capital base. Through this market-based approach, the World Bank has been able to borrow at relatively low cost, and to lend-on to member states who either lack access to large capital markets, or who would attract high borrowing charges. In its early decades of operation, enhancing the organisation's ability to tap into private capital markets became something of an institutional obsession. In the drive to gain and maintain a 'AAA' credit rating, the World Bank's Central Projects Staff sought to push the organisation toward low-risk investment opportunities that displayed clear capacity to generate the revenue flows needed to cover loan repayments. Eugene Black, the World Bank President from 1949-62 who joined from a senior leadership position at Chase National Bank, sought to embed a focus on sound lending at the heart of the organisation. Through this relatively cautious approach to investment during Black's tenure, the World Bank began to suffer from something of an embarrassment of riches, consistently generating a healthy surplus on its lending operations.[54]

Beyond this background structural dependence on private finance, two points of operational evolution are particularly noteworthy in the World Bank's journey toward financial sector development and market making. The first of these occurred early in the World Bank's existence, and came with the formation of its International Finance Corporation (IFC). The second occurred significantly later, and came with the use of policy-based lending to encourage members to establish institutions and frameworks capable of catalysing a deepening of financial markets as a means of expanding the volume of domestic resources available to support economic transformation.

It was in 1956 that the IFC was formed. Prior to the creation of the IFC, the World Bank was operationally constrained to lend only to its member governments. A very small volume of lending to private enterprises did take place within the World Bank's first decade, but operational rules meant that these loans ultimately required a member government to act as guarantor. Loans were, for example, arranged in 1949 and 1950 with the governments of the Netherlands and Finland respectively, which saw around US$325m being passed first to financial intermediaries, who then on-lent to a range of private corporations.[55] With World Bank senior management advocating the extension of this ability to support private enterprise, movement commenced toward the launch of a new operational unit. Initially the IFC was operationally and financially housed within the World Bank, and later became an autonomous entity under the broader World Bank Group umbrella.[56]

[54] For an overview of these early institutional foundations, see Clegg (2017: 44-8).
[55] For ease of comparability, all historic data on World Bank lending in this section have been converted to current values.
[56] For a history of the IFC, see Mason and Asher (1973: 359-65).

Across its history, the importance of financial market making has grown within IFC operations. From the IFC's launch up to 1971, around US$350m was transferred to financial institutions as seed capital to be on-lent to domestic private enterprises. However, this volume was dwarfed by the US$2.8bn that was directly lent by the IFC to manufacturing companies across the same time period. With the formation of a Capital Markets Department in the IFC during the 1970s, a pivot toward financial sector lending was embedded. The Capital Markets Department provided advisory services to financial institutions and state agencies involved in financial regulation, and the unit helped financial sector lending to become the largest component of IFC lending (Sethness 1988). By 2000 around 40 percent of IFC lending went to the financial sector, with manufacturing, the next largest sectoral recipient, receiving around 15 percent (Mavrotas 2002: 4). Currently, IFC financial sector lending constitutes around 50 percent of total outlays (IFC 2022: 63).

In parallel with these shifts in IFC operations, we have over recent years also seen something of a mainstreaming of financial market making within the World Bank's state-based lending. The turn towards financial sector engagement within World Bank lending to its member states came as part of its wider turn toward domestic policy and institutional reform in the 1990s. At the time, governments were often heavily involved in shaping financial sector behaviour and outcomes. The use of 'directed credit' was common, through which targets were set for the volumes of loans to particular economic sectors to be distributed by state-controlled or state-influenced financial institutions. Rates of interest levied through financial systems were also often shaped by governments. Under its policy-based lending of the 1990s and 2000s, the World Bank sought to support member states to introduce broad-based financial sector liberalisation. In the decade from 1992, 68 World Bank member states (around two-thirds of all states eligible to borrow for the World Bank) received at least one policy-based loan that included conditions relating to financial sector reform (Cull and Effron 2008). In its contemporary lending operations this focus on financial sector development remains quantitatively significant, with around 15 percent of the World Bank's lending to its member governments being targeted at this sector (World Bank 2022: 14).

Over recent years, the World Bank sought to codify aims of its financial sector engagements through its Maximising Finance for Development (MFD) agenda. Foundations of the MFD approach were laid in 2015, as the World Bank led a broad-based call for a ‘paradigm shift’ in development financing that would see public resources being used to crowd-in private investment, thereby increasing resource flows ‘from billions to trillions’ (African Development Bank et al 2015). As later codified within operational guidance, under MFD World Bank staff were encouraged to ask a simple question when designing a new lending arrangement: ‘Is there a private sector [financing] solution?’. If the answer to this question was deemed to be ‘Yes’, then the arrangement should be designed to support and catalyse private sector financing (World Bank 2017). Mechanisms for encouraging private financing into projects for sustainable development include various forms of ‘de-risking’, such as ‘first loss’ provisions and guarantees that would see a government sponsor agreeing to absorb costs of project failure, and assured long-term pricing and payments contracting under public-private partnerships (Dimakou et al 2021).

So, through these points of evolution within its IFC and core lending operations, the World Bank has undergone a notable pivot towards financial sector market making across its relatively recent history. As well as this broad-based shift, we can see that a prominent focus has also been placed on work to catalyse private climate financing in particular. Recently, the MFD agenda has moved increasingly towards one of Maximising Finance for Climate, as the World Bank has sought to fill the gap in bilateral and multilateral climate financing resources by turning to private sources (Galindo-Gutierrez 2024). And, most notably, we have seen the World Bank playing a key role in establishing and embedding the asset class of green bonds since the late 2000s.

The emergence of green bonds was inextricably linked to the ambition of multilateral development banks to accelerate the flows of private financing for sustainable transformation, with the European Investment Bank and World Bank leading with the creation of the new asset class. The back-story to the World Bank’s first green bond issue in 2008 is one of contingency, and capacity building. The World Bank Treasury reports that the process leading to its first green bond issue began with an out-of-the-blue telephone call from a consortium of Swedish pension funds, who wanted to deploy their investment capacity to support climate change transition. The pension fund representatives identified a missing piece of the sustainable investment puzzle; environmentally-motivated investors such as themselves were, unfortunately, unable to easily judge whether their bond investments were making a meaningfully beneficial contribution to the climate change cause. A solution hit upon through discussions with World Bank staff was to incorporate declarations over 'use of funds' within bond issue documentation to reassure investors over the eventual impact that would be generated by finances raised through the bond issue. It was also decided that investors' confidence in the attainment of climate-beneficial outcomes could be increased by contracting-in a third party monitoring agent, capable of providing authoritative judgement on the impact of activities being funded by a green bond issue. The Norwegian Centre for International Climate and Environmental Research (CICERO) was contracted by the World Bank to fulfil this monitoring role, as the organisation moved towards its first green bond launch.[57]

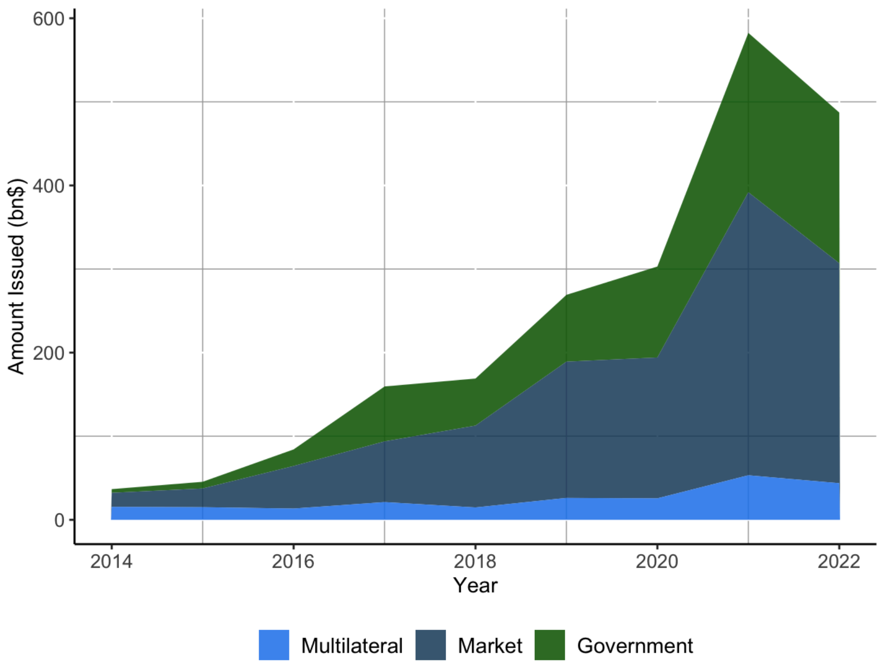
With CICERO's oversight, the World Bank's first green bond was issued on 12th November, 2008. With this placement on the Luxembourg Stock Exchange, the World Bank became the first institution to issue a financial asset labelled as a 'green bond'. The World Bank was able to raise around US$300m from investors, at an annual rate of interest of 3.5 percent. Within its green bond launch documentation, the World Bank committed to using the resources to deliver projects 'that promote the transition to low carbon and climate resilient growth in the recipient country'. Example projects listed in the documentation included solar and wind power infrastructure, energy efficient housing, and reforestation (World Bank 2008: 3).

Building from this initial offering, the World Bank increased its issuance of green bonds over subsequent years. Through 2009, over US$500m was raised through four green bond placements by the World Bank, and through 2010 an additional US$730m raised through around 20 issues.[58] In these later years of the 2000s and into the 2010s, global green bond issuance remained dominated by the

[57] See '10 Years of Green Bonds', World Bank website, available at https://www.worldbank.org/en/news/immersive-story/2019/03/18/10-years-of-green-bonds-creating-the-blueprint-for-sustainability-across-capital-markets. Accessed 21st March, 2024.
[58] Authors' analysis of World Bank bond issuance, World Bank website, available at https://finances.worldbank.org/Other/Green-Bonds-since-2008/ayy5-fxut/about_data. Accessed 9th April, 2024.

World Bank and other multilaterals, with these green bonds functioning as a mechanism to channel private investment into projects backed by state actors and agencies. However, from the mid-2010s, green bond issuance became an increasingly private sector game. A step-change in the green bond market came in 2013; the Swedish property company Vasakronan became the first corporate green bond issuer, and was rapidly joined by Apple, Credit Agricole, and Industrial and Commercial Bank of China launches. By 2014, the majority of green bond issuance was being undertaken by market actors and, as the volume of green bond issuance has increased over the subsequent years, this predominance has remained (see Figure 4.1).

Figure 4.1. Annual Green Bond issuance (US$bn)



Source: Authors' analysis of Climate Bond Initiative database.[59]

[59] See 'Green Bond database', Climate Bond Initiative website, available at https://www.climatebonds.net/market/data/. Accessed 9th April, 2024. To create the 'Market' category, we have amalgamated the entries for 'Financial corporate' and 'Non-financial corporate' entities.

Green bonds, then, have become a quantitatively significant source of market-based climate financing. What, though, has been the impact on green bond market performance from the World Bank's issuance of green bonds? Has the World Bank functioned as a market maker, with its own green bond issuance being associated with an increased size of issuance across other green bonds in a given market? To lay the groundwork for addressing this question, in the section below we turn to scholarship on bond market functioning, to gain insight into the range of market factors and characteristics that are likely to influence issuance.

4.2. Explaining performance variation - existing literature

A bond is a financial security through which an investor agrees to lend a given volume of resources to a government or company for a specified period of time, and in return receives regular interest payments. The rate of interest agreed between issuer and investor is often referred to as the coupon rate, derived from the historic practice of bond holders submitting a paper coupon to call-in an interest payment. In its most common form, when a bond reaches maturity at the end of the specified time period, the holder is repaid the face value of the asset. So, for example, where an investor purchases a US$10m bond with a 2 percent coupon rate and a maturity of ten years, they may accrue 10 annual payments of US$200,000 before then receiving back the initial US$10m face value.

In the early history of bond markets, states functioned as the sole type of issuer. With the consolidation of their fiscal capacities, governments could plausibly commit to repayment schedules stretching many years into the future. While there is significant debate over when the asset class first emerged, by the renaissance period Italian city states were commonly involved in bond issuance (Massa 2002). The first major act of the Bank of England following its establishment in 1694 was to issue what is generally regarded as the first modern sovereign bond, the purpose of which was to raise finance for the Anglo-French Nine Years' War.[60] Nowadays, with corporations, states, and international organisations all involved in bond issuance, and with a multitude of traders acting in

[60] See 'History', Bank of England website, available at https://www.bankofengland.co.uk/about/history. Accessed 27th June, 2024.

secondary markets, bonds provide a central pillar of financial markets. Each year new bond issuance reaches around US$8tn globally, with the total outstanding value of bonds thought to be well in excess of US$100tn.[61] Average issuance size varies greatly by market, issuer, and bond characteristics. Global bond listings from a midsummer's day in 2024, for example, show a range running from the Brazilian Banco Daycoval's US$9,271 issuance at a coupon rate of 9.8 percent over a two-year maturity, to the government of China's US$20.2bn issuance at 2.1 percent over seven years.[62]

The World Bank IEG (2020: Annex F) evaluation of the impact of the institution's green bond issuance provides a useful point of entry for identifying likely determinants of green bond issue size. The headline finding from the IEG study was the existence of a positive association between World Bank green bond issuance into a given market, and the average size of green bond issuance within that market. In markets where the World Bank had been active, other issuers tended toward larger green bond launches. The IEG study captured 721 green bonds issued from 2010-18 across 27 markets. The IEG specifically focused on the impact of IFC operations, identifying some 49 green bonds that had been issued by the IFC into domestic markets. The headline finding from the IEG was that, where a market had received an IFC green bond issue, the size of green bond issues within that market was on average around 100 percent larger.[63] However, the IEG is cautious about the extent to which causal inference should be made off the back of this analysis. Whereas it might be the case that IFC green bond issuance into a given market was causing an increase in the size of other green bond issues within that market by for example demonstrating to other issuers and investors the potential demand for sustainable finance products, the IEG acknowledged that it also might be the case that the IFC was simply being attracted to more vibrant green bond markets.

The central aim of the empirical work through this chapter is to update and extend the IEG analysis of the impact of World Bank green bond issuance. When seeking to identify additional empirical work to inform our expectations over the likely impact from World Bank green bond

[61] See 'Global Bond Markets', International Capital Markets Association website, available at https://www.icmagroup.org/market-practice-and-regulatory-policy/secondary-markets/bond-market-size/. Accessed 28th June, 2024.
[62] Authors' analysis of Refinitiv listings of all bonds issued on 24th June, 2024.
[63] The IEG analysis used a log transformed measure of Green Bond issue size in the core models. Table F.4 reports a coefficient effect of 1.012 from the dummy variable capturing IFC market presence; this 1.012 increase in the value of log transformed measure equates to an increase of around 101 percent.

issuance on wider market performance, there are no additional direct studies of multilateral organisation issuance to turn to beyond the IEG evaluation. However, we can look to broader scholarship on the political economy of bond markets for transferable insights.

Within scholarship on public issuance and wider bond market dynamics, there are prominent debates over 'crowding-out' and 'crowding-in'. In this context, the former term refers to an increase in private debt issuance and investment that follows on from public issuance and investment, and the latter refers to a reduction in private debt issuance and investment that follows on from public issuance and investment. Expressed using our preferred terminology, crowding-out can be seen as a form of market breaking, and crowding-in a form of market making.

A negative crowding-out impact could be generated through a direct pathway whereby the public issuance attracts capital that otherwise would be free to circulate to alternative private securities, or through an indirect pathway whereby public issuance bids-up the rate of interest required to attract a given investment quantum and so dampens the scale of private borrowing. Contributions to this scholarship on the crowding-out effect from public sector issuance include Baldacci and Kumar's (2010) long-term analysis of advanced and emerging-market performance across 1980-2008, Broner et al's (2014) analysis of sovereign issuance during the Eurozone crisis in the early 2010s, and Anyanwu et al's (2017) assessment of dynamics across 28 resource-rich economies from 1990-2012.

Contrastingly, we have a range of prominent studies that identify pathways for crowding-in. Friedman's (1978) suggestion that public issuance can in fact generate a crowding-in effect by laying foundations for improved growth trajectories and market confidence is mirrored by Ahmed and Miller's (2000) finding that debt-financed public investment on infrastructure can lead to an acceleration in private investment. Hatano (2010) and Andrade and Duarte (2016) offer broadly supportive findings from their respective analyses of Japan and Portugal. Public issuance can, then, exert beneficial effects on wider market performance. Given the prioritisation of infrastructure to support climate transition by World Bank green bonds,[64] it may plausibly be the case that these

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[64] For an overview of use of proceeds from World Bank Green Bonds see 'Climate Stories - Green Bonds', World Bank website, available at https://www.worldbank.org/en/news/feature/2023/04/10/from-india-to-indonesia-green-bonds-help-countries-move-toward-sustainability. Accessed 10th July, 2024.

issuances are meeting the growth-enhancing criteria laid out by Friedman, Ahmed and Miller, Hatano, and Andrade and Duarte, and therefore generating a market making effect. Complex time lags are involved in the realisation of these crowding-in or crowding-out effects. On the one hand, market participants may react rapidly to a World Bank green bond issue on the basis of the expected growth-enhancing impact of the issue. On the other, participants may await actual evidence of shifting economic fundamentals before shifting their expectations, evidence that would emerge only over the medium- or longer-term.

To move beyond this uncertainty over the likely impact of World Bank green bond market issuance on overall market performance, it is useful to turn to Dittmar and Yuan's (2008) study of the mechanics of crowding-in effects in emerging markets. Dittmar and Yuan specifically found sovereign issues to provide valuable information signals to investors; in markets where it can be a challenge for investors to calibrate expectations and assess investment prospects owing to information shortage, sovereign issues can provide a useful benchmark. Given the relative immaturity of green bond markets, it is possible that an equivalent signalling effect could be occurring from World Bank green bond issuance. A World Bank green bond issue, under this logic, may provide a benchmark of market appetite for this form of asset, around which future issuers and investors can more effectively and confidently calibrate their expectations and actions. Where Dittmar and Yuan found increasing willingness to invest in emerging market corporate bonds off the back of sovereign issuance, parallel dynamics may see World Bank green bond issuance exerting a demonstration effect that causes other green bond issuance in that market to be viewed in a more positive light.

The plausibility of World Bank green bond issuances exerting a form of demonstration effect is supported by wider scholarship on the political economy of international financial institutions. Focusing on the International Monetary Fund, Gehring and Lang (2020) find that countries with an active lending arrangement enjoy a more rapid improvement to perceived creditworthiness after a period of macroeconomic crisis. Reinsberg et al (2022) offer a confirmation of this relationship in the negative, showing that an IMF programme interruption undermines investor confidence in sovereign creditworthiness. While there is ongoing debate as to the actual impact on investment flows from the activation of an IMF programme (cf. Bird and Rowlands 2002, Breen and Egan 2019, Krahnke 2023), Arabaci and Ecer's (2014) analysis of shifts in yield spreads and maturities suggest an overall positive effect. The intuition is that IMF operations exert a catalytic effect on market actors, providing reassurance that the country is moving to a more positive trajectory than would otherwise be the case. International organisation financial engagements, then, have the potential to supply information that is supportive of future investment flows. Transferring these findings to the World Bank green bond market engagement, it may plausibly be the case that a World Bank presence provides a stamp of credibility on a given green bond market that reassures current and potential future investors and overall expands scope for larger subsequent corporate green bond issues.

Existing scholarship, then, presents nuanced insights into the likely effect of World Bank green bond issuance on wider domestic green bond market performance. However, taken together, the positive association found by the earlier IEG assessment of IFC green bond issuance, and the plausibility of a World Bank green bond issue generating a positive information signal that catalyses additional investment capacity, leads us toward the following hypothesis:

H_1: World Bank green bond issuance within a domestic market will lead to more effective market performance.

It is worth noting that, in line with Dittmar and Yuan (2008), we would expect the demonstration effect from a World Bank green bond effect to be stronger in markets with greater information shortages. We should, then, have a stronger expectation of a positive market making impact from green bond issuance in emerging and developing country green bond markets.

Exploring the impact from this World Bank institutional engagement in domestic green bond markets constitutes the primary point of interest in our empirical work. However, to enhance the strength of our analytic insights, we also derive a series of additional control variables from existing scholarship on the political economy of bond performance.

The first of our control variables relate to macroeconomic factors that we expect to exert a significant influence over the scale of green bond issuance. We know from Bierman (1966), Hotchkiss and Jostova (2017), and Barry et al (2008, 2009) that benchmark interest rates play a vital role in constraining or enabling larger bond issuance. As noted above, the coupon yield is the cost of borrowing levied on the issuer of a given bond. Where a coupon is higher, it would cost the issuer more to gain access to a given volume of investment funds. Given relatively fixed repayment capacity, a higher coupon rate translates into a smaller bond; if an issue is faced with a 10 percent rate of interest rather than a 5 percent rate of interest, the issuer will wish to take-on a reduced scale of debt. The individual coupon rate of a bond is heavily contextually determined, and in particular is shaped by the prevailing rate of interest within a given market. To attract investors, the bond issuer must offer a return that is in line with other securities that display broadly equivalent characteristics. As such, when seeking to explore determinants of the size of green bond yields, we control for the prevailing benchmark interest rate.

The second of our macroeconomic controls relates to the size of the economy that hosts a given financial market. We know that smaller economies are less able to host vibrant capital markets. The lower volume of domestic resources means that there is a limited pool of internal investment, and external finance is likely to just the costs of entering a shallow market where information flows may be limited and difficult to access to outweigh potential gains (Eichengreen and Luengnaruemitchai 2004, Bhattacharyay 2013, Presbitero et al 2016). The reduced financial innovation that is associated with lower levels of economic development provide an additional reason for expecting a reduced uptake of green bond issuance in smaller economies.[65]

Beyond these expected macroeconomic drivers of the size of green bond issuance, to close we consider expected effects from bond-level characteristics. While the relationship between bond issue size and maturity can be complex (Stohs and Mauer 1996), at a general level we expect to see an inverse relationship between issue size and maturity. Larger firms with stronger growth prospects hold an expectation that their cost of borrowing is likely to reduce over time, and so are less willing to issue bonds with longer maturity (Ozkan 2000). And finally, we expect to see a relationship

[65] For a review of the relationship between financial innovation and economic performance, see Qamruzzaman and Jianguo (2018).

between coupon type and issue size. Coupon type refers to the structure of interest payments that are assigned to a particular bond. The most common coupon type is referred to as a ‘plain vanilla’ coupon. Under this model, the rate of interest that will apply throughout the bond’s lifespan is fixed at the point of issue. A range of alternative coupon types exist under which the rate of interest paid to the bondholder is variable, including the ‘margin over index’ where the rate tracks a specified benchmark, and the ‘step-up’ where periodic increases are planned-in to the payment schedule. Overall, we expect there to be a negative relationship between a fixed-interest coupon and issue size, as smaller firms with less well-established reputations more typically use fixed-interest offerings as a tool to mitigate against their perceived riskiness (Amiram et al 2018).

4.3. Data and Method

Our overarching interest through this study is in understanding determinants of green bond issue size. To operationalise our outcome variable, we turned to the Refinitiv Government and Corporate Bonds Advanced Search application. We initially extracted data on around 10,800 green bonds listed in this source.[66] However, given our particular focus on domestic market making, when we exclude ‘eurobond’ issues in non-domestic currencies and issues that are not linked to a specified national market, we reduce the number of cases to around 7,300. Each case observation captures one green bond issue. On the Refinitiv database the value of individual issues are recorded in US dollars, and we take this figure as our outcome measure. With our interest in the behaviour of market actors, we focus on corporate green bond issues. This focus on corporate issuance reduces our overall cohort size to around 6,400. Our final dataset includes 4,297 corporate green bond observations, with case attrition largely explained by the loss of observations from 2023 owing to the lack of contemporary

[66] With data extracted in late-June 2024, in this initial sample the latest issue date was 21st June, 2024, and the earliest was 5th July, 2007.

GDP data. In order to ensure a normal distribution of the issue size outcome variable, we follow IEG (2020) by log transforming their dollar values.

We created two specifications to operationalise our main independent variable on 'World Bank green bond market presence'. Firstly, we follow IEG (2020) in creating a dummy variable for 'Any World Bank green bond market presence'. Through this first specification, we identify domestic markets into which the World Bank has issued a home-currency-denominated bond, through manual coding of the Refinitiv-generated cohort of green bonds.[67] Under this measure, we assigned a score of 1 to a green bond that was issued into a market that had, at one time or another, received a World Bank green bond issue. Under this specification, the World Bank green bond issue could have taken place at any point in time. A weakness of this specification is that corporate green bonds can be identified as being issued into a market with a World Bank presence even if the World Bank issue had not yet taken place. As such, this initial specification can end up testing for the effects of a treatment that was not actually received. We deploy this initial specifically to test the reproducibility of the IEG findings across a contemporary dataset,[68] but hold our alternative specification as our preferred operationalisation.

As an alternative specification of this main independent variable, we develop a measure that more effectively controls for temporal characteristics of the World Bank green bond issues. Given our specific interest in the impact of World Bank presence in a national green bond market, with our alternative specification we differentiate between Green Bond issues that were preceded by a World Bank green bond, and those that were not. In a given national market, we identified the date of the first World Bank green bond issue; where an observed green bond was issued into that market after this date we coded the observation 1, with a value of 0 for all other cases. We illustrate using the Canadian market. The first World Bank green bond issue into the Canadian market came on 24th September, 2019. Prior to this date, 10 Canadian corporate green bonds had been issued; these

[67] Within Refinitiv the IFC and IBRD have been assigned separate 'tickers'; acronyms used to identify specific companies, states, or state agencies involved in bond issuance. We have combined these tickers, to capture all Green Bonds issued by the World Bank.
[68] Whereas the World Bank IEG captures Green Bond issuance 2007-18, our dataset runs through to 2024.

observations were given a score of 0 to indicate that they were issued into a green bond market that had not received the World Bank treatment. There were 37 Canadian corporate green bonds that were issued after this date; these observations were given a score of 1 to indicate that they were issued into a green bond market that had received the World Bank treatment. In markets where there was no World Bank green bond issuance, all green bonds issued into that market were given a score of 0. With its capacity to more adequately capture important temporal nuance, this alternative specification of the World Bank green bond presence variable is our preferred measure.

To operationalise the benchmark rate of interest, we turned to the Refinitiv MacroMonitor suite to create a Policy Rate variable. The central bank policy measure constitutes a key baseline interest rate in a given market, representing the rate at which the central bank is willing to lend to financial institutions requiring short-term funds. We extracted daily information on the policy rate for each of the markets that featured on our cohort of green bonds, and matched these data points to each green bond observation according to the date and market of issue. So, for example, with the Commonwealth Bank of Australia's green bond issue of 31st March, 2017, the Policy Rate variable has a value of 1.5 because on this date the Australian central bank policy rate sat at 1.5 percent. To operationalise a variable to capture the size of the economy, we remain with the Refinitiv MacroMonitor suite. We specifically use the annual GDP figure for the host market, calibrated in constant US dollar values to support comparability of the measure across time.[69]

To control for the maturity features of individual green bonds, we used the issue and maturity dates listed in Refinitiv to generate a 'Lifespan' measure expressed in days. So, for example, a Green Bond issued on 21st April, 2017, and with a maturity of 21st April, 2027, would have a lifespan of 3,652 days. Finally, to control for the form of interest arrangement attached to a given green bond, we drew on the coupon type as listed in Refinitiv. We specifically created a dummy variable to differentiate between bonds with a vanilla coupon where the rate of interest is fixed at and stable from the point of issuance, and bonds with variable or dynamic rates of interest.

[69] These national GDP figures are expressed in 2021 US dollar values. With 2023 GDP values unavailable at the time of writing, our end year is fixed at 2022.

With the inclusion of this collection of variables, our total sample size of green bonds stands at 4,297. Observations form a cross-sectional dataset, with each case capturing the macroeconomic context and bond-level characteristics associated with a given issue. As noted above, by log transforming the issue size outcome variable, we have ensured that the normal distribution requirement for OLS regression is not violated, and so we deploy this estimator when running our models. In our models, to control for temporal and spatial heterogeneity, we include year and country fixed effects. Given that our cohort is constituted by units that have, in effect, been assigned a country-level treatment (i.e. the (non-)presence of World Bank Green Bond issuance within a national market), we deploy country-level clustering of standard errors across our models.[70]

4.4. Results and discussion

Our dataset covers 4,297 home currency-denominated corporate green bonds, issued between 2012 and 2022. Looking at our outcome variable, there is an extremely wide spread in the value of green bond issuance across the cohort (see Table 4.1). At one end of the scale we have the trivially small US$194 issue from the Brazilian Solfacil Securitizadora de Creditos Financeiros,[71] an entity established by the financial institution Travessia Securitizadora for the purpose of making bond placements.[72] We then have a further seven issues with values below US$50,000, all placed in the US, before we see the quantum increase significantly. At the top of the table we see over 90 green bonds with values in excess of US$1bn. The US$4.1bn Green Bond from Industrial Bank, China, leads the way on issue size. The five largest issues are constituted by three additional similar-sized placements from Chinese financial institutions, and a US$3.4bn placement from the Danish bank Nykredit Realkredit. The mean green bond issue value was US$187m.

[70] For a comprehensive review and statement of good practice, see Abadie et al (2022). For a brief overview of key points of debate, see McKenzie (2017).
[71] While we considered removing this observation as a potential case of erroneous data entry, the information on Refinitiv appears to be internally consistent.
[72] See 'Solfacil Securitizadora Credito Financeiro', CBonds website, available at https://cbonds.com/company/245571/. Accessed 3rd July, 2024.

Table 4.1: Descriptive statistics

Variable	Obs	Mean	Std dev.	Min.	Max.	Frequency
Outcome						
Green Bond value (US$m)	4,297	187.26	325.86	.00	4,146.85	~
Independent						
Any WB Green Bond issuance in market	4,297	.55	.50	0	1	0 (1,924), 1 (2,373)
Prior WB Green Bond issuance in market	4,297	.49	.50	0	1	0 (2,193), 1 (2,104)
Controls						
Policy rate	4,297	.98	1.71	-.75	16	~
GDP (US$tn)	4.297	7.98	7.43	.02	20.93	~
Lifespan	4,297	2,392.22	2,044.91	30	28,317	~
Vanilla	4,297	.78	.42	0	1	0 (958), 1 (3,339)

By delving further into dynamics across individual domestic markets, we see evidence of wide performance variation in terms of the number of corporate green bond placements and their average value (see Table 4.2). Denmark leads the way on average green bond issuance size at a value of US$1.2bn, while China leads the way on the quantity of corporate issues with over 1,000 placements listed as green bonds. In contrast, the average green bond issue size was under US$10m across the Kenyan, Malaysian, and Nigerian observations, and five markets returned just one green bond issue.

Table 4.2: National Green Bond market characteristics

Country	Corporate issues	Mean corporate Green	World Bank Green	Country	Corporate issues	Mean corporate Green	World Bank Green

		Bond value (USDm)	Bond issues			Bond value (USDm)	Bond issues
Australia	26	219.78	2	Mexico	20	141.95	0
Austria	27	112.87	0	Morocco	4	28.32	0
Belgium	14	253.38	0	Netherlands	1	541.05	0
Brazil	103	68.33	0	New Zealand	15	88.63	1
Canada	47	345.79	1	Nigeria	1	8.38	0
China	1004	201.29	1	Norway	194	96.18	0
Colombia	11	39.79	0	Peru	1	26.84	0
Denmark	17	1177.35	0	Philippines	5	113.91	0
Finland	8	208.31	0	Portugal	4	97.45	0
France	118	612.48	0	Romania	2	130.82	0
Germany	443	156.01	0	Russia	8	150.08	0
Hong Kong	3	129.53	0	Singapore	7	201.81	0
Hungary	23	59	0	Slovakia	5	266	0
Iceland	10	30.99	0	South Africa	20	38.33	1
India	35	28.25	1	South Korea	202	66.31	0
Indonesia	7	93.7	0	Spain	18	265.94	0
Italy	11	64.26	0	Sweden	405	54.09	1
Japan	284	67.48	0	Switzerland	82	179.61	0
Kazakhstan	1	78.76	0	Thailand	54	55.61	0
Kenya	1	6.05	0	Turkey	3	18.14	0
Latvia	6	71.15	0	United Kingdom	3	78.12	1
Lithuania	2	29.84	0	United States	510	446.71	70
Malaysia	223	6.49	0				

Turning to the initial specification of our independent variable, we see that there is a relatively even split between green bonds whose launch was within a market that, at one time or another, hosted a World Bank green bond placement, and green bonds issued into a market with no World Bank presence. 2,373 green bond observations displayed the former characteristic, and 1,924 observations displayed the latter. If we switch our unit of analysis from individual green bond issues to individual markets, the distribution of World Bank engagement is more skewed. As shown in Table 4.2, just 9 of the 45 green bond markets included in our dataset hosted a World Bank green bond placement.

We now move to our alternative specification of the independent variable, which captures whether a World Bank green bond occurred prior to a given corporate green bond issue. We see that notably fewer corporate green bonds meet this alternative specification for World Bank market presence, with 2,104 cases meeting this criteria and 2,193 cases not meeting this criteria. Overall, we consider the alternative specification superior. The initial specification marks some corporate green bonds as meeting the World Bank presence criteria even though the World Bank issue occurred after the corporate issue; our alternative specification avoids this problem.

From our macroeconomic control variables, we see a mean policy rate of .98 percent across all observations. Sweden and Switzerland provide the lowest policy rates, featuring negative values during the late 2010s and early 2020s. In contrast, Brazil, Kazakhstan, Nigeria, and Turkey all featured policy rates above 10 percent, with the Kazakh value of 16 percent in December 2022 leading the field. There is wide variation in market size across our cohort, from the Icelandic GDP of US$0.02tn in 2020 through to the US GDP of US$20.93tn in 2022. In terms of green bond characteristics, across our cohort we have an average lifespan of around 2,392 days, which equates to some 6.5 years between issue and maturity dates. Finally, we see that the heavy preponderance of green bond issues feature plain vanilla interest rate schedules, with 3,339 issues displaying this feature.

Through Model 1, test the initial IEG specification of our independent variable. Using this measure, we return results that are in line with the expectations of H_1, and in line with the findings from IEG (2020). Where a corporate green bond is issued into a market that, at some point in time, hosts a World Bank green bond, we see an associated uplift in the value of that green bond. With Model 1 overall explaining around 44 percent of the observed variation in green bond size, we see also a significant and positive impact on the outcome from green bond lifespan. We introduce our favoured specification of World Bank green bond presence in Model 2. Here, the specification captures only cases where the World Bank green bond issue occurred prior to the given corporate green bond issue. Under this specification the coefficient of the World Bank green bond market presence becomes negative, suggesting that World Bank issuance may in fact be crowding out corporate issuance. The explanatory power of Model 2 is equal to that of Model 1, and we continue to see a significant and positive association between green bond lifespan and value. Model 2 confounds H_1 by suggesting that the World Bank may be playing a market breaking role, with World Bank green bond issuance reducing subsequent market capacity. However, given the relatively low significance level from the World Bank market presence on corporate green bond values (p=.090), we interpret this result with a degree of caution.

4.3: Explaining variation in Green Bond size, all cases

	Model 1		Model 2	
Variable	Coefficient	Std error	Coefficient	Std error
Independent				
Any WB Green Bond issuance in market	4.071	.323***	~	~
Prior WB Green Bond issuance in market	~	~	-.367	.212+
Controls				
Policy rate	.037	.039	.045	.040
GDP	.037	.132	-.028	.130
Lifespan	.000	.000+	.000	.000+

Vanilla	.027	.117	.026	.117
Year FE		Y		Y
Country FE		Y		Y
Country clustered errors		Y		Y
n	4,297		4,297	
X^2	.438		.439	

Note: +p<0.1, * p<0.05,** p<0.01, ***p<0.001.

Through Models 3 and 4, we progress to consider specifically the World Bank impact on green bond markets across emerging and developing countries.[73] Focusing on this subset of markets, our total number of observations drops to 2,050. However, we retain significant variation under both specifications of the World Bank green bond market presence variable. With the initial IEG-informed initial specification, we see that 685 corporate green bonds were placed into markets that did not at any time feature a World Bank presence, while 1,365 placements entered into markets that did at some time feature a World Bank presence. Using our alternative more time sensitive measure, the respective figures are 688 and 1,362. Model 3 uses the initial specification, where we find no evidence of a significant relationship between World Bank market presence and corporate green bond size. Model 4 uses the alternative specification, under which we see evidence of a significant and negative relationship between a World Bank presence in a market prior to a given corporate green bond launch, and the size of that green bond launch. Overall the explanatory power of Model 4 is high, with its parameters capturing around 54 percent of the observed variation in green bond size.

4.4: Explaining variation in Green Bond size, emerging and developing markets

[73] We specifically deploy the Bank for International Settlements' Country Grouping Convention when identifying emerging and developing countries. The BIS provides a list of advanced industrialised states, and defines emerging and developing countries in the negative as any states not on this list. See 'BIS Country Grouping Convention', BIS website, available at https://www.bis.org/statistics/country_groupings.pdf. Accessed 8th July, 2024.

	Model 3		Model 4	
Variable	Coefficient	Std error	Coefficient	Std error
Independent				
Any WB Green Bond issuance in market	.035	.134	~	~
Prior WB Green Bond issuance in market	~	~	-.948	.089***
Controls				
Policy rate	.059	.009***	.059	.010***
GDP	-.088	.072	-.088	.072
Lifespan	.000	.000	.000	.000
Vanilla	.084	.083	.084	.083
Year FE	Y		Y	
Country FE	Y		Y	
Country clustered errors	Y		Y	
n	2,050		2,050	
X^2	.540		.540	

Note: +p<0.1, * p<0.05,** p<0.01, ***p<0.001.

Looking across the cohort of emerging and developing markets, then, we find no evidence to support H_1. In contrast, through Model 4 we see findings that appear to confound H_1. For emerging and developing countries, it seems that where a corporate green bond issue is preceded by a World Bank green bond issue, the corporate issue is significantly smaller. Off the back of the insights provided from Models 2 and 4, which deploy our preferred specification of World Bank market presence, it seems that the World Bank has functioned as an unintentional market breaker, with its own issuances attracting market capacity away from subsequent corporate issues. To help us to interpret these confounding findings, it is useful to return to the scholarship considered above on the political economy of crowding-in and crowding-out.

One pathway available theoretically for World Bank green bond issuance to generate a market making impact was through impacting other investors' and issues' expectations on future market performance. Extrapolating from work including Ahmed and Miller (2000), Hatano (2010) and Andrade and Duarte (2016), we hypothesised that one might witness either a rapid market making impact through this pathway, or a slower impact over the medium- or longer-term. A rapid impact would be generated if the expectations held by other investors and issuers of future market performance were immediately shifted by the very fact of a World Bank green bond issue. A more medium- to long-term impact would be generated if evidence of changed economic performance emerged that caused investors and issues to update their expectations. Our analysis features a cohort of treated corporate green bonds that run from issuance just days after a World Bank issue into the same market, through to a maximum lag of 13.5 years between the World Bank and observed corporate issue. On average, where a corporate green bond was launched in a treated market, the gap between World Bank issue and corporate issue was around 6.5 years. Given the breadth of the timeframe for capturing market impact, our findings raise questions over the consequence on wider market performance from World Bank green bond issuance across and short-term, and the medium- to long-term.

A second pathway available theoretically for World Bank green bond issuance to generate a market making impact was through a demonstration effect.[74] Under this pathway, a World Bank issue would function as a benchmark that would allow other market participants to calibrate their expectations, and to issue or invest with greater confidence. However, with the preponderance of green bond markets having been established without a World Bank presence, alongside the overall finding of a market breaking impact from World Bank issuance, we have notable reasons to reject the existence of this mechanism. With 36 of the 45 green bond markets captured by our dataset not featuring a World Bank presence, this demonstration effect is clearly a non-necessary factor behind the establishment of a market for green bonds.

[74] As noted above, this expectation was derived from insights provided by Dittmar and Yuan (2008), Gehring and Lang (2020), Reinsberg (2022), and others.

As noted above, the World Bank played a vital historic role in establishing green bonds, an asset class that has grown to become the most important source of climate financing. However, our analysis here raises the possibility that negative consequences may follow from World Bank domestic green bond issuance on the value of corporate green bonds subsequently issued. Given the quantitative importance of green bonds as a tool for supplying market-based climate financing, it is vitally important that further interrogation of these dynamics be undertaken.

4.5. Conclusion

As the multilateral development institution with the largest lending power and whose expertise shapes ideas over global best practice, World Bank operations are of high intrinsic importance. Focusing specifically on climate transition and climate financing, the organisation offers us a case with which to probe the interaction between public and private actors in supporting effective performance. While the most prominent aspects of World Bank operations involve lending to member states for the purpose of implementing projects and policies to support sustainable development, the World Bank has in fact always been an institution with important links to market actors and market finance. The World Bank has since its inception used borrowing from private capital markets to supplement its resource base, and since the launch of its International Finance Corporation has lent directly to commercial organisations including in particular financial institutions. This engagement with market-based finance further stepped-up from the 1990s, as World Bank interventions increasingly sought to catalyse financial sector development as a means of boosting domestic capacity to fund economic transformation.

The World Bank's dual focus on sustainable transformation and financial market engagement elides around green bonds. As a mechanism for releasing private finance to support climate transition, green bonds have grown to become the leading source of climate financing. Since its first green bond issue in 2008, the World Bank has been a key presence on green bond markets. Through this chapter, we have sought to probe the relationship between World Bank green bond issuance, and the wider effectiveness of national green bond markets. Our overarching expectation was that World Bank green bond issuance would have a positive effect on market performance, as measured by the size of corporate green bond issues. In line with existing scholarship and the IEG's (2020) assessment, our core hypothesis was that in markets where the World Bank had been active, we would see an uptick in the value of corporate green bond issues.

Our analysis of this relationship between World Bank green bond market presence and overall market effectiveness suggests, however, that the World Bank may be functioning as a market breaker rather than a market maker, with subsequent corporate issuance being scaled back after a World Bank issue. While we interpreted our finding of this market breaking dynamic across the whole cohort of country observations with caution, when we turned specifically to emerging and developing markets the results appeared more compelling. Informed by these findings, we suggest that the hypothesised demonstration effect from World Bank green bond issues likely does not exist. The fact that most green bond markets have been established without the assistance of a World Bank issue to benchmark market performance and expectations, alongside the headline findings from our analysis, lead us to this position. We also suggest that World Bank green bond issuance is not positively impacting on investor and issuer confidence over either the short-term, or medium- to longer-term. Given the vital importance of green bonds as a tool for raising the finance needed to deliver climate transition, further research into the interaction between multilateral development bank operations and the performance of green bond markets is urgently required, with a view to identifying and addressing any constraining effects.

5. Institutions for Regulating Private Climate Finance: The European Central Bank's 'Anti-Green Impact'

5.0. Introduction

Since at least the time of Walter Bagehot, the vitally-important role of central banks in stabilising financial systems has been widely acknowledged. In his seminal *Lombard Street*, Bagehot (1873: 2) describes the co-existence of 'the greatest combination of economical power and economical delicacy' within the late-Victorian-era London financial system. The financial system centred around Lombard Street and Threadneedle Street in the City of London had a concentration of power and level of domestic and international influence that was hitherto unparalleled. But, for Bagehot, there was a surprising level of fragility to this all-powerful beast. With little warning and at obtuse cause, the financial system was want to transmute from a Leviathan bringing order, to a Behemoth bringing chaos. The prime task of central banks and central bankers was, then, to tether financial markets to the former, bringer-of-order, version of themselves.

Bagehot's writings continue to be drawn upon as inspiration for creative and proactive central banking.[75] At core, Bagehot saw central banks as fulfilling their stabilisation mission most effectively by providing short-term loans at relatively high rates of interest to commercial banks.[76] The purpose of these short-term loans were to allow commercial banks to meet their immediate payment obligations. Bagehot was aware that running a commercial bank involved an extraordinarily complex act of cashflow management, where even fundamentally sound institutions could, from time to time,

[75] See, for example, Carre and Le Maux (2024).

[76] In line with Bagehot we use the term 'commercial bank' expansively, to refer to appropriately registered institutions engaged in formalised lending to and borrowing from individuals and corporate entities.

lack the ready money needed to meet payments that were expected and required by others. To forestall loss of confidence and situations of panic, central banks, for Bagehot, must develop tools for short-term stabilisation.

Over the years, Collateral Frameworks have become a central pillar of central banks' infrastructure for short-term stabilisation finance. When a commercial bank requires a short term cash injection, central bank Collateral Frameworks in effect list a range of financial assets the central bank is willing to purchase from the commercial bank. Collateral Frameworks provide details of the 'haircut' that a central bank will impose within such a transaction; where a commercial bank wishes to sell an asset that appears to have a current market value of US$100m, the central bank may demand a haircut of 5 percent and pay only US$95m, in order to reduce the risk that when the central bank re-sells the asset on the open market it will suffer a financial loss.

In addition to this market stabilising function, central banks have over the last decade come to consider climate transition and climate financing an increasingly important part of their mission.[77] There is a small but vibrant literature that seeks to explore the effectiveness with which central banks have assimilated this new climate change mission (e.g. Campiglio et al 2018, Bailey and Jackson 2024, Jackson et al 2024). Through this chapter, we probe interaction between Collateral Framework operations on the one hand, and this new central bank concern with climate change on the other. Looking specifically at the impact of European Central Bank liquidity operations on green bond market performance, our headline finding is that the Collateral Framework is imbued with what we term an 'anti-green impact'. The novelty of green bonds as an asset class, and the relative marginality of green bonds to mainstream financial system operations, seems to lead the ECB to systematically demand larger haircuts when purchasing green bonds within their Collateral Framework operations. Moreover, we show in turn that this anti-green impact from the ECB translates into depressed performance of the wider green bond market. It appears that, unintentionally, ECB 'business as usual'

[77] See, for example, Mark Carney, 'Breaking the tragedy of the horizon: Climate change and financial stability', Bank of England Chairman's speech to Lloyds of London, 29th Sept, 2015. Bank of England website, available at https://www.bankofengland.co.uk/speech/2015/breaking-the-tragedy-of-the-horizon-climate-change-and-financial-stability. Accessed 5th February, 2025.

Collateral Framework operations may provide headwinds that generate costs for green bond issuers and holders.

In elaborating this line of analysis, we present the chapter through the following structure. Through the first section below, we contextualise the study by reviewing the increased engagement from central banks with climate transition and climate financing over recent years. We then in the second section review literature on the political economy of central banking from which we can derive expectations on the interaction between Collateral Framework operations and green bond performance. In this section we present hypotheses relating to an expected negative impact from central bank operations on the performance of market-based green finance. We explain our approach to operationalising our empirical work in the fourth section of the chapter, providing information on data sources and analytic strategy. In the fifth section we present and discuss the results of our analysis, before concluding the chapter with a recap of key findings and reflection on their scholarly and policy relevance.

5.1. Central banks and climate finance - key dynamics

According to conventional wisdom, central banks are bastions of conservatism. By structure and by culture, central banks are thought of as institutions that hold tightly to their established principles and practices. Neoclassical colonnades and oak-panelled grand conference rooms serve, architecturally, to anchor central banks as institutions of tradition; institutions with history, and institutions within history. However, as recently argued by Manuela Moschella (2024), this conventional wisdom view may overlook central banks' capacity for change. Moschella makes the claim that central banks should in fact be considered 'unexpected revolutionaries' who have, over the last five decades, both made and unmade 'economic orthodoxy'. While our study introduces questions over the extent to which central banks have become climate revolutionaries, it is nonetheless the case that there has been a significant re-framing of central bank purpose and practice to align with the exegesis of climate change.

With their survey of the mandates provided by national governments to their central banks, Dikau and Volz (2021) supply a valuable snapshot of the state of the art of modern monetary policy. Drawing on the International Monetary Fund's Central Bank Legislation Database, Dikau and Volz demonstrate the surprising frequency with which central bank mandates create a relatively direct link between these institutions and a concern with climate change and climate policy. Over half of the 135 institutions surveyed had a mandate that required the central bank to support sustainable economic performance. On a small number of occasions the central bank mandate explicitly mentioned a need for operational alignment with climate transition, and more commonly the central bank mandate required the central bank to support fundamental policy commitments made by their national government (which in most cases includes a ratified commitment to Paris Agreement principles). The remaining central banks have mandates more tightly linked to price stability and economic growth. However, given the fundamental tension between unabated climate change and the predictable achievement of such outcomes, Dikau and Volz suggest that these remaining central banks' mandates supply an implied need for close engagement with climate change transition. Overall, the position laid out by Dikau and Volz is that all central banks, under their current mandates, are required to ensure that their operational practice supports the increased flow of climate financing.

As central banks' operational focus on climate change has begun to emerge, it has become common for the institutions to develop a twin-track focus. On the one hand, central banks have sought to understand the nature of 'physical risk' from climate change; to anticipate the negative impact on economic activity from increasingly extreme weather events, so as to more actively and effectively support *ex ante* mitigations and *ex post* response. On the other hand, central banks have sought to interrogate potential sources of 'transition risk'; that is, to identify potentially negative impacts on growth and employment trajectories from national climate change strategic plans (Grippa et al 2019).

Central banks' efforts to orient decision-making processes and operational practice around climate change-informed themes have required some reconfiguration of core ideas. As noted by Boneva et al (2022: 773-7), frameworks for inflation targeting are having to be revised, to allow for informative modelling of (monetary policy responses to) climate shocks. Typically, extreme weather events are conceptualised as a 'supply-side exogenous shock'; an event that is not directly driven by underlying economic conditions, and which are likely to decrease output and increase prices. Analytically, central bank economists are trying to generate frameworks for clarifying when such climate change shocks should be 'looked through' (i.e. when central banks should allow for an unimpeded period of inflation and economic adjustment after a climate shock), and when such shocks should be responded to (i.e. when central banks should raise interest rates to depress demand and reduce the magnitude of climate shock-driven inflation). An additional pressure on central banks is a need to alter the temporal parameters around their decision-making. Whereas the institutions are used to thinking in terms of short-term financial market behaviours and responses to monetary policy as manifest across the medium term as measured by periods of months, taking climate change serious involves considering the impact on decarbonisation pathways ten or more years into the future from contemporary action (Thiemann et al 2023).

Beyond these ideational reconfigurations, surveys of central bank operational practice have started to reveal evidence of some climate change-informed evolution to operational practice. Campiglio et al (2018) differentiate between informationally-oriented change, and market intervention-oriented change.

In terms of informationally-oriented change, we see that central banks are increasingly supporting and requiring larger market actors to monitor and report on both their exposure to climate change risk, and their strategies for managing and reducing such risk. Given that the majority of companies lack the knowledge base or established processes for accomplishing delivering such tasks, central banks have themselves assumed positions of leadership. In the late 2010s, the Bank for International Settlements' Financial Stability Board,[78] for example, issued its *Task Force for Climate-Related Disclosures* recommendations (FSB 2017). The intention behind the FSB report was to provide a framework to guide national practice, with the recommendations including sector-specific advice to improve the accuracy with which financial sector lenders and insurers understood the nature of their climate risk exposure. The French Energy Transition Law (2015) is seen as an early example of legislative change to require disclosure on climate risk and climate impact, whose implementation is being supported by the efforts of the FSB and beyond.[79]

In terms of market intervention-oriented change, we see some evidence of central bank reform over recent years. Vibrant debate over the climate change implications of quantitative easing (QE) has, in particular, informed change in this area. QE featured as a core component of central bank responses to the Global Financial Crisis of 2007/8. Through the late-2000s, many central banks made vast purchases of assets from financial institutions, in order to inject cash into financial systems and allow these financial institutions' payments and disbursals to remain relatively stable through a time of deep market uncertainty. QE was typically designed with the aim of avoiding the creation of a 'distorting' impact on the market. In practice, this meant that central banks would favour assets that were performing relatively strongly, and be reluctant to purchase assets that seemed to have weaker market demand. It is, however, possible that the market-neutral intentionality behind QE generated unintentionally negative climate outcomes. By using the purchasing power of central banks to reproduce today's prevailingly fossil fuel-reliant economic processes, the argument goes, QE may well

[78] The Bank for International Settlements (BIS) functions as a venue for cooperation on financial system supervision and regulation. The BIS is headquartered in Basel, Switzerland, and its core membership is composed of representatives from 63 central banks. The Financial Stability Board, hosted and funded by the BIS, makes recommendations on good practice for monitoring and managing systemic financial risk. See Financial Stability Board website, available at https://www.fsb.org/. Accessed 10th February, 2025.
[79] See Climate Change Laws website, available at https://climate-laws.org/document/law-no-2015-992-on-energy-transition-for-green-growth-energy-transition-law_aea3. Accessed 10th February, 2025. Climate Change Laws is a collaborative enterprise between the London School of Economics Grantham Institute and the Columbia Law School Sabin Center.

have passed up an opportunity to prioritise funding for decarbonisation and climate transition (Ferret Mas 2023). Indeed, Schreiber et al (2020) suggest that the European Central Bank's preference for assets linked to oil- and gas-based productive processes were such that QE came, in climate impact terms, to be a 'dirty secret'.

Central banks are said to commonly be 'nervous' about moving away from the principle of market neutrality. While reflecting prevailing market sentiment within QE operations fits with central bankers' technocratic self-image, moving towards acting as catalysts for climate transition can be seen as a form of politicisation. Nonetheless, some central banks have taken operational steps toward reconstituting QE as a tool for supporting climate transition. The Bank of England, for instance, in 2021 introduced a series of measures to 'green' its Corporate Bond Purchase Scheme. While continuing to only purchase investment grade corporate assets, the Bank of England committed to 'tilt purchases toward stronger climate performers'. This outcome was to be achieved by incorporating metrics on corporations' carbon disclosure practices, carbon intensity of production, and achieved emissions reductions, into the Bank of England's decision-making processes on corporate bond purchases.[80] Also in the early 2020s a broadly parallel reconfiguration of ECB bond market interventions was announced by its President Christine Lagarde.[81]

There are reasons to consider, then, that changing ideas within central banks have led to an increased prioritisation of climate transition, and an awareness that greater consideration could be given to the impact of central bank operations on national decarbonisation performance. Given that, since 2008, the world's central banks have poured in excess of US$13tn into the purchase of financial assets,[82] it is important to enhance our understanding of the impact from considerable material power on climate financing trajectories. In essence, the empirical analysis that we present through this chapter targets this issue. As is detailed below, we assess whether the European Central Bank's

[80] See Bank of England website, 'Greening our Corporate Bond Purchase Scheme', available at https://www.bankofengland.co.uk/markets/greening-the-corporate-bond-purchase-scheme. Accessed 10th February, 2025. See also Bank of England (2021).
[81] For information on this aspect of ECB greening, see DiLeo et al (2023).
[82] See Logan and Bindseil (2019: 8).

everyday Collateral Framework market interventions serve to maintain and reproduce carbon-intensive economies, or to tilt toward support for climate transition.

5.2. Explaining performance variation - existing literature

The analytic focus of this chapter is on the interaction between central bank liquidity interventions, on the one hand, and (green) bond performance, on the other. In order to outline the expectations we derive from existing scholarship into this relationship, it is necessary to begin by explaining some detail on Collateral Framework operations. After covering this detail, we then briefly present the expectations on the drivers of variation (green) bond performance that we derive from existing scholarship.

As noted above, Collateral Frameworks are an important mechanism through which central banks intervene in financial markets. Collateral Frameworks are used to ensure that short-term challenges constraining an otherwise healthy institution's ability to meet its obligations do not spill over into broken contracts, loss of confidence in that institution, and wider market distress. When a financial institution requires access to cash from a central bank, Collateral Framework operations allow the financial institution to sell an asset to the central bank in exchange for cash. Central banks rely on a 'marks-to-market' approach to valuing the asset, using contemporaneous market pricing as the core input when deciding how much cash to pay for the asset. However, the valuation process involves central bank personnel making judgements about the perceived riskiness of the asset. Where an asset is thought to be more risky - that is, where the asset is thought to be more likely to loose value during the period of time that the central bank would be holding the asset - the central bank will pay a smaller proportion of the asset's current market value when purchasing. The difference between the market value and the purchase price paid by the central bank is known as the 'haircut'; where an asset has a market value of, for example, US$100m and a central bank pays US$95m, a haircut of five percent has been applied.

We know from existing studies that financial asset structural characteristics impact on the size of haircut applied through Collateral Framework purchases. Where the asset has a shorter maturity,[83] the haircut is typically lowered. The form of asset is of significance, with bonds (whose repayment comes directly from the initial issuer) typically receiving lower haircuts relative to asset-backed securities and other instruments where repayment is contingent on more complex flows of finance between a larger network of actors. We also see systematic effects from issuer type and credit rating, with sovereign issuers and more issuers with stronger credit ratings receiving lower haircuts (ECB 2015, Cassola and Koulischer 2019).

Given the lack of existing literature specifically exploring interactions between green bond performance and central bank Collateral Framework operations, to derive expectations into this relationship we turn creatively to proximate work. We specifically seek insight from scholarship more generally on the conservative and anti-green effect of central bank collateral operations.

In terms of this general anti-green effect from collateral operations, existing scholarship has probed the unintended consequences of the principle of market neutrality. According to this principle, central banks want to avoid exerting influence on financial market behaviour or outcomes when delivering their stabilisation function. The market neutrality principle has, however, been criticised widely for its conservative and anti-green effects. When undertaking any market intervention, it is easier to avoid distorting a larger market than a smaller market. The metaphor likening central bank transactions to the throwing of stones into water can be used to illustrate these dynamics. If one were to toss a pebble into a sea, the impact remains essentially invisible; toss an identical pebble into a small pond, and the impact would be impossible to ignore. The anti-green impact comes from the fact that markets for carbon intensive and fossil fuel sector-based financial

[83] Maturity refers to the length of time before the date at which the bondholder will receive their final interest payment or/and full repayment of the principal.

assets are relatively large and mature, and as such central banks feel freer to intervene without fear of causing disruption. These carbon-intensive asset markets are the metaphorical sea into which central banks can relaxedly toss pebbles, purchasing and re-selling assets without concern for the trivially invisible ripples (Boneva et al 2022, Jourdan and Kalinowski 2019, Monnin 2018). In their empirical testing of these dynamics, Dafermos et al (2021) find that ECB decision-makers see carbon-intensive financial assets, with their larger and more mature markets, as representing a lower risk option. Under its quantitative easing programme in the late 2010s, haircuts given by the ECB to assets from carbon-intensive sectors were lower by between 60 and 366 basis points relative to equivalent assets from non-carbon-intensive sectors. ECB collateral operations, in other words, were displaying preference for carbon-intensive sectors, and thereby putting non-carbon-intensive sectors at a disadvantage (given that sellers would by the ECB be paid a lower proportion of the market value for non-carbon-intensive assets).

Informed by these studies of anti-green impact from collateral operations, we adopt the following hypothesis to be tested empirically:

> *H_1: Less favourable treatment will be given to green bonds under the ECB collateral framework.*

The aim of our empirical work is to go beyond existing scholarship in the area, by specifically testing whether this outcome is driven by subjective factors. Existing scholarship suggests that structural characteristics of carbon-intensive bonds cause preferential treatment, with for example their typically larger issue size and greater market liquidity contributing to these outcomes (Dafermos et al 2021). By controlling for core structural characteristics in our empirical work, we seek to identify whether the subjective ‘green bond’ label is driving variation in treatment under the ECB Collateral Framework.

Our intuition here is that subjective decision-making within the ECB is contributing to systematic anti-green impact. While subjective bias is not commonly assessed in studies of bond performance or central bank market operations, we gain a strong inference for the possibility of such a dynamic from wider literature on the behavioural foundations of financial markets. Shefrin's foundational study attracted analytic attention to the micro-level drivers of financial market behaviour; by rejecting a priori neoclassical assumptions about investor behaviour, Shefrin (2008: 2-5) instead pushed for theoretically-informed empirical interrogation of actual practice. Drawing on Chappell et al (2008), we posit that central bank evaluations of instruments from a relatively novel asset class may be influenced by the constrained access to historic data on market performance. Chappell et al suggest that financial analysts' ways of 'making sense' of markets contain a built-in conservatism against novel asset classes. A novel asset class, in short, does not have an established data trail against which to inform judgements of the likely performance of an individual product from within this class. By aiming to control for the influence of a green bond label on ECB haircut decisions, we seek to isolate and probe the existence of this constraint on novel asset class performance.

We also include a second-stage hypothesis, whose inclusion in our empirical work is conditional on a finding of positive support in our first-stage modelling for H_1. This second-stage hypothesis will probe the market impact from an anti-green bond impact in ECB haircut application. In line with Nyborg (2016), who suggests that central banks' application of larger haircuts to an asset serves to reduce demand in the secondary market for that asset, we adopt the following second-stage hypothesis:

> *H_2: Systematically larger haircuts from the central bank to green bonds will be associated with worsened secondary market performance from these assets.*

Our intuition here is that larger central bank haircuts would send an important signal into the secondary market, with investors responding to a perception of the asset's elevated risk profile with reduced demand.

5.3. Data and method

In the first stage of our analysis, we focus on the comparative haircut levied by the ECB on green and non-green bonds. For our outcome variable in this first stage of analysis, we turn to data on haircut values. The ECB List of Eligible Marketable Assets dataset provides information on the financial assets that, on a given day, the organisation was willing to purchase.[84] This List of Marketable Assets also notes the quantum of haircut that would be applied, should a holder wish to sell a listed asset to the ECB. The haircut value is expressed as the proportional difference between an asset's full market value, and the value that would be paid by the ECB on a given day. So if, for example, on a given day a bond had a face value of US$100,000 and the ECB was willing to purchase the bond for US$97,500, the ECB's haircut would be 2.5 percent. To test for the expected anti-green bond impact, we create a dummy variable based on the individual bond's classification as green or non-green, as captured by the Refinitiv Government and Corporate Bonds Advanced Search function. Overall, we construct a time series cross sectional dataset, with repeated daily observations across the cohort of green and non-green bonds from October 2017 through to May 2022.[85]

To identify additional factors that are likely to impact on these haircut values, we turn to ECB framework documents. Ultimately, we draw on these factors as either control variables or matching variables within our empirical work.[86] The Guideline of the European Central Bank (2015),[87] known colloquially as the ECB 'Haircut Guideline', sets out tariffs to be applied to a financial asset by employees when making an asset purchase. Under this Haircut Guideline, the haircut tariff is generated by taking account of market liquidity for the asset, and its coupon, size, lifespan, and

[84] See 'List of eligible marketable assets', ECB website, available at https://www.ecb.europa.eu/mopo/coll/assets/html/index.en.html. Accessed 18th March, 2025.
[85] The start date reflects the issuance of the first sovereign green bond, while the end date represents the most contemporaneous data available when empirical work was conducted.
[86] We explain the 'matching' element of our empirical strategy, which may be relatively unfamiliar to readers with little prior engagement with Financial Economics literature, below.
[87] See 'Guideline (EU) 2016/65 of the European Central Bank, 18th November, 2015', European Union EU-Lex website, available at https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02015O0035-20230629. Accessed 19th March, 2015.

current years to maturity. While the ECB would factor in information on coupon type into its haircut decision-making process, given that we focus only on plain vanilla bonds there is no variation in coupon type within our sample. We also, in line with scholarship on Financial Economics, also take account of issuer type, and credit rating.[88]

Liquidity refers to the extent to which demand for an asset broadly matches supply of an asset, with a market being said to be illiquid where sellers are unable to offload the asset at a price they find tolerable. The focus on liquidity reflects the ECB desire to avoid being stranded with a financial asset with a declining value, and to operationalise the liquidity of the market for a given bond we turn to the 'bid-ask spread'. The bid-ask spread captures the difference between a bond's ask price and bid price, with a higher value representing a more illiquid market. With a bond with repayment flows of US$1.1m, an ask price of US$1.06m, and bid price of US$1.04m would give an ask yield of 3.77 percent, a bid yield of 5.77 percent, and a bid-ask spread of 200 bps. Given the same repayment flow but reduced demand, we may see an ask price of US$1.06m but a reduction in the bid price to US$1.02m. These new market conditions would generate an unchanged ask yield of 3.77, a new bid yield of 7.84, and a new and higher bid-ask spread of 407 bps. A higher bid-ask spread captures greater illiquidity of the market for the bond, with more intense mismatch between the price being sought by sellers and price being offered by buyers. We calculate bonds' bid-ask spreads using data from the Refinitiv platform.

The coupon rate captures the perceived riskiness of the bond at its point of issuance, and data on a bond's coupon rate is available through the Refinitiv platform. For issue size, lifespan, and days to maturity, we expect to see a negative impact on haircut size. Other things being equal, we would expect a larger bond with a longer lifespan and more days to maturity to all decrease the perceived risk of the ECB purchasing the bond, as larger bonds typically have higher market liquidity and longer lifespans and days of trading to maturity give possibility for the ECB to hold the bond through a temporary downturn in sentiment, if required. Beyond these core structural characteristics

[88] See, for example, Bachelet et al (2019), and Simeth (2022).

of bonds, we also controlled for issuer type using dummy variables to differentiate between sovereign and corporate issuer.

To prepare for the second stage of our analysis, whose execution is contingent on a positive stage one finding, we collated an outcome variable to allow us to comparatively analyse bond yields. Again using the Refinitiv Government and Corporate Bonds Advanced Search application, we gathered daily observations of mid-yield.[89] The mid-yield measure represents a valuation of a bond's true rate of return, at a given point in time on the secondary market. An increase to the mid-yield value of a bond represents a worsening of market sentiment against that bond. Where on day one we have a mid-price of US$1.05m on a bond with a repayment flow of US$1.1m, we have a mid-yield of 4.76. If market sentiment turns against a bond on day two to pull the mid-price to US$1.04m but where we still have a future repayment flow of US$1.1m, the mid-yield has risen to 5.76.

Following contributions from Bachelet et al (2019), Gianfrate and Peri (2019), Hyun et al (2021), Simeth (2022), and Zerbib (2019) at the leading edge of Financial Economics, we adopt a 'matching' approach as part of our empirical strategy. Matching increases the confidence we have in the robustness of research findings, by allowing us to structure our analysis of real world financial market data in order to find 'hidden randomised experiments'. In this case, we identify pairs of bonds with closely matched characteristics, save for one characteristic we are particularly interested in exploring the effect of. Given our analytic interest in the potentially different treatment being given to green and non-green bonds, we have created 111 pairs of green and non-green bonds that match on amount issued, coupon, issue date, maturity, currency, issuer, currency, and credit rating. Table 5.1 provides an overview of the matching characteristics and parameters, which are conventional to comparative analysis of bond performance. The analytic approach supports efficient estimation of the average treatment effect on the treated (ATT), by removing observations from both the control and treatment groups that have no close matches on covariates (Iacus et al 2012, Imai et al 2023). By deploying this approach to probe the expectation that the ECB is systematically forcing larger haircuts

[89] See our Glossary at the start of the book for a definition of mid-yield, and a worked example.

on green bonds, the treatment group becomes the collection of 111 green bonds, and the control group becomes the collection of 111 conventional bonds that display matching characteristics. Alongside this matching, we also plan in our empirical strategy for regression analyses to further probe variation in haircuts, and to explore the impact from variation in haircuts on secondary market performance.

Table 5.1. Bond matching characteristics and parameters

Bond characteristic	Matching parameters
Amount issued	±400%
Coupon	±0.25%
Issue Date	±4 years
Maturity	±4 years
Currency	Same
Issuer	Same
Credit rating	Same

5.4. Results and discussion

Overall, our data collection process generates a total cohort of 111 pairs of matched bonds, constituted by 111 green bonds and 111 conventional bonds with aligned characteristics. We are interested in understanding the drivers of daily movements in haircut values and, if we see systematically higher haircuts being applied to green bonds, in the subsequent impact from haircut values on daily secondary market performance. Consequently, within our universe of observations we have 43,565 daily observations pertaining to the cohort of 111 green bonds, and 43,565 daily observations pertaining to the cohort of 111 of non-green bonds (see Tables 5.2 and 5.3). In total, we have 87,130 daily observations in our dataset.

Table 5.2. Descriptive statistics (green bonds)

Green Bonds	N	Mean	St. Dev.	Min	25th Perc.	75th Perc.	Max
Haircut	43,565	9.52	6.85	0.90	3.60	14.90	30.50
Yield	43,565	1.95	1.63	-0.86	0.32	3.32	6.95
Liquidity	43,565	0.35	0.27	-1.02	0.16	0.44	3.72
Coupon	43,565	0.90	1.01	0	0.01	1.2	5
Size (US$m)	43,565	2149.85	3728.48	11.23	561.35	1122.70	17615.58

Maturity	43,565	9.04	4.63	2.00	5.42	10.01	30.02
Days to Maturity	43,565	2,737.67	1,702.11	61	1,540	3,404	10,573

Table 5.3. Descriptive statistics (conventional bonds)

Conventional Bonds	N	Mean	St. Dev.	Min	25th Perc.	75th Perc.	Max
Haircut	43,565	8.83	7.24	0.00	3.00	13.20	31.50
Yield	43,565	1.97	1.63	-0.83	0.35	3.34	7.08
Liquidity	43,565	0.34	0.29	-1.99	0.15	0.43	4.96
Coupon	43,565	0.87	1.01	0.00	0.01	1.30	5.25
Size ($ MM)	43,565	3320.19	7290.78	11.23	561.35	1295.48	31435.60
Maturity	43,565	9.08	4.62	2.00	5.25	10.51	30.96
Days to Maturity	43,565	2,647.91	1,704.94	32	1,469	3,297	11,304

When reviewing the descriptive statistics, we see an interesting difference in the mean haircut applied to the green bond cohort as compared to the non-green cohort. Under the ECB collateral framework, the average green bond receives a haircut of 9.52 percent, whereas the average conventional bond receives a haircut of just 8.83 percent. Relying on a two-sample t-test to probe this difference across the green and conventional bond cohorts, we see a statistically significant difference in mean haircut values (p>.001), with green bonds having a higher mean haircut of 9.52 percent relative to the conventional bond mean haircut value of 8.83 percent. Figure 5.1 presents these results, providing initial evidence in support of the H_1 expectation of systematic anti-green bond impact from ECB collateral operations. Table 5.4 reports the results from an OLS regression, which confirms the finding from the two-sample t-test of the existence of a systematic anti-green bond bias from the ECB. When controlling for a range of bond characteristics and market conditions, we see that the ECB is applying an elevated haircut of between 66 and 69 basis points.

Figure 5.1. Difference in mean haircuts, 95 percent confidence intervals

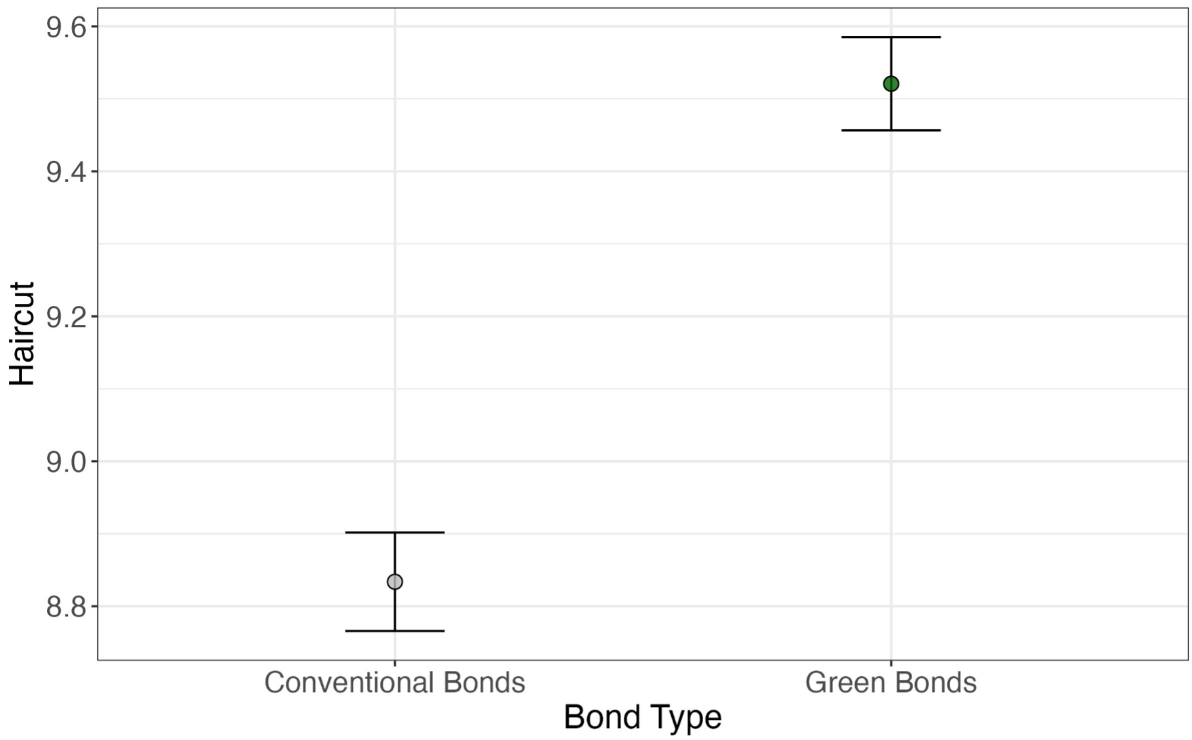


Table 5.4. Impact of green bond label on ECB haircut value

	Haircut Value		
	(1)	**(2)**	**(3)**
Green Bond	0.664**	0.661**	0.687**
	(0.285)	(0.283)	(0.309)
Issuer FE	✓	✓	✓
Year FE	✓	✓	✓
Liquidity	✓	✓	✗
Full controls	✓	✗	✗
Observations	87,130	87,130	87,130
Bonds	222	222	222
R2	0.21	0.06	0.02

Note: Cluster robust standard errors in parentheses. * p<0.1, ** p<0.05, ***p<0.01

Through this first stage of our analytic journey, then, we see support for the H_1 expectation of an anti-green bond bias from the ECB. To further probe this finding, we test for the impact from a re-specification of matching parameters. Given that a bond's maturity is incorporated into ECB haircut calculation processes, we re-run our analysis with a tightened parameter such that bonds were matched only where maturity values were within two years. We also tighten the matching specification for the amount issued to ±200 percent, for the coupon to ±0.2 percent, and for the issue date to ±2 years. Given that the initial results remain significantly and substantively similar to those attained under the initial specification, we take these findings to offer a confirmation of H_1. This confirmation of H_1 leads us to execute the second stage of our empirical strategy, and test for evidence of the market impact from the higher haircut level being applied to green bonds. The expectation from H_2 is that higher haircuts on green bonds will translate into constrained performance in secondary markets, as higher haircut values depresses market sentiment against green bonds. To assess market performance against these expectations from H_2, we turn to a comparative analysis of mid-yield values across our cohorts of green and conventional bonds.

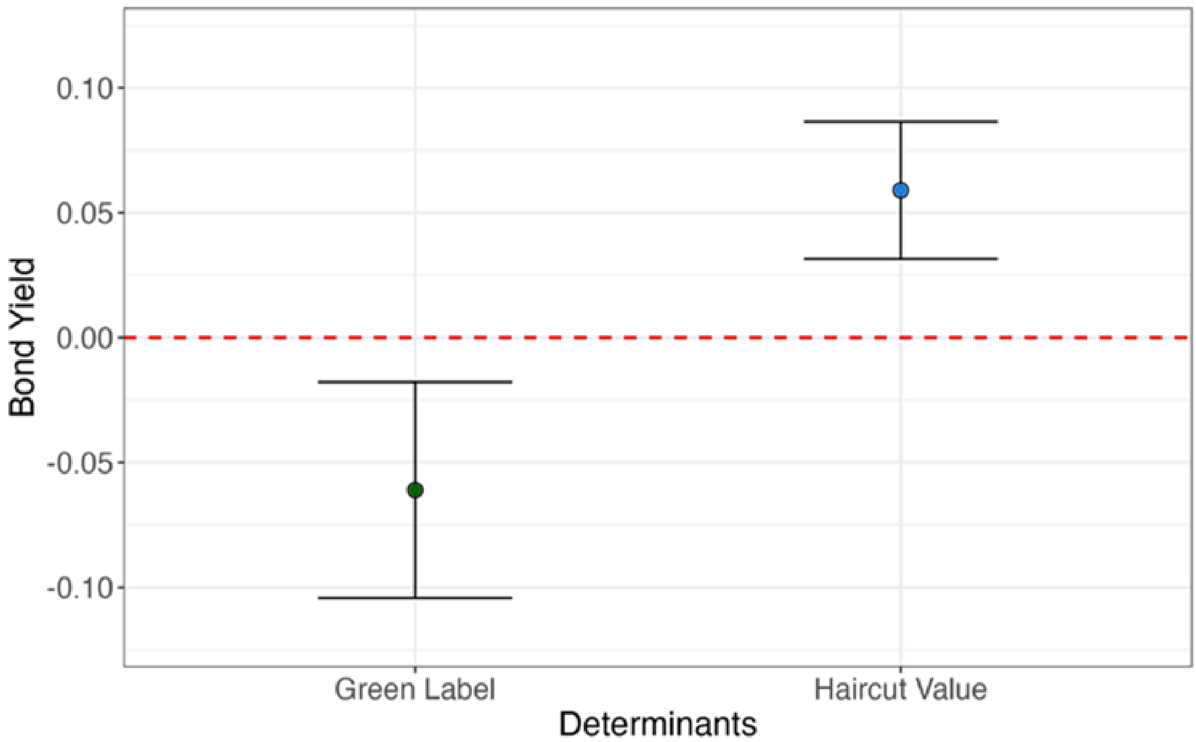
In assessing the market impact from higher haircuts levied against green bonds, we focus our analytic attention on the yield spread between green and conventional bonds. When holding all other variables constant, we see that our cohort of green bonds enjoy a greenium across our secondary market performance observations. Compared to the matched cohort of 111 conventional bonds, green bonds trade at lower yields, reflecting stronger demand for this asset class (Table 5.5). However, we also see that ECB haircut values have an impact on market performance, with higher haircuts pushing up yields as demand weakens in the face of enhanced perception of risk. Whereas the greenium effect sees a 6.1 bps reduction in green bond yields relative to their conventional counterparts, we see that an increase in haircut value of one percentage point translates into a 5.9 bps increase in the bond's yield.

Table 5.5. Explaining variation in bond yield

	Bond Yield		
	(1)	**(2)**	**(3)**
Haircut	0.059***	0.066***	0.067***
	(0.014)	(0.013)	(0.012)
Green Bond	-0.061***	-0.057***	-0.057***
	(0.022)	(0.022)	(0.022)
Liquidity	-0.019	0.048	
	(0.089)	(0.094)	
Issue Size	-0.084*		
	(0.048)		
Maturity	0.013**		
	(0.006)		
Coupon	0.096***		
	(0.025)		
Issuer FE	✓	✓	✓
Year FE	✓	✓	✓
Observations	87,130	87,130	87,130
Bonds	222	222	222
R2	0.07	0.06	0.06

Note: Cluster Robust standard errors in parentheses. * $p<0.1$, ** $p<0.05$, *** $p<0.01$

Figure 5.2. Marginal effects on bond yields, 95 percent confidence intervals



To gain a sense of the real-world impact from the ECB's anti-green bond impact, we can close by bringing together core findings from stage one and stage two of our analysis. The stage one analysis demonstrated that green bonds attracted haircuts that were around 66 bps higher than conventional counterparts. The stage two analysis demonstrated that a 100 bps increase in haircut translated into a 5.9 bps increase in yield on the secondary market. Consequently, the overall insight from the empirical work undertaken through this chapter suggests that, for the average green bond, the ECB anti-green bond impact will translate into a yield increase of around 3.9 bps. When we consider the quantum of financial flows involved in Collateral Framework operations and in the wider green bond market, the consequences from the ECB impact can be seen to be notable. To give a sense of the potential market impact from a 3.9 bps increase to borrowing costs, we can compare its impact on the coupon rate of a theoretical green bond with mean characteristics from across our cohort. At a coupon rate of 90 bps, an issuer of a bond with a face value at US$2.1bn and a 9 year lifespan would make interest payments of some US$170.1m. If we raise the coupon rate to 94 bps, interest payments increase to some US$245.7m. In order to ensure that issuers of green bonds are not unfairly penalised through its collateral framework operations, it is important that future analysis probes ECB impact on green bonds and, where evidence of negative impact is found, action is taken to amend performance.

5.5. Conclusion

Since the emergence of modern financial systems, central banks have come to play a vital stabilising function. Financial institutions' tendency to create mis-matches between their cash holdings and their payment obligations gives rise, periodically, to a need for central banks to step-in to supply short-term liquidity. Where an institution lacks the ready cash to meet payments that are due, central banks can arrange to purchase assets from the institution to provide a needed cash boost. Through this chapter, we have advanced scholarship on such central bank collateral framework operations by exploring their intersection with climate finance.

While there has been a significant pivot toward climate from central banks over the last decade or so, few analyses have systematically probed this relationship between central banks' everyday liquidity management and the performance of climate finance-linked assets. In exploring this relationship, we specifically probed the treatment of green bonds under the European Central Bank's Collateral Framework. Using a bond matching strategy, we assessed the comparative treatment of green bonds and their conventional equivalents under the ECB market intervention framework. We found, overall, that the ECB demonstrated a greater reluctance to hold green bonds relative to their conventional equivalents, levying a higher haircut on such bonds. Where a financial institution wished to sell a green bond to the ECB, other factors being equal they would receive a lower proportion of the market value of the asset than if they were selling a conventional bond. We also found evidence that these higher haircuts on green bonds were impacting secondary market dynamics, with the negative ECB signal being translated into depressed demand and consequent adjustment of pricing and yield.

Our headline empirical finding of a systematically differential treatment by the ECB of green bonds was in line with expectations derived from existing literature. Chappell et al (2008) suggest that financial market actors are driven towards more conservative valuations of novel asset types, with the limited availability of time series data on market performance from new forms of financial asset translating into elevated risk perception. The timeframe covered by our analysis covered 2017-22, a period during which green bonds may still have suffered from this cost on novel financial instruments. Given the central role of green bonds in financing climate transition, it is important that any residual anti-green impact in ECB collateral operations continues to be both tested for and, if present, ameliorated. The ECB and central banks more widely have come to position themselves as institutions working to support the effectiveness of climate finance. To meet these ambitions, everyday operations must align more tightly with strategic commitments and principles.

6. Institutions for Informing Private Climate Finance: The Bond Underwriter 'Home-Turf Effect'

6.0. Introduction

At a foundational level, capital markets fulfil an important 'match-making' function. On one side of the equation we have individuals and organisations with an excess of money, money that they would ideally like to put to productive purpose so that it begets more money. On the other side of the equation we have individuals and organisations who think they have a potentially productive activity, but who lack the money required to kick-off the cycle of additional money-making. Capital markets today function at a speed and scale far beyond the levels reached just two or three decades ago, and light-years beyond their hazy historic foundations.[90]

Since 19th century industrialisation, bonds have constituted an important component of the assets traded on capital markets.[91] A bond is a form of promissory note; investors pass money to the issuer to use for productive investment, and in return receive a commitment from the issuer that they will make repayments according to a set schedule. And, as we have seen across Chapters Four and Five, over the last decade or so bonds have become an important mechanism for redirecting market-based finance towards supporting the cost of climate change transition. Green bonds have

[90] Keynes (1930: 11-12) pointed us back some 12,000 years when reflecting on foundations of contemporary financial systems. For a more contemporary foundation, see de Roover's (1946) history of the Medici bank, which is often taken to represent an important staging post in the deepening of transnationalised financial flows.

[91] Governments, it should be noted, have issued bonds for many centuries; in 1694 the Bank of England was, for example, founded to issue bonds on the government's behalf, in large part to finance the Second Hundred Years War. See 'Our History', Bank of England website, available at https://www.bankofengland.co.uk/about/history#:~:text=It%20was%20primarily%20founded%20to,and%20Benefit%20of%20our%20People. Accessed 20th March, 2024.

moved from being an entirely new asset class in the late-2000s, to being a fairly mainstream asset class by the early-2020s. With green bonds, the issuer makes the same type of repayment guarantee that would be made by an issuer of a standard bond, but in addition commits to using the monies raised to support environmentally-sustainable activity.

Through this chapter, we turn our attention to links between market-based institutional structures and green bond performance.[92] Success in effectively aligning private finance with climate transition efforts is, to a large extent, conditional on the capacity of green bonds to translate resources from capital markets into climate action. Existing scholarship on green bond performance sheds some light on the role of institutions in shaping the effectiveness of this branch of climate financing. We advance this scholarship by probing, in particular, the impact of bond underwriters on green bond performance. Our headline contribution comes through our identification of a 'home-turf effect' from bond underwriters. Where an underwriter is supporting a green bond issue within their home market, we see a boost to the green bond's performance. Across the cohort of European green bonds we analyse, the average issuer saves around US$27.1m in interest payments from this underwriter home-turf effect.

In presenting our analysis of economic institutions and market-based climate financing flows, we develop the chapter through the following structure. In the first section below, we briefly review the contemporary expansion of green bond markets, building on the introduction to green bonds provided in Chapters Four and Five. We then progress, in the second section of the chapter, to derive insights into expected institutional drivers of variation in green bond performance from existing scholarship. We explain decisions taken over data sources and analytic approach in the third section of the chapter, before in the fourth discussing results. To conclude, we recap on headline insights, and reflect on implications from our empirical work.

[92] In this chapter, when discussing green bond performance, we use the terms 'efficiency' and 'borrowing cost' as place-holders for our credit spread gap and primary-to-secondary spread outcome variable specifications. We take a lower spread to indicate higher efficiency of issuance that ultimately has enabled the green bond issuer to access a lower cost of borrowing.

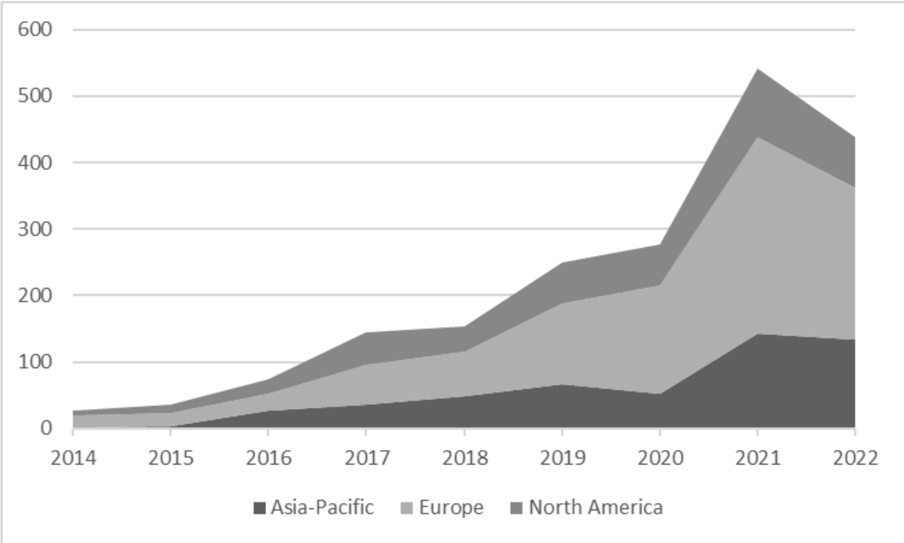
6.1. Green Bonds and market actors - key dynamics

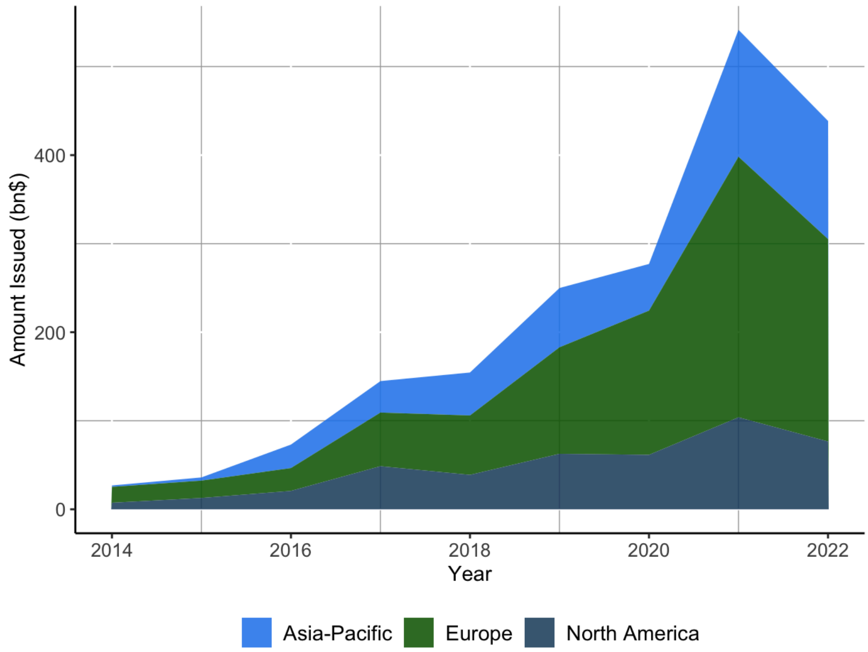
In economic terminology, an 'externality' refers to a market transaction in which the cost paid by the transacting individual is below the societal cost generated by the underlying activity. Air travel provides a tidy example. While environmental externalities are difficult to measure precisely (Ryley 2014), the £43 cost of a flight from London to Berlin is unlikely to effectively price-in the future harm caused by the associated greenhouse gas emissions.[93] Green bonds can be seen as an attempt to turn the externality challenge on its head. By committing to use funds raised for environmentally-beneficial activities, issuers are in effect locking-in enhancements to the societal benefit from the underlying activity. And, in theory, green bonds may present a win-win scenario. The hope is that environmentally-motivated investors are sufficiently keen to place their money in a sustainable asset that green bonds attract enhanced demand, which leads to a reduced cost of borrowing for issuers.[94]

As noted in previous chapters, Green Bonds stepped increasingly into the mainstream of financial markets from the mid-2010s, when private sector issuance began to outstrip issuance from states and international organisations and overall issuance rapidly began to rise. In addition to this increased importance of market actors as issuers of green bonds, since 2014 we have seen the emergence of three major trading centres (see Figure 5.1). According to the Climate Bond Initiative's green bond market analysis,[95] the Asia-Pacific region by the early-2020s accounted for in the region of US$100bn of green bond launches annually, mainly driven by the strength of the Chinese market. The North American market grew to account for green bond issues worth some US$130bn. The European region continues to constitute the largest source of green bond issuance under the CBI classification, with total European issuance through the early 2020s exceeding the combined value from the Asia-Pacific and North American regions. Across Europe, the German, Dutch, and French green bond markets came to constitute the dominant forces, with combined issuance of around US$100bn in 2022 (CBI 2023: 9).

[93] Price attained through skyscanner.net search. Accessed 21st March, 2024.
[94] As we see when we turn to scholarship on Green Bond performance, messy real-world outcomes do not necessarily follow this tidy theory.
[95] The CBI's relatively strict typology for classifying green bonds generates estimations of green bond market depth and issuance that are below the estimations from other sources.

Figure 6.1. Annual Green Bond issue by region (US$bn)





Source: Authors' analysis of Climate Bond Initiative Database.[96]

[96] See 'Green Bond database', Climate Bond Initiative website, available at https://www.climatebonds.net/market/data/. Accessed 9th April, 2024.

Through the process of green bond market consolidation, expectations and protocols surrounding issuance have solidified. Across the major Chinese, European, and US markets, around half of green bond issues are accompanied by third-party accreditation. In the Chinese case, 936 from a total of 1,873 Green Bond issuances on record included external verification. The equivalent figures for the US and Europe stood at 385 from a total of 857, and 1,255 from 2,150, respectively.[97] Green Bond documentation will typically include details on use of proceeds, which in the case of the EU and US issues often include commitments to following International Capital Market Association Green Bond Principles. With Chinese issuance, the China Securities Regulatory Commission's Guiding Options for Supporting Green Bonds provides a prominent framework document (Zhang 2020).[98]

Green bonds now form a relatively mainstream asset class across financial markets. Across the EU, for example, green bonds by 2022 had come to account for around 9 percent of all bond issues.[99] Indeed, it is widely accepted that green bonds constitute an essential pillar for effectively achieving climate change transition plans. The IPCC (2022: 1550) has noted that Green Bonds allow for the 'smooth integration of Paris-aligned investment opportunities into existing asset allocation models', while Bigger and Millington (2022: 607) suggest that green bonds are the single most significant mechanism for unlocking additional climate finance.

Debates are ongoing as to the impact on issuers' environmental performance from green bond launches. High information costs and a lack of enforcement mechanisms present significant barriers for investors, who are unlikely to be willing or able to effectively monitor an issuer's compliance with their use of funds declarations, and in any case would lack means of redress to punish any non-compliance (Agostini 2023). As explored in Chapter Seven, these structural weaknesses are being counterbalanced by the emergence of overarching regulatory frameworks. While standards such as the ICMA Green Bond Principles remain voluntary, the EU in particular is advancing an approach with enforcement powers. Through the agreement on the European Green Bond standard, EU institutions

[97] Authors' analysis of the Refinitiv database, conducted on 9th April, 2024.
[98] For exploration of emerging dynamics on green bond definition and regulation, see Chapter Seven, Section 7.2, below.
[99] See European Environment Agency website, 'Green Bonds', available at https://www.eea.europa.eu/en/analysis/indicators/green-bonds-8th-eap#:~:text=Green%20bond%20issuance%20increased%20significantly,8.9%25%20of%20total%20bonds%20issued. Accessed 9th February, 2024.

have moved toward ensuring that assets labelled as green bonds are in alignment with the EU taxonomy for sustainable activity.[100] Evidence of improved environmental performance off the back of Green Bond issuance comes from Shi et al (2023), who report that, following a launch, corporations display an enhanced propensity to issue environmental patents. Zheng et al (2023) also point towards a positive green bond placement effect, with significant improvements in companies' sustainability metrics coming in the wake of an issue.

The most active strand of scholarship on green bonds explores whether the asset class is able to generate additional financing. Contributions have sought to unpick whether issuers are able to borrow more cheaply through a green bond issue relative to the issue of a standard bond, and to identify particular factors that are associated with improved green bond performance. Given the quantitative importance of green bonds to overall climate financing flows, it is vital that our understanding of the institutional factors and mechanisms associated with effective green bond performance are more fully understood. Through the empirical focus of this chapter on the underwriter effect, we turn our attention to the influence of market-based institutions on green bond performance.

6.2. Explaining performance variation - existing literature

Institutions matter for bond performance. Indeed, financial markets can be seen to function as complex organisational fields, in which interactions between a range of organisations shape system outcomes. In this regard, characteristics and past performance of the bond issuer, judgements from credit rating agencies, and regulatory requirements that (dis-)incentivise investments in particular assets, all exert a push and pull on bond performance (Duffee 2013). While the noise of day-to-day 'market sentiment' may move the prices received by investors and yields paid by issuers, underlying regularities emerge off the back of organisational structures that exert sufficient coherence and influence to shape market

[100] See European Parliament website, 'Green Bonds: More Transparency, No Greenwashing', available at https://www.europarl.europa.eu/topics/en/article/20230928STO06003/green-bonds-more-transparency-no-greenwashing#:~:text=The%20European%20green%20bond%20standard%20would%20allow%20better%20regulation%20of,a%20practice%20known%20as%20greenwashing. Accessed 10th April, 2024.

outcomes. As detailed below, existing scholarship provides valuable insights into interactions between market-based institutions and green bond performance. However, we see a notable gap in this existing scholarship. Whereas in wider Financial Economics literature we see that bond underwriter characteristics can play an important role in supporting the efficiency of a bond launch, within studies of green bonds we see little attention being placed on the influence from this intervening institutional factor.

There are reasons to think that underwriters may play a particularly important role in green bond performance, which makes this gap in our understanding particularly worthy of attention. From Partridge and Medda (2020: 52), we know that there is a notable discrepancy between green bond performance in primary and secondary markets. Given the importance of underwriters in shaping the initial issue into the primary market, it is plausible that underwriter characteristics may constitute an institutional feature that is salient to effective green bond performance. To explain this potential intervening role from underwriters on green bond performance, it is useful to briefly review dynamics surrounding primary and secondary bond markets.

For the most commonly-launched types of green bond, primary markets determine the rate of interest that will be paid by the issuer. Bond underwriters draw on their knowledge of prevailing market expectations to advise the issuer on the rate of interest to offer to primary market investors. The underwriter works as an advertising agent for the bond, drawing the attention of potential investors toward an upcoming issue. The underwriter also provides a backstop function, committing to purchase any residual that remains after an under-subscribed launch. After this initial bond launch, action passes into the secondary market. Here, investors re-sell their claims on future repayments from the bond issuer. The price they are willing to pay is determined by many factors, including their confidence in the ability of the issuer to make the agreed payments, and the returns on investment that might be achieved through investment in alternative financial assets.

The discrepancy identified by Partridge and Medda is a persistent gap between relatively high green bond coupon yields set within primary markets, as against the lower rates demanded within secondary markets. It seems that the higher demand exhibited in secondary markets is not effectively filtering through into primary markets, resulting in unnecessarily high interest rates being signed-up to by green bond issuers. With green bonds, there may be a pool of investors who are willing to hold the asset for a lower return that is being locked in at the primary market issue. Insights into underwriter characteristics that may influence the effectiveness of green bond launch can be gained through a combination of Financial Economics scholarship that has explored underwriter effects on the performance of standard bond markets, and from wider Political Economy scholarship that sheds light on the role of 'financial ecologies'.

Within the Financial Economics scholarship, we have seen that underwriter location can exert a significant effect on bond performance. Insights come, in particular, from Kollo (2005) and Butler (2008). Kollo provides a study of European bond market performance that clarifies the relationship between underwriter location and bond performance. The study specifically explores the underwriter location impact on bond gross spreads. Gross spread is the difference between the price received by the issuing entity and the price that is offered to investors, with a lower gross spread meaning that less resources are extracted by the underwriter. Kollo finds that local underwriting exerts a significant and, from the issuer perspective, beneficial independent influence by lowering the gross spread. Kollo suggests that, as gross yield is not set at the point when issuer contracts-in an underwriter, issuers are not selecting local underwriters on account of this factor. Rather, underwriters are selected on the basis of factors including perceived market expertise, experience, and capacity to generate investor interest in an issue. An underwriter's understanding of the local market, and their established networks of local investors, may by this measure be supporting an effective bond launch. From Butler, we see that where a US municipal bond is underwritten by an institution with an active presence in the issuing state's territory, a reduced rate of interest is achieved. Our intuition here is that these efficiency gains from having a local underwriter may transpose to green bond issuance, with an underwriter 'home-turf effect' lowering the cost of this form of climate financing.

Additional evidence in support of this expected home-turf effect can be drawn from Political Economy scholarship on financial ecologies. The term 'financial ecology' directs our attention to the importance of place in financial system performance. The concept has been defined by Grafe and Meig (2019: 502) as referring to:

> … a social structure in which actors, locations and their relations form geographically distinct constellations of knowledges, practices and subjectivities that enable the provision of financial services. These smaller, anchored financial ecologies form links with other financial ecologies, constituting the wider financial system.

The relationships and networks that constitute a financial ecology can, for Grafe and Meig, support efficient 'match-making' between actors in search of investment funds on the one hand, and those searching for investment opportunities on the other. From Grafe et al (2023) and Tripathy (2022), we gain insight into the role of financial ecologies in supporting the emergence of new forms of climate financing specifically. Off the back of this Financial Economics and Political Economy work, then, we derive the first hypothesis to be tested through our empirical work on green bonds:

H_1: Where a green bond is supported by a local underwriter, the cost of borrowing will be reduced.

The intuition here is that a local underwriter constitutes an economic institutional structure that will benefit green bond performance. The local underwriter is likely to bring more extensive links with a pool of investors interested in investing in their national territory, and to hold more fine-grained understanding of local market sentiment including in the realm of green and sustainable investment. As such, we expect a local underwriter to be well-placed to achieve a lower cost of borrowing for the green bond issuer.

To develop our second hypothesised relationship between economic institutions and green bond climate financing performance, we remain focused on underwriters. Previous studies suggest that underwriter market share may be positively related to efficiency of launch. From Carbo-Valverde et al (2021), we see that larger underwriters are able to secure a primary market rate of interest that is more closely aligned to the secondary market yield. The causal pathway that likely explains this outcome comes from the plausibly higher 'convening power' enjoyed by larger underwriters; other things being equal, large underwriters are able to deliver a higher level of investor engagement at the point of issuance, with this stronger demand leading to lowered primary borrowing costs. Parallel findings are presented by Su et al (2023), who explore the impact of underwriter size on green bond issuance in China. For Su et al, larger underwriters are shown, on average, to drive down the cost of borrowing for issuers. From these studies of underwriter size and bond issuance, we derive our second hypothesis to be tested in our empirical work:

> *H2: Where a green bond is supported by a larger underwriter, the cost of borrowing will be reduced.*

The expectation here is that the patterns observed by Carbo-Valverde et al and Su et al will be replicated in our analysis.

Beyond these institutional impacts on Green Bond performance, and as touched on in Chapter Five, we know from the literature that characteristics such as bond maturity, the issue size, the type of issuer, and credit rating can have a significant effect on bond spreads (Zerbib 2019). Additionally, we know that conditions including liquidity in the secondary market can shape performance (Adrian and Shin 2010). In our models, we operationalise variables that allow us to control for such factors.

6.3. Data and method

Through the empirical work below, we are seeking to probe the influence of underwriter characteristics on the efficiency of green bond issuance. We have a particular interest in the potential underwriter home-turf effect on Green Bond launches, with which we expect to see a positive relationship between the involvement of a local underwriter and the overall efficiency of the green bond issuance. To pursue this thematic interest, we follow Kollo (2005) in focusing on European financial markets in our study. The European bond market provides an excellent opportunity for studying the influence of home-turf underwriting, as variation in the country of origin within the 'home-turf' underwriter group allows for fixed country effects to be controlled for. If the focus was on the US market, for instance, it may be unclear whether a home-turf effect was being driven by the underwriter location within the US, or by characteristics that were shared across US underwriters or indeed the wider US bond market. By focusing on the European market, the home-turf effect can be observed in underwriters based in Austria, France, Germany, or indeed any other European country; this feature of the European market means that we can be more confident that it is the home-turf effect, and not some other country-level properties, that is being captured.

In terms of timeframe, we target our study on European green bond issuance across the period 2014-23. As detailed above, prior to 2014 the green bond market remained immature, with low overall issue volumes and limited issuance by market actors. By considering observed dynamics since moves towards market maturity, we avoid the very early years of green bond trading during which stable institutional effects are unlikely to have become established. 2023 constitutes, at the time of writing, the most recent year for which Green Bond performance data is available.

With our headline interest in green bond effectiveness, we turn to bond underpricing as our outcome variable. A bond is said to be underpriced when the price set by the underwriter for the primary market issue is revealed to be below the price attained in the secondary market. Expressed in terms of borrower costs to the bond issuer, a condition of underpricing can be said to exist where the coupon yield set in the primary market is higher than the yield set in the secondary market. A brief worked example can help to clarify the nature of the outcome variable. Should the underwriter support a primary market launch with a coupon yield of 2 percent, but when the bond begins to be re-sold between investors in the secondary market the yield drops to 1.75 percent, we would say that the bond was underpriced by 25 basis points (bps). Real-world bond pricing results from the interplay between a complex array of factors. Consequently, to probe green bond underpricing, we developed a more refined measure.

In developing our measure of green bond underpricing, we proceed towards a 'credit spread gap' operationalisation through two stages. First, we developed a measure of differences in perception of the credit risk of a bond between the underwriter in the initial offering, and the overall market's perception in subsequent trading. To do this, we estimate the credit spread of a bond at issuance, and the credit spread in the secondary market. The credit spread is the difference between the mid-yield of a given bond and the yield of risk-free bonds which, for Euro-denominated bonds, are typically German Treasuries of similar maturities. If, for example, a given green bond with a 10-year maturity is issued with a coupon rate of 2% and the yield for the 10-year on-the-run German treasury is 1%, then the credit spread on the given green bond would be 100bps. In general terms, lower spreads mean lower associated credit risk. As such, we estimate the credit spread as the difference between the original yield to maturity and the yield of an on-the-run treasury security with similar maturity, and the credit spread. As the second step in creating our credit spread gap outcome variable, in the spirit of Dick-Nielsen et al (2021: 3) we measure underpricing as the difference between the credit spread of a bond at issuance and the average of the credit spread of the same bond across its first ten days of trading. As such, our outcome variable measures the difference in the perception of the value and riskiness of a green bond that exists between its underwriter (in the initial issuance) and the overall market (in the subsequent trading days). The benefit of taking this mean approach to the secondary price is that daily volatility from unrelated factors is smoothed out.[101]

To test for the presence of an underwriter home-turf effect on underpricing as specified by H_1, we develop an explanatory variable to capture the local-ness of the underwriters involved in the green bond Issue. Drawing on publicly listed underwriter location data, we construct this 'home turf' variable to capture underwriters that are headquartered in the country of issuance. Because underwriting services can be delivered either by a single financial institution or jointly delivered by two or more underwriters acting in syndication, we measure our 'home turf' variable as the proportion of domestic underwriters in the underwriting group. Additionally, existing literature suggests that issuing through syndication affects the pricing of bonds because syndicates have access to larger distribution networks that can lower the issuance cost for borrowing firms (Carbo-Valverde et al 2021, Dick-Nielsen et al 2021, Narayanan et al 2004, Shivdasani & Song 2011, Song 2004, Yasuda 2005). As such, to account for the potential that this 'home turf' effect can be moderated by underwriting in

[101] As a supplementary outcome variable specification that we later deploy when extending our analysis, we directly deploy Dick-Nielsen et al's (2021: 3) approach of measuring underpricing as the difference between primary yield and the mean transaction price across the first ten days. This alternative specification removes the focus on benchmark performance. As noted below, our results are robust to this alternative specification.

syndication, we interact our outcome variable with a dummy variable that takes the value of 1 if a bond was issued through a syndicate of underwriters and 0 otherwise.

The second hypothesised relationship between underwriting and Green Bond performance comes through a 'big-hitter' effect, whereby we expect larger underwriters to be associated with more effective green bond issuance. To explore whether dominant market players do indeed attain lowered costs of borrowing for green bond issuers, we developed a measure of underwriter dominance across the European green bond market. Owing to the relatively concentrated nature of the European green bond underwriting market, we borrowed from the approach taken to identifying large underwriters in the similarly concentrated US market, classifying the three largest European green bond underwriters in a given year as 'big hitters'.[102] We created the measure based on green bond market share on the assumption that the demonstrated history of working at scale on green bonds may impact positively on issue performance. To test for the interaction between the local-ness and market dominance of an underwriter, we created a dummy where a value of 1 was given to cases that exhibited both the home-turf and big-hitter effect, and a value of 0 elsewhere.

The main source of data is the Refinitiv database. We used the Refinitiv Government and Corporate Bonds Advanced Search application from which we extracted an initial sample of all green bonds issued across Europe between 2014 and 2023. We first filtered the dataset to exclude any bonds without fixed coupons, to avoid the uncertainty that non-constant coupons have on prices at issuance (Gianfrate & Peri 2019).

To control for the impact that a lack of credit rating might have on a bond's price, we excluded unrated bonds and constructed a credit rating score for all other observations following the common practice in the literature where categorical values are converted into a continuous scale. The safest assets are given a score of 1 (Moody's Aaa rating, Fitch's and Standard and Poor' AAA), with the score increasing in increments of one unit up to a high of 20 where the bond is deemed to present

[102] Overall, we see a relatively high level of stability in European Green Bond underwriter market dominance across our timeframe. BNP features amongst the top three in all years, Credit Agricole in nine of the 10 years, and then Barclays and Commerzbank appear twice, and Bayerische, Citigroup, Goldman Sachs, Mizuho, Morgan Stanley, Natwest, and Nomura all make one appearance.

an excessive speculation risk (Moody's Ca rating, Fitch's and Standard and Poor' CCC) (see Afonso et al 2007, Cantor & Packer 1996, Yildirim et al 2021). For the other control variables, we used Refinitiv data on issue and settlement dates to capture green bond lifespan from issuance through to maturity. The same source provided data on issue size (expressed in US dollar values), and was used to identify whether the issuer was an agency, corporate, or sovereign entity. Finally, to control for the effects of liquidity in the secondary market, we collected daily observations of the bid and ask prices, as the bid-ask spread is the most common proxy for a bond's liquidity (Adrian & Shin 2010).

In total, around 750 plain vanilla green bonds were issued across Europe between 2014 and 2023. However, data on bond underwriters is present in only 352 cases. A further 91 cases were lost owing to the lack of data on secondary market performance, leaving us with our core cohort of 261 cases. Table 5.1 summarises the variables we use, and provides descriptive overviews.

Table 6.1. Description of variables

Variable	Description
Credit Spread Gap	The gap between the spread in the primary and secondary markets
Home Turf	Proportion of underwriting provided by a local institution
Syndicated	Dummy if a bond is issued in syndication.
Size	Amount issued in US dollars
Original Yield	Coupon rate of the bonds
Maturity	Maturity of a bond in days
Big-Hitter	Large underwriter
Rating	Credit rating (Aaa = 1; Ca = 20)
Liquidity	Bid-Ask spread

Our empirical strategy has been designed to suit key characteristics of our dataset, to produce efficient estimates and give confidence in the results obtained. Given the cross-sectional nature of our data set and the structure of our dependent variable, a linear regression is the most efficient method to estimate the parameters of our argument. Diagnostic tests reveal heteroskedasticity in the data and a non-normal distribution of residuals. As such, we employ a generalised least squares (GLM) estimator in our empirical models, which relaxes the assumption of normally distributed errors and allows us to deal with heteroskedasticity in the data by letting us specify the covariance structure of the errors (Dunteman & Ho, 2006).

In addition to this, to control for any unobserved heterogeneity introduced by economic and domestic market effects, we introduce in our model yearly and country fixed effects. Finally, to avoid the false perception that, in a cross-sectional context where we can account for the clusters in the data using only fixed effects there is no need to cluster standard errors (Abadie et al 2022, Arellano 1987), we cluster the standard errors at the issuer level in all of our specifications to account for within-issuer variation.

Table 6.2. Descriptive statistics

Variable	N	Mean	St. Dev.	Min	Max	Freq.
Spread Gap	261	-0.018	0.235	-1.567	0.964	-
Home-turf	261	0.549	0.310	0.000	1.000	-
Big-hitter	261	0.510	0.501	0.000	1.000	128(0), 132(1)
Syndicate	261	0.835	0.501	0	1	43(0), 218(1)
Credit Rating	261	4.732	2.784	1	15	-
Maturity	261	3,489.2	2,140.2	731	18,263	-
Liquidity	261	0.448	0.471	0.027	6.750	-
Amount ($bn)	261	1.293	3.034	.05	35.966	-

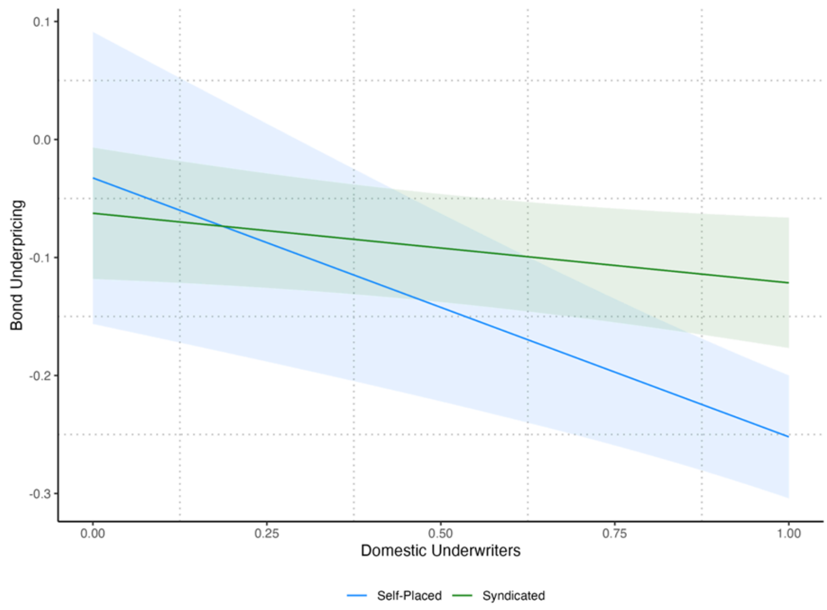
6.4. Results and discussion

Across the 261 green bond observations that constitute our core cohort, we see that the spread between the primary coupon and secondary yield had a mean value of -.018 (see Table 6.2). On average, the valuation attained in the secondary market was 2 bps closer to the benchmark rate than that attained at issue in the primary market. This provides an initial suggestion that, overall, some latent secondary demand existed that was not effectively fed through into the primary pricing. There was notable performance variation across the cohort. We observed the largest spread gap of 96 bps in Aareal Bank's 2022 issue. At the other end of the spectrum, placements achieving a negative spread gap were led by Hurtigruten's 2022 issue for which the primary coupon spread was some 156 bps lower than the mean secondary spread over the first two weeks of trading. In terms of our independent variable of primary interest, we see that on average half of the members of underwriting teams were domestic institutions.

Turning to the control variables, the mean credit rating score of 4.7 sits at the boundary between 'high' and 'high-medium' asset grading. There was significant variation across the cohort, with a relatively even spread between grades one-to-nine, and a small tail with two green bonds graded at 10, and one at 15. Regarding issuer type, we have 30 issues from governments, 40 from public agencies, and 191 from corporate entities. The mean issue size was around US$1.3bn, with the maximum of US$36bn coming from the French government's 2017 issue, and the minimum of US$54.7m coming from Smakraft's 2021 issue. The mean green bond maturity was around 9.5 years (or 3,494 days), with a maximum lifespan of 50 years from Societe du Grand Paris' 2019 issue, and a minimum of two years from Raifeissen Bank's 2022 issue. Our green bond 'big hitter' variable, which captures the top three underwriters as measured by European green bond market share, shows that just over half of the issuances in the sample involved market-leading underwriters. Turning finally to our control for the big-hitter-and-home-turf interaction effect, we see that 80 green bonds were launched with an underwriter exhibiting both of these characteristics.

Figure 6.3 below predicts the values of the underpricing of a bond depending on the proportion of local members in an underwriting team, conditional on whether the bond is issued through syndication or not. This prediction is based on Model 1, as reported in Table 6.3 below. We see that home-turf underwriting is significantly and negatively associated with issue spread and, as such, minimises the underpricing of a bond in the primary market, relative to the market trading in the secondary market. On average, a green bond led by a local underwriter attracted a spread reduction of 22 bps relative to an issue by an external underwriter. In support of H_1, it seems that home underwriters demonstrate greater placement efficiency, ensuring a higher degree of similarity between primary and secondary market performance. Given a mean issue value of US$1.3bn and mean maturity of 9.5 years, this 22 bps efficiency gain translates into a total saving of around US$27.1m in interest payments across the green bond lifespan. This effect is moderated by its syndicated status, indicating that this effect is less pronounced for bonds underwritten by a syndicate of financial institutions. However, the slope of the curve is still negative, which indicates that syndicates benefit from having a higher proportion of domestic members.

Figure 6.3. Effects on the credit gap spread by the number of local underwriters



In Model 1, we see significant effects from bond maturity and from corporate issuance. Notably, we also in Model 1 see a significant effect from the involvement of a green bond underwriting market 'big-hitter' on the primary-to-secondary yield spread. However, contrary to the expectations presented in H_2, large underwriters are found on average to be less efficient, associated with an increased spread of some 9 bps. While surprising, this finding can be seen to fit with our headline insight regarding the role of local expertise and networks in supporting green bond launches. It is plausible that big-hitters may, in general, have less extensive local expertise and networks. Country of issue and year fixed effects are controlled for in the model. Overall, Model 1 is significant, and explains around 34 percent of observed variation in this measure of green bond spread. Through Model 2 we re-ran our estimator, but introducing a one-year lag into the big-hitter controls. The home-turf, maturity, and big-hitter effects remain unchanged. Under this new specification we do find that the big-hitter and home-turf interaction effect explains some of the green bond spread, with the impact from syndication seeming to override the home-turf benefit to issuers.[103]

Table 6.3. Explaining variation in Green Bond credit gap spreads

[103] Please see Annex Table A6.1 for the full results tables from Models (1) and (2).

	Credit Spread Gap	
	Model 1	**Model 2**
Home Turf	-0.22**	-0.29***
	(0.08)	(0.07)
Syndicated	-0.03	-0.09
	(0.09)	(0.07)
HT×Syndicate	0.16+	0.25**
	(0.09)	(0.08)
Country FE	Yes	Yes
Year FE	Yes	Yes
Full controls	Yes	Yes
Observations	261	261
R2	0.360	0.344

Note: Clustered standard errors in parentheses. +p<0.1, * p<0.05,** p<0.01, ***p<0.001

To test the robustness of these findings, we re-ran our models using an alternative outcome variable specification. Here, we directly apply Dick-Nielsen et al's (2021: 3) approach of measuring underpricing as the difference between primary coupon rate and the average mid-yield over the first ten days' secondary trading. Model 3 runs the contemporaneous version of the big-hitter variable, and Model 4 runs the lagged version. As reported in the Annex, findings on the home-turf effect from this extension are consistent with the results from Models 1 and 2 using the original outcome variable specification.[104]

[104] Please see Annex Table A6.2 for the full results tables from Models (3) and (4).

6.5. Conclusion

Through our empirical modelling of market-based institutional drivers of variation in green bond performance, we can have confidence in the presence of the underwriter 'home-turf effect' across the cohort of European green bond issues launched from 2014-23. With this insight, we present an enhanced understanding of the interaction between market institutions and effective climate finance performance. Earlier scholarship has suggested that local underwriters can have a beneficial impact on conventional bond issuance, with their enhanced expertise on national market conditions and established network of actors seeking national investment opportunities plausibly supporting these improved outcomes (Kollo 2005, Butler 2008). There are additional reasons to expect such local underwriter effects to be present in the realm of sustainable finance, with inter-personal links and networks having been seen to play important roles in facilitating the mainstreaming of green bonds and similar climate financing products (Grafe et al 2023, Tripathy 2022).

Our finding of an underwriter 'home-turf effect' suggests that local underwriters may constitute significant hubs within place-based climate finance institutional ecologies. The finding from our analysis that 'big-hitter' green bond underwriters were associated with less effective issuance was somewhat surprising. However, this finding can also be interpreted in line with this prioritisation of the importance of place-based networks, given that these market-dominant players may lower levels of embeddedness in local networks than their peers.

Over recent years, within the Financial Economics scholarship on green bond performance, there has been a useful turn towards institutions. It is increasingly widely accepted that institutions, in the sense of stable organisational units exhibiting a meaningful influence over market outcomes, can play an important role in supporting green bond efficiency and effectiveness. Given the importance of green bonds to the overall envelope of global climate financing, it is vital that further work is done to understand other links between economic institutions and green bond market outcomes.

7. Institutions for Climate Finance - Emerging Trends

7.0. Introduction

In the previous chapters of the book, we have shown how institutions systematically impact the effectiveness of climate finance. We have gone beyond existing scholarship by exploring the influence of institutions on a range of state- and market-based climate financing channels, from domestic climate transition spending by national governments, to bilateral climate aid transfers, to green bond markets across emerging economies and advanced-industrialised states. We have adopted this holistic approach and considered these multiple channels of climate financing to acknowledge the reality of the contemporary climate change challenge; effective financing for climate transition requires action from, and positive interaction between, states and markets. We have shown how political parties, electoral competition and election cycles, multilateral development bank operations, central bank market interventions, and financial institutions' underwriting operations all impact on the effectiveness of climate finance.

We have, across these previous chapters, adopted a data-driven approach to understanding institutions and effective climate financing. Although the empirical chapters considered recent trends and dynamics, they are by nature backward-looking. The datasets we drew on captured behaviour of state and market actors, in the main, through the 2010s and early 2020s. Through this concluding chapter, we shift our focus from explaining the recent past of climate financing, to considering the emergent future for climate finance. Insights generated from previous chapters inform our interpretations, as we shift our gaze to the future.

When turning our attention to these emerging trends for climate finance, we continue to follow the thematic orientation we used to structure the book as a whole. We remain interested in institutional dynamics shaping state-led climate financing, shaping the impact of state agencies and regulation on market-based climate financing, and shaping interaction between market-based agents. Consequently, we structure this concluding chapter according to these organising themes. We consider emerging dynamics shaping state-based climate financing, shaping state-market interactions in the realm of climate financing, and shaping market-based trajectories of climate financing.

In the first section below, we focus on patterns of politicisation and de-politicisation of state-based climate financing, which show potential for significant shifts in national-level leadership and performance. We review dynamics within the three largest global emitters to reveal a mixed picture of politicisation and volatility across the US and EU, and a depoliticised relative stability across China's state capitalism. In the second section below, we explore the use of state regulatory capacity and state investment power to support market-based climate finance. While the rise of Sustainability Taxonomies illustrate an important lever for state regulation to accelerate market-based climate financing, moves to align Sovereign Wealth Fund (SWF) investment with climate goals and accelerate the market-based delivery of climate transition remain in a more nascent stage. In the third section of the chapter, we review market-based initiatives to supply information to support market-based resource allocation to more effectively take account of climate impact and transition. The Carbon Disclosure Project provides an embedded framework to support the greening of investment practices and acceleration of corporate decarbonisation, while the Partnership for Carbon Accounting Financials has potential to mainstream climate focus within financial sector decision-making. We close the chapter, and the book, with a recap of key insights, and a call for what we term 'evidence-based optimism' on the future of climate financing.

7.1. Politicisation, Depoliticisation, and State-based Climate Financing

To shed light on emerging patterns in state-based climate financing, we here review dynamics across the three largest emitters with greatest capacity to shape regime-level dynamics.[105] Across the US, EU, and China, we see contrasting patterns of politicisation and depoliticisation of climate action and financing. Dynamics shown in the more open political systems of the US and EU parallel findings from Chapters Two and Three, where electoral cycles and party politics were shown to exert significant influence. In the US and EU, we see ongoing politicisation with contemporary electoral success from right-leaning parties with resistance to climate action generating increased volatility on spending commitments.[106] The dramatic reversals on climate financing from the US government is counterposed by more moderated movement across the EU. In the case of China, where long-term policy-making is depoliticised by virtue of its insulation from competitive inter-party dynamics, we see stability of financing commitment, and potential for enhanced regime leadership.[107] Below, we review each of these three cases in turn.

In the US, while partisan division has been a consistent feature of national climate change politics and financing, more extreme inter-administration volatility in climate spending has become evident in recent years. In headline terms, the persistent importance of partisan politics over the US approach to climate change can be seen from the history of federal government engagement with, and disengagement from, climate treaties. With presidential primacy over the international affairs of

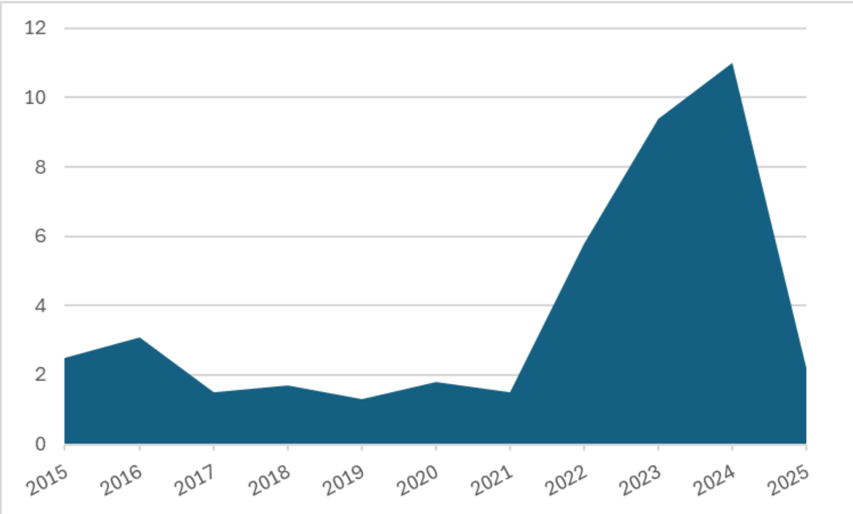
[105] To identify the three largest emitters, we aggregated the annual emissions as reported by the European Commission Emissions Database for Global Atmospheric Research (EDGAR) 2000-2020. See 'GHG emissions of all world countries', European Commission website, available at https://edgar.jrc.ec.europa.eu/report_2023. Accessed 27th May, 2025. Measured in annual emissions, EDGAR reports that India replaced the EU as the third-largest emitter around 2020.
[106] We adopt an understanding of politicisation based on the extent to which a policy area is insulated from direct party political influence.
[107] This trend of more closed political systems displaying potential for a depoliticised and more stable commitment to climate transition is particularly associated with Beeson's (2010) work on the emergence of 'environmental authoritarianism'.

the federal government, changes in the partisan orientation of the presidency have brought with them reversals on external climate commitment. The Clinton administration's signing of the Kyoto Protocol in 1998, which committed the US to a small reduction in greenhouse gas emissions against a 1990 baseline to be delivered by the early 2010s, was followed by the Bush administration's refusal to engage with the treaty obligations and its declaration in March, 2001 that the Protocol was 'dead' (Borger 2001). The Obama administration's joining of the Paris Agreement in 2016, which committed the US to set, report on, and ratchet-up emissions reduction targets that aligned with a global commitment to keep warming under 2°C, was followed by the Trump administration's announcement of an intention to withdraw in June, 2017.[108] This polarised approach to international climate change engagement has continued through to the present time. The Biden administration moved to rejoin the Paris Agreement on the Democratic party president's first day in office in January 2020, and four years later an Executive Order from Donald Trump on the Republic party president's first day back in office signalled the US' second Paris Agreement withdrawal (Noor 2025).

Below the high politics of this diplomatic dance, we know that national-level partisan politics in the US has an impact on climate financing. The relationships between partisanship and climate financing are complex, given the interplay between the US executive, Senate, and House of Representatives in determining federal spending.[109] Nonetheless, with significant levels of presidential discretion over aid spending, we see evidence of clear step changes in the volume of climate aid occurring with shifts in presidential partisan orientation (Figure 7.1). Between 2016, the final year of the Obama administration, and 2017, the first year of the first Trump administration, we saw a reduction in climate aid of around 50 percent. On current trend, a similar or even larger proportional reduction may occur between climate aid of some US$11bn in the final year of the Biden administration in 2024, and climate aid in the first year of the second Trump administration.

[108] The withdrawal did not take effect until November 4th, 2020, owing to a Paris Agreement treaty clause that locked signatories in for a period of four years from the date at which a signature was initially added (Ward and Bowen 2020).
[109] For useful investigations of these intricacies of US domestic and international financing, see Berry et al (2010) and Lavelle (2011) respectively.

Figure 7.1. US bilateral climate aid (US$bn)



Source: Harvey (2025).[110]

Climate aid constitutes a small component of total public spending on climate change, with domestic spending far exceeding the volume of climate aid. While domestic federal government spending on climate change is not systematically tracked in the US, a review from the Government Accounting Office (GAO) showed relatively consistent increases in spending from the mid-1990s through to the later 2010s (GAO 2018: 58), suggesting a limited impact from partisan orientation on domestic spending in support of decarbonisation. The emerging trend, though, is toward increased politicisation, with Democrat-led acceleration of climate spending and Republican-led reversal.

The Biden administration delivered a rapid acceleration of federal climate financing under the Inflation Reduction Act (IRA) of 2022. The IRA emerged from a context of macroeconomic shock

[110] The 2025 figure is a projection based on the Trump administration's targeted reduction of the volume of US aid by 80 percent. Given the high level of policy uncertainty, and unclarity over the relationship between administration aims and policy and spending output, the figure should be treated with a caution.

induced by the Coronavirus pandemic and the Ukraine war, and was ushered through Congress off the back of a small Democrat majority. To accelerate decarbonisation of the US economy, the IRA prioritised support for expanded renewable generation of electricity, and electrification of transportation including through subsidy for purchases of US-made electric vehicles. Amongst its headline provisions, the IRA included a programme of tax credits that would allow US$7,500 of the cost of a new electric vehicle to be recovered, providing local content and manufacturing requirements were met. Overall, it was estimated that the IRA's suite of clean technology tax credits stood cumulatively to run to between US$244bn and over US$1tn by the early 2030s. The IRA also included provision for around US$80bn of support for thousands of projects to increase renewable generation, grid capacity, and electric vehicle and battery manufacture, with around US$9bn being earmarked for improvements in home energy efficiency and electrification of domestic heating (Milman and Uteuova 2025).[111] Bistline et al's (2023) systematic review suggests that interventions supported by the IRA had the potential to exert a significant and positive impact on US decarbonisation: prior to the IRA the US was through to be on track to attain a reduction in annual greenhouse gas emissions of around 28 percent by 2030 against a 2005 baseline; with the full implementation of the IRA, this reduction was calculated to likely lead to improved performance of around 8.5 percentage points.[112]

As IRA disbursals began to be made, expenditure flows from the Act were heavily tilted towards seven 'swing states' where electoral success was seen as a necessary component of gaining control over the executive branch of government. When examined at the Congressional district

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[111] For comprehensive tracking of the IRA implementation, see Climate Program Portal, available at https://climateprogramportal.org/. Accessed 24th April, 2025.

[112] Bistline et al (2023) specifically found established models to suggest that without the IRA a reduction of 25-31 percent would be achieved by 2030, and that with the IRA a reduction of 33-40 percent would be achieved. Our figure for the accelerated performance associated with the IRA captures the difference between the mid-point values from these two ranges. Full information on the assumptions and outputs from the impact models included in the systematic review are available in Bistline et al.

level,[113] by mid-2025 a heavy preponderance of the disbursed and earmarked IRA spending had gone towards Republican-controlled areas. This dynamic reflected Republican presence in rural areas where clean energy generation and transmission infrastructure would be sited, alongside presence in some more urban areas where historic automobile manufacturing stood to gain from support for expanded electric vehicle construction capacity (Milman et al 2025).

This interesting case, through the IRA, of 'reverse pork-barelling' was not,[114] however, sufficient to insulate associated spending against Republican row-back. On his first day back in office in 2025, President Trump ordered a cessation to any further disbursals of IRA funds. Constraints imposed through the legislative branch of government have reduced the initial direct IRA spending power of some US$80bn by around half (TIGTA 2025: 1). Drawn-out political and judicial contestation will determine the extent of the eventual reduction in the total IRA spend on clean energy and technology.[115] In addition to the eventual reduction in expenditure, the uncertainty introduced by the Trump administration has also exerted a chilling effect on domestic action. While the IRA was designed to crowd-in private sector investment, we now see that investors in clean energy and clean tech are losing confidence to break the ground on projects given fears that, at a future date, funding or tax credit eligibility may be rescinded (Britton and Runyon 2025). Given the depth to which partisan orientation impacts elite and popular attitudes to climate action in the US, with Republican politicians and voters holding entrenched resistance to spending on this issue area (Bogado 2024, Egan and Mullin 2024, Merry and Payne 2024), the US looks set over the medium term to retain its trend of dramatically fluctuating commitment to deploying government spending power to support climate transition.

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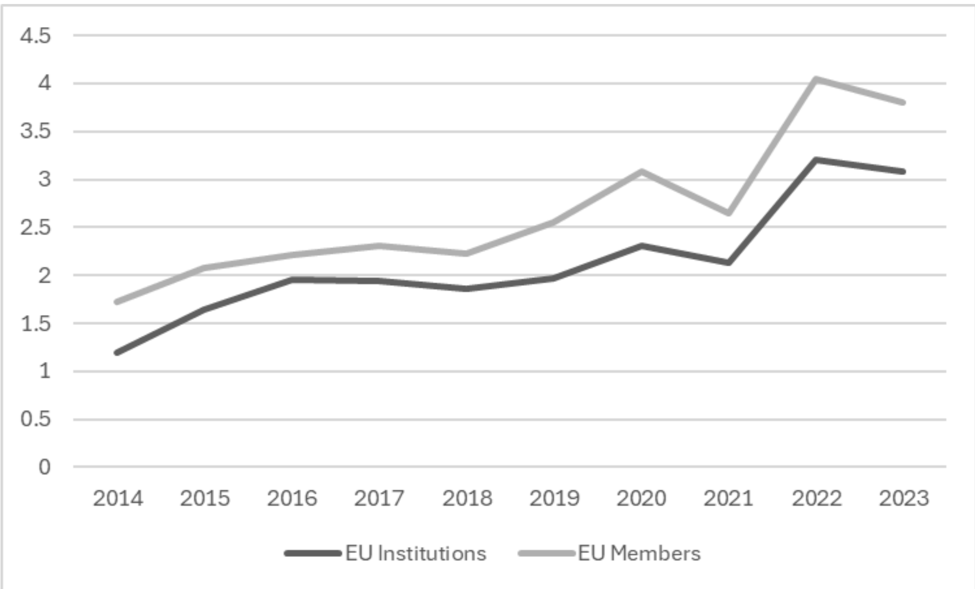
[113] Members of the House of Representatives are elected by geographic areas referred to as Congressional Districts, with individual states having between one and 52 Districts as determined by population. See 'Directory of Representatives', United States House of Representatives website, available at https://www.house.gov/representatives. Accessed 1st May, 2025.

[114] We use the term given that typically 'pork barrelling' refers to attempts by the party in power to direct spending to politically-aligned areas. See, for example, Pedron and Wessel (2025) and Clegg and Davies (2024).

[115] For comprehensive monitoring of climate policy positions and actions from the Trump administration, see Columbia Law School Sabin Centre for Climate Change Law 'Climate Backtracker', available at https://climate.law.columbia.edu/content/climate-backtracker. Accessed 1st May, 2025.

Whereas the inconstancy of the US engagement with climate transition and climate financing is well established, the European Union over the last decade or so has come to be viewed as something of a climate leader. EU institutions and political elites have become prominent advocates of accelerated climate effort, and externally EU institutions and member states have demonstrated stable commitment to climate aid (Figure 7.2). However, current trends suggest emerging volatility in EU climate performance and financing commitment. While EU positioning and spending commitments result from a complex interaction between the EU President and Commissioners, the European Council, and European Parliament, we see evidence across this nexus of politicisation and push-back on existing climate leadership. Before turning to this pushback, we first cover the earlier consolidation of leadership.

Figure 7.2. EU climate aid (US$bn)



Source: OECD Data Explorer.[116]

[116] See 'OECD Data Explorer', OECD website, available at https://data-explorer.oecd.org/vis?lc=en&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_RIOMRKR%40DF_RIOMARKERS&df[ag]=OECD.DCD.FSD&dq=4EU001%2BDAC_EC..1000..2.20%2B30.2%2B1..Q._T..&to[TIME_PERIOD]=false&pd=2014%2C2023&vw=tb. In line with our approach in Chapter 3, we have included all disbursals marked as primarily or significantly directed to climate adaptation and mitigation (USD 2023 values). We should note

2014 represented an important year in the EU's movement towards a position of climate leadership. Support for strengthened action had grown across the EU in preceding years, and the EU Parliamentary elections of 2014 laid foundations for enhanced climate action by providing centre-left supporters of this focus with increased influence. In this context, the Commission under the leadership of Jean-Claude Juncker, the EU Parliament, and Heads of State were sufficiently united behind an agenda for accelerated climate action to agree to the issuance of a new EU-wide climate commitment by the close of 2014. A target was agreed of a 40 percent reduction in greenhouse gas emissions across the continent relative to 1990 levels, with an integration of national and EU processes to ensure that individual member state actions and regional interventions together met this aim.[117] Driven by this consensus, the Commission introduced a 'climate mainstreaming' approach to EU budget setting and monitoring processes. In addition to requiring EU institutions to systematically report on the volume of spending that was being devoted to climate action, the Commission declared that, over the 2014-19 EU Parliamentary session, at least 20 percent of the EU budget would be ring-fenced for climate (Holzleitner et al 2019). Overall, around 21 percent of the EU budget for this period was targeted towards climate transition. While this represented some US$250bn, the Commission was explicit in outlining the need for significant expansion of public and private investment to reach emissions reduction targets (European Commission 2017: 3).

The European Parliamentary elections of 2019 created a political window of opportunity for a further acceleration of this EU climate commitment and financing drive. Across previous Parliamentary sessions, together the European Peoples' Parties (EPP) and the Progressive Alliance of Socialists and Democrats (S&D), the two main 'centrist' groupings, had been able to dominate over

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that the figures on EU climate aid are not directly comparable to the figures above on US climate aid, with the US not reporting under the OECD data framework.

[117] See 'A policy framework for climate and energy in the period from 2020 to 2030: Communication from the Commission to the Parliament, Council, Social Committee, and Committee for the Regions', EU Access to European Union Law website, available at https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM:2014:15:FIN. Accessed 6th May, 2025.

the institution's agenda, powers of appointment, and budget. 2019 marked a significant watershed; for the first time, the EPP and S&D failed to achieve a majority share of seats. In this context, the Green electoral success, which saw the European Green Party achieve a record high of 55 seats and the wider Greens/European Free Alliance grouping a combined total of 75 seats, provided Green leadership with the opportunity to exert significant leverage over the political centre (Pearson and Rudig 2020). The opportunity for Green strategic influence occurred contemporaneously with the emergence of a mass mobilisation of young and climate-focused individuals across Europe, with the Greta Thunberg-initiated 'Fridays for Future' climate strikes. This shifting centre of gravity contributed to EU institutions and national governments of the left and right coalescing behind a 'European Green Deal', an initiative that sought to transform climate change from being an 'urgent challenge into a unique opportunity' for global leadership from the EU (Bloomfield and Steward 2020: 770-1, European Commission 2019: 1).

Launched in December, 2019, the European Green Deal outlined a roadmap for the EU to reach a position of 'climate neutrality' by 2050. The European Green Deal also locked-in an accelerated pace of emissions reductions over the shorter-term, ramping up the targeted reduction to be achieved by 2030 of 55 percent relative to 1990 levels. Overall, the associated agenda for change prioritised actions including decarbonisation of electricity generation, embedding principles of 'repair, reuse, and recycling' across a circular economy model, reducing harmful emissions and waste products from industrial processes, and transforming transportation systems away from individualised use of internal combustion engine vehicles (Koundouri et al 2024).[118] As well as a programme of legislative activity to align EU regulatory capacity with the accelerated timeframe for climate transformation, the Commission pledged to mobilise over US$1.25tn through the 2020s to support the European Green Deal. The EU financial mobilisation was constituted by a commitment

[118] See also 'European Green Deal', European Commission website, available at https://www.consilium.europa.eu/en/policies/european-green-deal/#:~:text=The%20vision%20of%20the%20European,balance%20in%20nature%20and%20ecosystems. Accessed 7th May, 2025.

that at least 30 percent of the core EU budget, and at least 37 percent of the post-pandemic Next Generation EU Recovery and Resilience Fund (RRF), be directed to climate transition.[119]

The 2024 European Parliamentary elections brought with them a realignment to the right, with some success for groupings of Eurosceptic and more nationalist parties.[120] With polarisation on climate change and the emergence of 'climate backlash' being a feature of public opinion across many European states (Caldwell et al 2024), in the aftermath of these elections fears were articulated by its advocates that the European Green Deal may be 'at high risk of being killed off'.[121] While it appears unlikely there will be immediate row-back on headline regulatory commitments such as the targeted increase in the share of renewables in energy production to 42 percent by 2030, it is likely that there will be increased volatility in the supply of public finance to support this and other headline commitments. Meeting the EU goal of a 55 percent emissions reduction by 2030 implies increased annual climate-related investment from around US$775bn to around US$1.2tn per year.[122] With the first tranche of European Green Deal funding rapidly drawing down, resistance at the EU and national levels may become an increasingly significant barrier to future action.

As noted by Pisani-Ferry and Tagliapietra (2024), significant headwinds need to be tacked against to achieve the required public investment for the EU climate transition ambitions. There is the lack of clarity over a follow-on EU mechanism to the RRF, which is due to release its final funding in 2026. There is the limitation from the EU fiscal framework on the use of sovereign borrowing to finance green investment. Under EU rules, a debt ceiling of 60 percent GDP functions as an absolute benchmark for members, with no mechanism to provide special treatment for climate transition

[119] See 'Finance and the Green Deal', European Commission website, available at https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal/finance-and-green-deal_en. Accessed 7th May, 2025.
[120] Most notably, the Identity and Democracy/Patriots for Europe group expanded its presence by 34 seats, and the new Europe of Sovereign States gained 25 seats. See Mudde (2024: 126).
[121] Belgian Member of the European Parliament and Co-Chair of the Green grouping Philippe Lamberts, as quoted in O'Carroll (2024).
[122] See Guntram Wolff, Simone Tagliapietra, and Klaas Lenaerts, 'How much investment do we need to reach net zero?', 25th August, 2021, Bruegel website, available at https://www.bruegel.org/blog-post/how-much-investment-do-we-need-reach-net-zero#:~:text=In%20the%20EU%2C%20the%20European,of%20two%20(Figure%203). Accessed 27th May, 2025.

investment.[123] While it may be possible for a member state with an above-benchmark debt level to argue that additional green public investment is 'growth enhancing' (and so should, under intricacies of EU rules, be treated more leniently),[124] the Commission's willingness to accept this rationale may need to be established on a case-by-case basis by member states progressing through the Excessive Debt Procedure.

In the contemporary EU context, then, we have reduced support for climate action in the 2024 Parliament, evidence of climate backlash amongst European citizens, and the existence of fiscal processes and rules that may inhibit state capacity to release public investment for climate transition. Current geopolitical and trade policy uncertainty seems to reduce the likelihood of sufficient government climate finance being released across the EU; commitments being made on enhanced military spending across the continent may squeeze the scale of financing available for climate change, and rising tariffs may generate macroeconomic challenge and depressed fiscal capacity with which to resource climate commitments.[125]

We have shown in Chapters Two and Three that party politics and election outcomes exert important influence over government action in this area. Some evidence of underpinning resilience to climate commitment and financing across the EU comes from survey findings from Eichorn and Grabbe (2025), who suggest that the cohort of climate-motivated voters across the EU may be relatively stable. With effective mobilisation, this cohort of climate-motivated voters may coalesce into a significant voting bloc in national and regional elections. Given public prioritisation of concerns with climate, economic security, and defence, there may be space for parties across traditional

[123] For further information on the debt and deficit targets and monitoring and adjustment mechanisms within the EU see 'Excessive Deficit Procedure', European Commission website, available at https://www.consilium.europa.eu/en/policies/excessive-deficit-procedure/. Accessed 8th May, 2025.
[124] Under EU rules, where debt is classified as 'growth enhancing' then a member with a debt level above 60 percent GDP may be able to switch from the standard four-year adjustment period for meeting the benchmark to a seven-year adjustment period. See Pisani-Ferry and Tagliaietra (2024).
[125] For reflection on the potential tension between rising military spending and climate financing see 'Every Euro spent on the military carries a carbon cost to the climate and to our future security', Conflict and Environment Observatory website, available at https://ceobs.org/military-climate-action-has-never-been-more-urgent-heres-why/#:~:text=In%20addition%20to%20directly%20increasing,innovation%2C%20disrupting%20the%20green%20transition. Accessed 8th May, 2025.

left-right divides to construct electoral and policy platforms that highlight payoffs to household-level finances and employment prospects, and geopolitical gains from reduced reliance on hostile fossil fuel exporting states, than can follow accelerated climate financing and action. While a more volatile and constrained environment for government spending on climate transition appears to be emerging across Europe, there will likely be countervailing pressures against row-back on climate commitment.

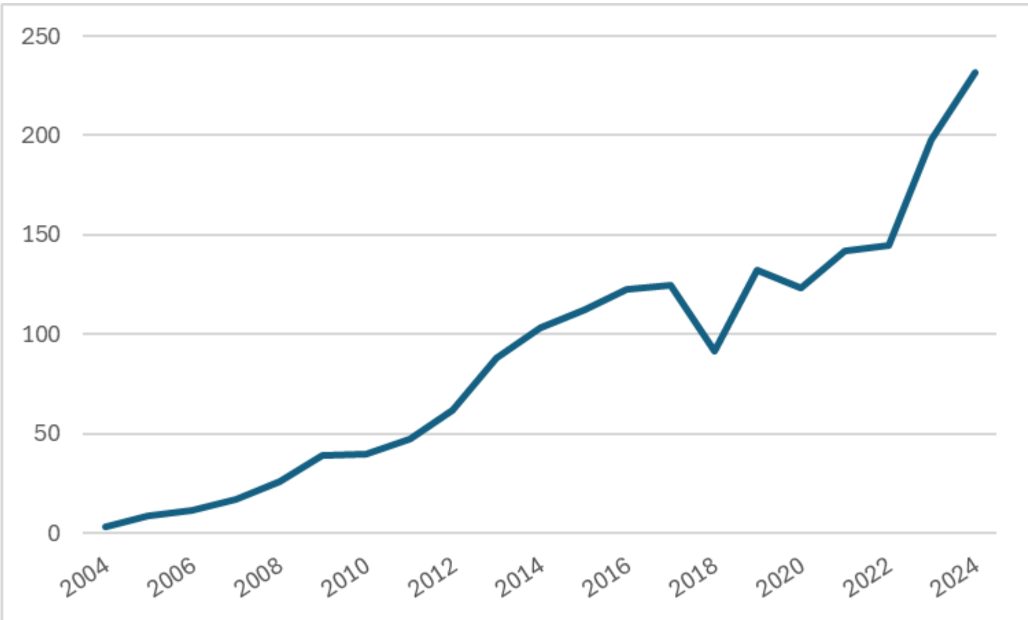
To close this review of emerging trends in state-based climate financing, we turn to depoliticised dynamics in China. China is the single largest national greenhouse gas producer, responsible for around one-third of global annual emissions.[126] Under its model of one-party 'state-led capitalism', public and quasi-public authorities plan a central role in guiding and financing high-level transformation. Five-Year Plans constitute a core mechanism through which transformation is designed and implemented, with the Communist Party Central Committee issuing strategic objectives, its State Council supplying detail and coordinating the drafting of the Plan, and the National People's Congress formally approving. With a heavy presence from state-owned enterprises in both highly polluting sectors and in clean energy and clean technology sectors, and with state control over the largest banks whose lines of credit facilitate sectoral expansion and reconfiguration, the dividing line between state and non-state financing for climate transition can be difficult to draw in the sand. Nonetheless, by exploring the nature of the managed transition towards clean energy and technology, we can gain a sense of the scale of state resources being directed to support climate transition.

The passage of the 12th Five-Year Plan (2011-15) has been identified as an important moment in the government of China's accelerated climate transition. Li and Wang (2012) suggest that the 12th Plan introduced a 'paradigm shift' on clean energy. The sustained upturn in investment in

[126] Measured in per capita or cumulative terms, Chinese greenhouse gas emissions sit at around 50 percent of the quantum of US emissions, with the US sitting amongst the leaders under the former and leading under the latter. See 'Analysis: Which countries are historically responsible for climate change', Carbon Brief website, available at https://www.carbonbrief.org/analysis-which-countries-are-historically-responsible-for-climate-change/. Accessed 8th May, 2025.

renewables from this time illustrates the scale of financial commitment, rising from around US$60bn in 2012 to US$230bn by 2024 (Figure 7.2). With renewable energy expansion representing an essential pillar of the Chinese government commitment to reach 'peak carbon' before 2030 and carbon neutrality by 2060,[127] this prioritisation of investment in the sector is set to continue across the medium term.

Figure 7.3. Annual investment in renewable energy, China (US$bn)



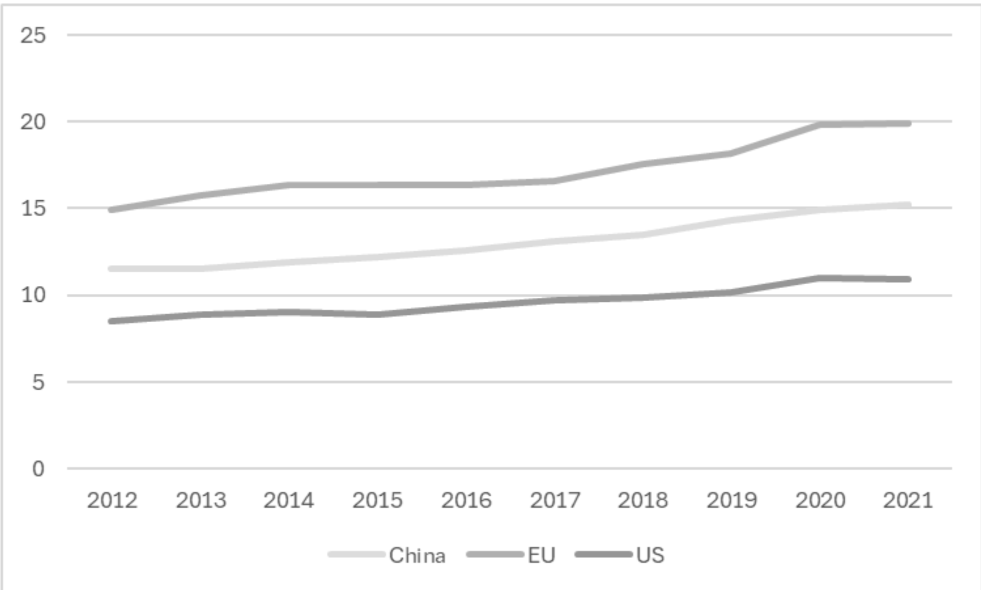
Source: Evans (2016) and Keenan (2025).

With this spending on clean energy and attendant policy prioritisation, moves to decarbonise an electricity grid that remains dominated by coal have become embedded in China. At around 15 percent, the share of renewables within the overall energy mix in China sits between the US figure of 11 percent and the EU figure of 20 percent (Figure 7.3). Owing to the scale of the national grid, the

[127] Debates are ongoing as to whether China has already reached its point of peak carbon emissions, and given the sensitivity of analyses to modelling assumptions these debates will continue for some time. For an overview, see You (2024).

total renewable energy capacity in China sits at over 250 percent of the capacity of the EU, and over 400 percent of the capacity across the US.[128] Buoyed by the ongoing financial commitments that have been planned-in by the government of China, the scale of Chinese leadership in renewable capacity is set to accelerate over coming years (Figure 7.4). Dynamics in China, where the one-party state apparatus has been able to lock-in financing and policy support for clean energy over the medium term, appear to reflect the broader findings from from Wang et al (2024) that political stability is associated with strengthened renewable energy performance. While the pace of change is accelerating, the speed of the Chinese transition away from coal-based energy generation has been heavily criticised (Patel 2025), and significant limitations identified in the scope for local participation in the planning processes surrounding large-scale renewable and other infrastructure projects that provide the backbone of energy sector decarbonisation (Chu et al 2022).

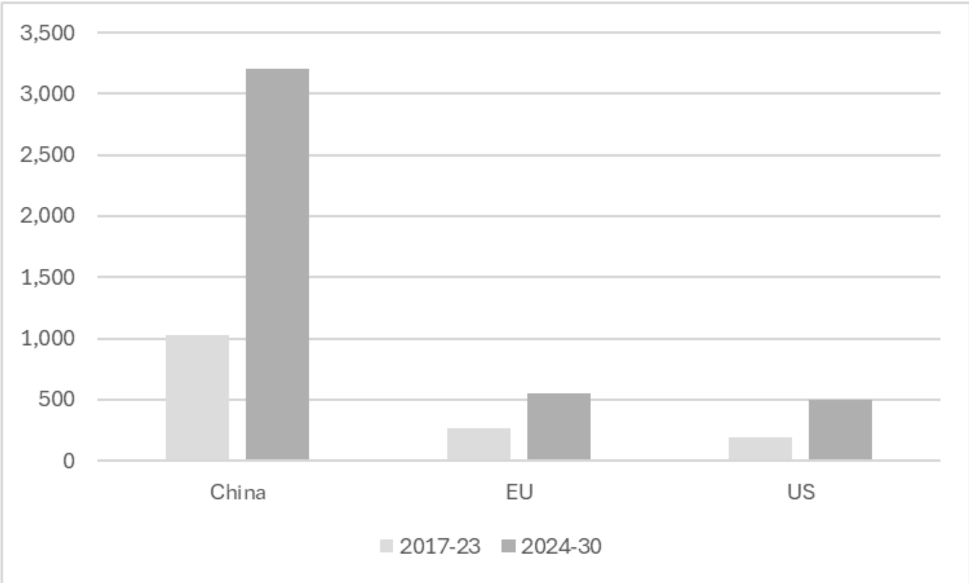
Figure 7.4. Share of renewables within total electricity generation



[128] See ‘Renewable energy capacity worldwide (2024) - Lucia Fernandez’, Statista website, available at https://www.statista.com/statistics/267233/renewable-energy-capacity-worldwide-by-country/. Accessed 9th May, 2025.

Source: World Bank Open Data.[129]

Figure 7.5. New renewable capacity (GigaWatt Hours)



Source: International Energy Agency.[130]

In addition to increased domestic clean energy capacity, the government of China's prioritisation of investment in renewables has helped the sector to emerge as a contributor to export growth. Around 60 percent of the world's capacity for manufacturing solar cells, wind turbines, and battery storage sits within China, with Chinese suppliers dominating these export markets.[131] Concerns with 'energy security', alongside a contemporary turn toward enhanced economic

[129] See 'Renewable energy consumption (percentage of total final consumption)', World Bank Open Data website, available at Source: https://data.worldbank.org/indicator/EG.FEC.RNEW.ZS?end=2021&locations=CN-US-EU&start=2012. Accessed 9th May, 2025.

[130] See 'Renewable electricity capacity growth', International Energy Agency website, available at https://www.iea.org/data-and-statistics/charts/renewable-electricity-capacity-growth-by-country-region-main-case-2017-2030. Accessed 9th May, 2025.

[131] Bian et al (2024: 11) project that preponderance of the solar and battery exports through to 2030 will be supplied by Chinese manufacturers.

nationalism,[132] may see increased reluctance on the part of some governments and national energy suppliers to rely on externally-supplied renewable infrastructure. However, the potential for longer-term export gains from investment in renewable capacity is set to provide an additional incentive for ongoing public investment support.

While the scale of renewable energy generation expansion in China and the importance of this overall energy system transformation to overall global climate transition is widely acknowledged, the role of the government of China as a source of bileratal climate finance remains contested. Classified as a 'developing' country and therefore eligible to itself receive bilateral climate aid,[133] China nonetheless has become a net exporter of ODA. Kitano and Miyabayashi (2023) suggest that, since the early 2010s, the annual volume of bilateral assistance being committed committed through government departmental budgets has consistently been around US$3bn, with an additional US$1bn to US$1.5bn being supplied through multilateral channels. Focusing specifically on climate aid, Chichoka and Mitchell's (2024: 19) analysis suggests that from 2015-21, around US$0.7bn has been distributed annually.[134] From 2015-20, the volume of development assistance for climate finance was surpassed comfortably by the volume of Chinese development finance being used to expand fossil fuel power generation capacity before, in 2021, the government committed to cease additional funding for this purpose. As such, while there is evidence that financing for climate transition and energy decarbonisation is being prioritised domestically, the government of China occupies an ambiguous position in relation to financing for climate aid.

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[132] On the rise of economic nationalism, see Letzing (2025).

[133] Under the OECD Development Assistance Committee framework for recording aid flows, financing can be recorded as aid providing the aim is to contribute to development and welfare enhancement within a country that has been identified as Overseas Development Assistance-eligible. See 'Overseas Development Assistance', OECD website, available at https://www.oecd.org/en/topics/official-development-assistance-oda.html#:~:text=Defining%20which%20countries%20and%20territories,years%20by%20the%20OECD's%20DAC. Accessed 11th May, 2025.

[134] This figure includes outflows that take the form of grants and concessional loans, and excludes non-concessional loans (which account for around US$2.5bn per year across this period, albeit with significant year-on-year fluctuation).

7.2. Public Regulation, Investment Flows, and Market-based Climate Financing

When considering interaction between states and markets in the sphere of climate financing, we turn to emerging dynamics surrounding the regulation of sustainable finance and green bonds on the one hand, and the targeting of public investment funds towards climate transition, on the other. In relation to the former, we explore the increasingly widespread use of financial regulatory capacity to enhance the monitoring and reporting requirements for sustainable financial products in general and green bonds in particular, with the rise of Sustainability Taxonomies representing a key intervention in this regard. In relation to the latter, we explore the emerging attempts by Sovereign Wealth Funds (SWFs) to deploy their vast investment capacity in support of climate aims, a course of action that involves public funds being mediated through financial markets toward enhanced climate performance. While the climate pivot from SWFs remains nascent, with around US$11tn of assets under management SWFs represent a potentially potent tool for translating the flow of public finance through financial markets and into accelerated climate transition.[135]

Through Chapters Four and Five, we highlighted the challenge of effectively deploying state capacity to enhance the performance of market-based climate finance, and the potential for unintended negative consequences from regulatory intervention. Here, while the pivot from SWFs remains in its early phase, we present some evidence of positive impact from green bond regulation, demonstrating the potential gains to be had from effective deployment of state regulatory capacity.

[135] With a little over half of assets in bonds and other fixed income products and one-third equities, SWF operations have direct lines of engagement into financial markets. See 'Trends in sovereign wealth fund asset allocation', International Forum of Sovereign Wealth Funds website, available at https://www.ifswf.org/trends-sovereign-wealth-funds-asset-allocation-over-time-survey#:~:text=As%20shown%20below%2C%20the%20sovereign,funds%20are%20comparatively%20much%20smaller. Accessed 12th May, 2025.

As was noted across earlier chapters in the book, market-based sustainable finance represents a crucial element of the overall climate financing mix. The World Bank Treasury recently suggested that, across 2024, total issuance of labelled sustainable bonds reached US$1.1tn.[136] The scale of this issuance can be contextualised by placing it alongside a figure for annual public sector climate-relevant investment across the OECD of around US$137bn.[137] However, when we delve into market-based sustainable finance, we are immediately presented with an important question: what, exactly, is it about 'sustainable finance' that makes it 'sustainable'? Through the earlier phase of green bond markets and sustainability labelling, to help them identify a 'green' or 'sustainable' bond, investors were reliant on either third party verification from private standard setters, or on issuer self-declaration. Under external verification, a civil society organisation or corporate actor would be contracted in by the bond issuer to provide a judgement on the validity of the issuer's claims to be using proceeds raised to deliver improved sustainability performance or climate change outcomes. The issuer would then highlight the presence of this external verification within its issuance paperwork, and ensure that a 'green' label was affixed to the bond's Order Book listing.[138] Under self-declaration, the issuer would include claims relating to their green or sustainability credentials within their issuance paperwork, and thereby on the basis of their own evaluation of the bond's sustainability credentials ensure that a 'green' label was attached to its Order Book listing.[139]

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[136] See 'Labelled bond quarterly newsletter: February 2025', World Bank website, available at https://thedocs.worldbank.org/en/doc/cd82b4033281dab2cb1a1c71eeb691e4-0340012025/original/Labeled-Bond-Quarterly-Newsletter-Issue-No-10.pdf. Accessed 12th May, 2025.

[137] See 'Subnational and national government climate finance', OECD website, available at https://www.oecd.org/en/about/programmes/subnational-government-climate-finance-hub.html. Accessed 12th May, 2025. The US$137bn figure is the aggregate of climate-relevant government investment reported by 33 members of the OECD for 2019, the most recent year for which such data is available. At around US$470bn the figure for these states' aggregate climate-relevant spending is considerably higher but, given the sustainable bond purpose of raising funds for investment, the US$137bn value provides a more meaningful point of comparison.

[138] A description of the bond and issuer's core characteristics are included in an Order Book trading platform, which a potential investor can then use to identify bonds or bond issuers of interest. Where an investor is interested in sustainable financial products, she will be able to use the Order Book platform to search specifically for instruments listed within the platform as 'green bonds'.

[139] For additional information see 'What is the International Order Book?', London Stock Exchange Group website, available at https://www.londonstockexchange.com/equities-trading/asset-classes/shares-trading/international-order-book. Accessed 12th May, 2025.

Recently, though, an increasing range of states and agencies are seeking to deploy public regulatory power to increase the robustness of and confidence in such sustainability labels.

As noted in Chapter Four, a history of green bond labeling can be traced back to World Bank collaboration with Cicero in the late 2000s.[140] In 2010, Climate Bonds Initiative, an 'investor-focused not for profit' organisation that seeks to 'mobilise capital for climate action', made an early intervention to systematise green bond accreditation with the launch of its Climate Bonds Taxonomy. The Taxonomy identifies a range of bond features that ultimately shape whether an individual bond is classifiable as 'green' with, for example, a bond supporting the installation of solar capacity automatically qualifying as 'green', a bond supporting investment in the use of an oceanic heating or cooling facility potentially qualifying subject to a review of additional project detail, and a bond supporting fossil fuel-based energy generation automatically being locked-out of gaining accreditation (CBI 2021). The International Capital Market Association's publication, in 2014, of its Green Bond Principles represented a high-profile market-led intervention into the governance of sustainability accreditation. However, while providing a template for enhancing issuers' capacity to launch 'credible' green bonds, these ICMA Principles did not attempt to substantively define performance parameters that should be met to warrant the green label. When seeking out foundations of state-led regulation of green bonds, an important building block was laid in 2015 within the Chinese financial regulatory framework.

In December of 2015, the People's Bank of China and the National Development and Reform Committee jointly published its Green Bond Endorsed Project Catalogue. The Catalogue functions as a list of bond issuances that are judged to meet sustainability criteria, with projects aimed at energy saving, clean transportation, and clean energy being identified for potential inclusion. With around 90 percent of green bond issuance in China being backed by the state and purchased by state-controlled banks, the combined influence from its central bank and its core agency for

[140] I rely on Bishop (2019) in the following overview of the governance history relating to green and sustainability bond standards and accreditation.

coordinating and approving major infrastructure investment on market performance is extremely high. The more recent inclusion by the Peoples' Bank of China of green bonds as gaining preferential treatment under its collateral framework has served to ratchet-up the importance of this Catalogue (Macaire and Nef 2021). With some US$130bn of bond issuance meeting this national criteria in 2023, the Endorsed Project Catalogue oversaw a larger pool of green-labelled bonds than any other national market. However, the government of China continues to face calls for a strengthening of its regulatory framework. While the removal of 'clean coal' power generation from the revised China Green Bond Standard (CGBS) in 2021 was welcomed by external investors seeking reassurance on the strength of certified bonds' green credentials,[141] China's domestic standards remain notably more permissive than the Climate Bond Initiative benchmark. The gap between CGBS is narrowing over time,[142] but ongoing weakness in the Chinese regulatory framework is evident in the fact that, through 2023, only around 65 percent of CGBS-accredited bonds were judged to have met the Climate Bond Initiative benchmark (CBI 2024: 5).

The movement toward an EU Green Bond Standard (EUGBS) commenced significantly later than that of the Chinese authorities. The launch of the EU framework came in 2023, and represented a major shift in green bond regulation within this major market. Seeds of the EUGBS were sewn with the approval of the EU Sustainability Taxonomy in 2020. The Taxonomy was a key pillar within the commitment to accelerated climate action with the EU Green Deal, setting out as it did an approach for identifying projects and activities that aligned with the EU commitment to net zero by 2050. The Sustainability Taxonomy defines activities targeted at climate mitigation and adaptation, moves to a circular economy, sustainable water use, pollution prevention, and restoration of ecosystems as being sustainable and aligned with climate goals.[143] Under the EUGBS, in order for bond issues to be

[141] See 'Analysis of China's green bond principles', ICMA website, available at https://www.icmagroup.org/assets/Analysis-of-Chinas-Green-Bond-Principles.pdf. Accessed 13th May, 2025.
[142] Part of the reason for the narrowing gap relates to Climate Bond Initiative following the lead of the CGBS in now permitting within its framework the inclusion of 'transition' green bonds covering 'hard to abate' sectors such as cement and steel production, and extractive industry operations. See CBI (2024: 6).
[143] See 'EU taxonomy for sustainable activities', European Commission website, available at https://finance.ec.europa.eu/sustainable-finance/tools-and-standards/eu-taxonomy-sustainable-activities_en. Accessed 13th May, 2025.

certified as a ‘European Green Bond’, the issuer must commit to use at least 85 percent of the proceeds to fund activities that align with the Sustainability Taxonomy. Under the framework, bond issuers must gain third-party verification of the alignment between their activities and the EUGBS prior to the launch of the bond and also after all proceeds from the bond have been used, with reports being made publicly available. From 2026, these third-party verifiers must be registered with the European Securities and Markets Association.[144]

Recent analysis suggests that the implementation of the EUGBS had an immediate and positive impact on market behaviour. The intention behind the introduction of the EUGBS was to reduce space for ‘greenwashing’; by mandating external verification against robust criteria, the EUGBS was designed to ensure that issuers could not egregiously secure a ‘green’ label for activities that in reality were not aligned with sustainable transformation. By focusing bond market participants’ attention on the credibility of green bonds, the introduction of the EUGBS was followed by movements towards issuers with stronger environmental credentials (Galindo-Gutierrez 2024). This market reaction to the creation of the EUGBS illustrates that public regulation of a green bond standard can incentivise investors to move towards green bond issuers with relatively strong environmental credentials, thereby rewarding issuers with stronger green commitment and limiting space for greenwashing by issuers with lower climate commitment.

Beyond catalysing a positive market toward green bonds with higher sustainability credentials, there is potential for national green bond regulatory frameworks to be given additional bite. Firstly, with a regulator has a sustainability standard in place, it becomes possible to more effectively direct monetary policy tools toward climate transition goals. The Peoples’ Bank of China’s granting of preferential treatment to green bonds under its collateral framework gives one example to follow in this regard. Second, with a regulatory standard in place, it also becomes possible to

[144] See ‘External reviewers of European Green Bonds’, European Securities and Markets Association website, available at https://www.esma.europa.eu/esmas-activities/investors-and-issuers/external-reviewers-european-green-bonds. Accessed 13th May, 2025.

require banks to monitor and report on their 'green asset ratio' (GAR). The EU requirement that European banks report on their GAR gives an example to follow in this regard.[145] Under this reporting requirement, we see that on average only around 3 percent of banks' assets are aligned with EU Sustainability Taxonomy requirements. The goal is that, through a combination of shareholder and government pressure, this proportion of green investment will raise significantly over time.[146]

Whereas there is potential for a wide range of governments to explore the use of green bond regulatory standards to accelerate the greening of financial markets, a sub-section of states have an additional direct mechanism for using intervention in financial markets to catalyse climate action. Where a state holds a Sovereign Wealth Fund (SWF), there can be significant potential to use a deep pool of national resources to shape market behaviour. The term Sovereign Wealth Fund is used to describe a state agency that typically operates at arms-length from the national government, and which oversees a portfolio of market-based investments using public resources. The largest SWFs typically reside in states with control over large reserves of natural resources, and draw on revenue flows from extraction as a foundation for their financial base.[147] As well as aiming to use market investment to generate a healthy return, SWFs are often directed by their national authority to achieve broader social or economic goals. Fini and Rethel (2013), for example, show how the Norwegian Pension Fund and the Malaysian Khazanah Nasional Berhad drew on their power as major shareholders to push for improved corporate governance from financial sector institutions, in the years following the global financial crisis. More recently, a turn toward environmental and sustainability goals has taken place across SWFs.

[145] For detail on methodologies used to calculate banks' green asset ratios, see Bruhl (2024).
[146] See Marine Leleux, 'Taxonomy Disclosures: A slow start, but a start nonetheless', ING website, available at https://think.ing.com/articles/hold-taxonomy-disclosures-a-low-start-but-a-start-nonetheless/#:~:text=While%20the%20two%20previous%20policies,institution's%20alignment%20with%20the%20Taxonomy. Accessed 13th May, 2025.
[147] For an overview of the largest Sovereign Wealth Funds as measured by assets under management, see '100 Largest Sovereign Wealth Fund Ranking', Sovereign Wealth Fund Institute website, available at https://www.swfinstitute.org/fund-rankings/sovereign-wealth-fund. Accessed 13th May, 2025.

With around US$5.8tn in bond holdings and $3.6tn in equities under management,[148] SWFs are entities with extremely concentrated control over investment flows through financial markets. Reflecting their high potential to drive change, SWFs have become incorporated into the governance nexus surrounding climate financing. Following the conclusion of the Paris Agreement in 2015, the French government embarked on a series of follow-up One World Summit events focused on the creation of new financing mechanisms to support climate transformation. At the 2017 One World Summit, with high-level leadership from President Emmanuel Macron of France, UN Secretary General Antonio Guterres, and World Bank President Jim Kim, and participation from a wide range of international political leaders, a linkage between SWFs and climate transition was embedded through the creation of a One Planet Sovereign Wealth Fund (OPSWF) Working Group. OPSWF was founded with the stated goal of better aligning SWF investment operations with the need for accelerated climate finance. With the Abu Dhabi Investment Authority, Kuwait Investment Authority, New Zealand Super Fund, Norwegian Pension Fund, Saudi Public Investment Fund, and Qatar Investment Authority as founding members, the One Planet Sovereign Wealth Funds Framework included pledges to build climate considerations into decision-making frameworks, align portfolios with climate change considerations, and use leverage to accelerate climate action from companies within their portfolios (OPSWF 2018: 7).

The OPSWF acknowledges the diverse political and economic contexts that its member SWFs exist within, and relies on a voluntary approach to the greening of SWF practice. The OPSWF relies on peer learning and information sharing, with sessions at the fringe of the World Bank-IMF Annual Meeting and its high-level Annual Meeting providing regular opportunities for identification of current and emerging effective practice. Through these socialisation-based mechanisms, OPSWF claims successes in the shape of supporting members' first issuance of green bonds, their completion

[148] See 'Trends in sovereign wealth fund asset allocation', International Forum of Sovereign Wealth Funds website, available at https://www.ifswf.org/trends-sovereign-wealth-funds-asset-allocation-over-time-survey#:~:text=As%20shown%20below%2C%20the%20sovereign,funds%20are%20comparatively%20much%20smaller. Accessed 13th May, 2025.

of portfolio carbon footprint assessments, and ring-fencing of funds for climate transition and clean energy (OPSWF 2025: 18-20).

Beyond OPSWF's prominent communication of examples of members' strong performance, there is some evidence that over time SWF commitment to sustainability is systematically expanding. Since the mid-2010s, the organisation Global Sovereign Wealth Fund (GSWF) has tracked a range of performance indicators for SWFs, to gauge changes in governance structures and mission. The GSWF ranking grades SWFs along a sustainability metric, based on factors including the incorporation of sustainability principles into organisation mission, presence of a dedicated ESG team within the organisation, and strength of processes for identifying and managing ESG risk and performance within their portfolios. In aggregate terms there has been a steady increase in the strength of SWF engagement with sustainability issues over recent years, as demonstrated by the rise in the average sustainability score from 4.2 on a ten-point scale in 2021, to 5.1 in 2024.[149] While there is considerable variation in the gradings of OPSWF members, we see strong sustainability scores from two of the founding OPSWF members particularly, with the Norwegian Pension Fund and the Saudi Arabian Public Investment Fund respectively receiving scores of ten and nine on this sustainability performance criteria.[150]

Given our understanding of links between an organisation's communication of its mission and its structure, on the one hand, and its operational performance, on the other,[151] SWFs' increased incorporation of climate concerns into their mission and organisational structure may lay foundations for future strengthening of climate financing and impact. A recent study on SWFs and sustainability by Bortolotti et al (2023) provides initial support for this possibility. Bortolotti et al bring together the GSWF sustainability ranking and the Sovereign Wealth Fund Global Transaction Database, which

[149] See 'Governance Sustainability and Resilience Scorecard: 2024', Global Sovereign Wealth Fund website, available at https://globalswf.com/reports/2024gsr. Accessed 13th May, 2025.
[150] The Global Sovereign Wealth Fund sustainability metric has, in total, 10 criteria.
[151] Perhaps most prominently, Higgins et al (2014) explore dynamics of 'walking the talk', which sees organisation action aligning with external and internal communication over time. There is a wide range of scholarship on international organisations on 'norm lifecycles', which shows a tendency for discursive shifts to become embedded in structure and performance (e.g. Park and Vetterlein 2010, Clegg 2013).

tracks specific financial market purchases made by SWFs. The SWF Global Transaction Database follows a broad definition of 'sustainable investment', including SWF asset purchases that are aligned with any of the UN Sustainable Development Goals including agricultural production, climate action, education, gender equality, and healthcare (with healthcare representing the single largest class of sustainable investments observed). Results from Bortolotti et al suggest that stronger public commitment to sustainability goals by an SWF is associated with stronger sustainability performance, in terms of the number of and value of purchases within sustainability-aligned sectors. Although Bortolotti et al focus on a multi-dimensional conceptualisation of 'sustainability', the strengthened commitments to climate performance currently being promoted through OPSWF may, in time, support accelerated financing to climate positive sectors and activities.

7.3. Private Information for Market-based Climate Financing

The final strand of emerging practice we explore covers private initiatives aimed at improving information flows to facilitate the effectiveness of market-based climate financing. We review dynamics surrounding the Carbon Disclosure Project (CDP), and the Partnership for Carbon Accounting Financials (PCAF). The former represents a leading attempt to create information- and network-based institutional structures to facilitate equity investors' translation of their financial power into accelerated climate action, and the latter represents a more emergent mechanism to facilitate accelerated climate action across the banking and insurance sectors. Our insights from Chapter Six highlighted the potential importance of institutional networks in catalysing the effectiveness of green bond performance, and here we probe the potential for positive impact from information-based institutional structures on the performance of wider climate-focused financing.

The Carbon Disclosure Project grew from an attempt in the early 2000s to use investor activism to achieve greater climate change focus within large, publicly listed corporations. The CDP was founded by Tessa Tennant and Paul Dickinson, who are widely identified as pioneers of green finance. By the early 1990s, consumer and civil society pressure had contributed to the establishment of a focus on 'corporate social responsibility' within some larger corporations, particularly those operating in clothing and textiles where production operations had increasingly extended to lower cost emerging and lower-income countries. This corporate social responsibility agenda saw corporations issuing public commitments to baseline labour and environmental standards, with the creation of reporting processes that commonly evolved to incorporate some level of third-party oversight.

The goal of Tennant and Dickinson was to create a Carbon Disclosure Project framework for unified shareholder action. Rather than individual corporations unilaterally setting their own environmental priorities, the CDP was to provide a common template for corporate reporting specifically on climate impact. By developing an approach that could be promoted by shareholders to a wide range of corporations, the CDP aimed to embed corporate reporting that would generate readily comparable performance data. Across the first iteration of the Carbon Disclosure Project reporting cycle in 2003, with 35 investors using their position as shareholders to push for corporate engagement and supported by lobbying from CDP staff,[152] around 250 companies provided data submissions (Knox-Hayes and Levy 2011).

During its early phase of operations, the performance and utility of the CDP framework was given mixed assessments. On the one hand, the Project's overarching goal of supplying comparable data on corporate emissions and climate change impact was lauded as potentially revolutionary. Speaking in 2010, Christina Figueres, the then Executive Secretary of the UN Framework Convention on Climate Change, noted that:

[152] Funding in the early years of CDP operation was provided by environment ministries of the UK, US, Netherlands, and Sweden, as well as financial institutions including AXA, Merrill Lynch, and Pictet Asset Management (Gorham 2008: 8).

> The Carbon Disclosure Project is to the future of business what the X-ray was to the then future of medicine: without it, we would never see the inside of a patient's health. You can change regulatory and accounting standards to better price climate risk and you can launch new, innovative forms of climate investment assets but, unless you know the accurate carbon footprint of the investment objective, you will miss the hidden risk.[153]

However, CDP was also criticised for a lack of rigour through its initial cycles; by requesting but not requiring corporations to follow a Greenhouse Gas Protocol as the methodology for calculating emissions, the true comparability of companies' reported emissions remained unclear. More substantively, the CDP sought to assess the level of transparency displayed by a given corporation, rather than focusing on the actuality of its carbon footprint (Gorham 2008: 13-14). Although the CDP does request and collect emissions data from responding corporations, it ultimately grades according to criteria including whether this emissions data has been provided to CDP, whether this information is publicly available, and whether corporations have a published strategy that outlines a programme of activity to support emissions reduction that is demonstrably aligned with Paris Agreement targets. A heavily polluting corporation with an effective reporting process would, then, be graded more highly than a less polluting corporation with less robust reporting capacity. The CDP reporting guidance document runs to 360 pages, displaying the extensity of information that is collated and fed into the grading process. As a headline, CDP provides responding organisations with an A-D grade, and also grants responding organisations access to a database that covers content including all respondents' emissions figures.[154]

[153] See 'Speech by Christiana Figueres at global launch of Carbon Disclosure Project', UN Climate Change website, available at https://unfccc.int/news/speech-by-christiana-figueres-at-global-launch-of-carbon-disclosure-project. Accessed 20th May, 2025.

[154] See 'CDP Climate Change Scoring Methodology', CDP website, available at https://guidance.cdp.net/en/guidance?cid=46&ctype=theme&idtype=ThemeID&incchild=1&microsite=0&otype=ScoringMethodology&page=1&tags=TAG-605%2CTAG-13071. Accessed 21st May, 2025.

The CDP expanded its scope of coverage across the 2010s, and by the mid-2020s over 22,000 corporations provided the CDP with a return. With the CDP targeting engagement with the globally largest corporations, its respondents held a listed value of around US$67tn, representing around two-thirds of total global capitalisation.[155] In an attempt to accelerate its impact on corporate behaviour, the CDP has recently enhanced its focus on corporations' climate transition plans and their assessment of the adequacy of these plans to the scale of mitigation required to meet Paris Agreement goals (CDP 2024).

While there is some push-back against the onerous nature of the CDP reporting requirements and ongoing debate over the efficacy of the framework,[156] recent analysis from Cohen et al (2023) suggests that the reporting and information-sharing requirements from the CDP may be associated with strengthened corporate climate performance. Specifically, Cohen et al test whether investors who have publicly backed the CDP are able to translate their financial power into changed corporate behaviour. Cohen et al's study finds that where such investors have a financial holding in a company, that company displays a raised propensity to engage with the CDP. Moreover, this corporate engagement is in turn found to be associated with accelerated de-carbonisation, as measured by annual changes in emissions reported to CDP. With echoes of our findings in Chapter Six on the benefits to green bond borrowing costs from home-country underwriter networks and information sharing capacity, it seems that here market-based institutional networks and information flow nexuses are being translated into improved outcomes. In the case of the CDP, Cohen et al (2023: 2) suggest that investors can draw on CDP data to push for emissions improvements, and ultimately to threaten and deploy 'exit' to incentivise action.

[155] See 'Record 23,000+ companies disclosure climate impact through CDP', CDP website, available at https://www.cdp.net/en/press-releases/record-23-000-companies-disclose-environmental-impact-through-cdp-with-urgency-for-action-clear-in-wake-of-unprecedented-global-temperatures. Accessed 21st May, 2025.
[156] See 'CDP slashes workforce to address rising costs of upgrades and innovation', Greenbiz website, available at https://trellis.net/article/cdp-workforce-reduction-innovation-funding/. Accessed 21st May, 2025.

There are similarities in the aims of the CDP, and the Partnership for Carbon Accounting Financials to which we now turn attention. Both represent institutionalised attempts to generate information-based networks that improve the impact from climate-motivated finance. However, while the former seeks to translate equity-based financial market investment into stronger climate outcomes, the latter is more squarely targeted at embedding and mainstreaming a climate change and emissions reduction logic within banks' and insurers' capital allocation processes.

PCAF grew out of the Partnership for Sustainable Insurance (PSI), a collaborative initiative between the United Nations Environment Programme and insurance market actors that itself was founded in 2012. As its core mission, the PSI aimed to support the insurance industry toward 'identifying, assessing, managing and monitoring risks and opportunities associated with environmental, social and governance issues'.[157] PSI workstrands included the exploration of accounting methodologies to allow the relationship between capital allocation and greenhouse gas-emitting activity to be effectively mapped and understood.

It was in 2019 that PCAF was launched, to bring together the accounting foundations being provided by the PSI and a range of similar initiatives being developed elsewhere. The PCAF Steering Committee was constituted by representatives of internationally-active insurance companies including Allianz, Lloyds, Sompo, and Zurich (PCAF 2022: 4). Over subsequent years membership has grown to incorporate around 100 insurance companies, 350 banks, and 100 asset managers, with members' total asset base registering at over US$100tn.[158] PCAF can be viewed, to borrow the terminology of Seabrooke and Stenstrom (2025: 285), as an emerging component of climate governance 'infrastructure', with the potential to develop a streamlined process for climate change

[157] See 'Principles for Sustainable Insurance', UN Environment Programme website, available at https://www.unepfi.org/insurance/insurance/. Accessed 16th May, 2025.
[158] See 'Overview of financial institutions', Partnership for Carbon Accounting Financials website, available at https://carbonaccountingfinancials.com/en/financial-institutions-taking-action#overview-of-financial-institutions. Accessed 16th May, 2025.

mainstreaming that can be translated to organisations across the banking and insurance sectors and become accepted as a taken-for-granted component of operational practice.

At its core, PCAF seeks to gain traction over 'financed emissions', defined as 'the impact of lending decisions in the portfolio of financial institutions' (Nelson 2022: 2). The PCAF Global Greenhouse Gas Accounting and Reporting Standard was published in 2022, and provides financial institutions with a globally standardised approach. The PCAF Standard specifically describes methods to be followed by banks to impute emissions associated with their listed equity and bond portfolios, business loans, project loans, commercial real estate, mortgages, motor vehicle loans, and sovereign debt. So, for example, for a bank to impute the carbon footprint of a loan to a given corporation, the PCAF Standard template requires that the bank collate data on the loan value, the corporation's 'enterprise value including cash' (EVIC),[159] and the corporation's annual emissions. The bank would then denominate the loan value relative to EVIC to generate an 'attribution factor', which would then determine the share of total corporate emissions attributable to the loan (PCAF 2022: 70). A US$1m loan to a corporation with an EVIC of £1bn and annual emissions of 140tCO_2 would give an attribution factor of .001, and an attribution value of .14tCO_2 for the loan.[160]

The PCAF methodology has recently been evaluated by Granoff and Lee (2024), who note that the prioritisation of mark-to-market valuation can generate significant swings in the attribution value of a given loan that are unrelated to real-world climate performance. An increase in company value, driven by wider market dynamics, would push down the attribution factor and correspondingly decrease the carbon footprint associated with a given loan. Granoff and Lee also highlight inconsistencies across prominent PCAF backers in their financed emissions reporting, with some

[159] EVIC is a common measure of total company value at a given point in time. See 'Financial Conduct Authority Technical Standards Handbook', FCA website, available at https://www.handbook.fca.org.uk/techstandards/BMR/2020/reg_del_2020_1818_oj/chapter-i/002.html. Accessed 22nd May, 2025.

[160] With the loan representing 1/1000th of the company's market value, 1/1000th of the company's annual emissions are attributed to the loan.

banks basing their loan attribution CO_2 values on a company's direct emissions, rather than a combined measure of direct emissions and indirect emissions produced by production inputs.

Notwithstanding these issues, PCAF represents an initiative with potential to become more deeply embedded in banks' decision-making processes through, in part, the engagement of Financial Technology (FinTech) companies. For FinTech companies, potential returns are to be had by creating a subscription platform to support integrated PCAF Standard reporting. FinTech accountancy software providers increasingly offer subscribers carbon accounting add-ins that deploy 'spend-based methodology' to refine the calculation of a company's emissions footprint. Drawing on a company's financial transaction data trail, the spend-based approach aims to supply a CO_2 attribution value for payments made by a company, based on the payment recipient's own profile or an estimate from characteristics such as the recipient's sector and size. A vision of FinTech-informed banking would see financial institutions' own IT systems trawling through data on potential borrowers' emissions profiles and trajectories, with a view to pricing-in an increased borrowing charge for higher emitters.[161] Loukoianova et al (2024: 1) offer a pragmatic assessment of the future contribution of FinTech to accelerate financial sector alignment with global climate goals through mechanisms that are aligned with this PCAF approach. While the potential is significant, we remain at an early stage in this mainstreaming of emissions data within lending institutions' operational decision-making frameworks.

7.4. Conclusion

In the earlier years of the 21st century, discussion over climate financing were, by and large, founded on an assumption that governments would lead the way in supporting required transformations. By the

[161] For an overview of the mediating pathways through which FinTech has the potential to accelerate decarbonisation, see Aziz et al (2024: 3).

time of the Paris Agreement in 2015, a focus on ensuring that public spending was structured to 'crowd-in' private financing was already well established on the international agenda. Now, with conventional wisdom holding that sluggish recovery from the Global Financial Crisis of 2008 and the costs from the Coronavirus pandemic response has left governments' financial capacity sorely depleted, there is broad-based belief that market finance will have to lead us toward more sustainable ways of living and working. Our goal through this book has been to review institutional foundations for effectiveness across each of these three strands of climate financing: government spending and investment, state- and international organisation-led efforts to catalyse market financing, and private initiatives to accelerate the flows of market financing in support of climate transition.

In working towards this overarching goal, we have presented a holistic analysis covering these three strands of climate financing, each of which provides a necessary contribution to the pool of resources needed for climate transition. Public spending and investment, government- and international organisation-led efforts to catalyse market financing, and private initiatives to accelerate the flow of market finance are all vital components of overall climate financing efforts. Through this book, we have added value to existing scholarship by systematically probing institutional foundations of effectiveness across all three areas.

Through our individual empirical chapters, we have extended existing understanding of the institutional drivers of variation in climate finance performance. We have clarified institutional factors that help explain increased government domestic spending on climate transition, increased government provision of external climate aid flows, expansion of green bond markets, and stronger demand for green bonds and lowered borrowing costs for green bond issuers. In Chapter Two, we showed how political institutions shaped government climate spending commitments, with Green party electoral success and inter-party competition accelerating overall volumes of climate-relevant spending. In Chapter Three, we showed that social institutions, in the form of post-material national values and trust in foreign nationals, helped support climate aid commitments. We also showed here that political institutional context played a significant role, with electorally vulnerable right-leaning governments in particular tending to dial-down their climate aid generosity. In Chapter Four, we identified a need for refinement to the World Bank's intervention in emerging green bond markets. While the World Bank has been a central figure in moving green bonds towards the mainstream of financial markets, we found World Bank green bond issuance into a domestic market to have a constraining effect on the volume of subsequent green bond issues. Through Chapter Five, we again identified institutional foundations requiring critical review, given the negative impact that was found from the ECB Collateral Framework on green bond performance. Chapter Six returned to positive institutional foundations, with our finding that bond issuers who employ an underwriter working in their home market enjoy a reduced cost of borrowing. This underwriter 'home-turf effect' reduced total repayments made by the average green bond issuer by around US$27m. In headline terms, across the book as a whole, we have identified a range of political, social, and economic institutions that can provide foundations for effective climate financing. In this final chapter, we have turned attention to a range of emergent trends and dynamics in climate financing, drawing on our earlier empirical work to inform our reflection on constraints and opportunities for accelerated improvement.

To close this work, we distill our forgoing analysis into a call for what we term 'evidence-based optimism' on the future of climate financing. We now have an extended understanding of how electoral politics can nudge states toward stronger action, of areas for improvement in how international organisations and central banks impact on green bond market development and performance, and of how market-based networks and information flows can catalyse stronger market action. This enhanced understanding of existing and potential additional foundations for effectiveness give one pillar for evidence-based optimism; by better understanding existing institutional strengths and weaknesses, we can take action to support the former and address the latter. And a second pillar for evidence-based optimism comes from the fundamental under-determination of the future of climate financing. Our empirical work highlights the factors that we can measure and explain, and in relief shows the importance of factors that are not observed or incorporated into our models. Unobserved factors include the space that exists for individuals, social groups, economic organisations, and political actors to act on the basis of informed reflection, and to align behaviour with an understanding of planetary limits and the conditions we pass on to future generations.

Although timeframes for climate action become tighter with every day that passes, the future of climate financing and climate change performance has yet to be written. And the beauty of living with a future that has yet to be written is that there is opportunity, through concerted collective and individual action informed by understanding of foundations for effectiveness, to make a meaningful difference.

Annex

A2.1: Explaining variation in climate investment across political cycles

	Model 9		Model 10		Model 11		Model 12	
Variable	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error
Independents								
Green seats (share)	-.008	.005	~	~	-.011	.008	~	~
Green seats (any)	~	~	.023	.015	~	~	.015	.021
Green coalition	.002	.014	.000	.014	.004	.015	.004	.015
Green vote share	.012	.005***	.004	.002*	.016	.006*	.006	.003*
EU membership	-.015	.022	-.014	.022	~	~	~	~
Partisan orientation	.003	.004	.003	.004	.006	.005	.004	.005
Controls								
Economic growth	.001	.000	.001	.001	.001	.001	.001	.001
Climate vulnerability	-.008	.009	-.010	.009	-.003	.011	-.005	.011
Decentralisation	.006	.012	.010	.012	.006	.005	.004	.005
n	432		432		347		347	
X^2	.514***		.504***		.488***		.486***	

Note: *p≤0.05, **p≤0.01, ***p≤0.001

A2.1: Explaining variation in climate investment across post-election years

	Model 13		Model 14		Model 15		Model 16	
Variable	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error	Coefficient	Std error
Independents								

Green seats (share)	-.007	.006	~	~	-.010	.012	~	~
Green seats (any)	~	~	.022	.040	~	~	-.043	.051
Green coalition	-.080	.040*	-.084	.040*	-.100	.044*	-.100	.044*
Green vote share	.012	.007	.004	.004	.020	.012	.015	.006*
EU membership	.226	.037***	.208	.035***	~	~	~	~
Partisan orientation	-.009	.010	-.008	.010	.002	.013	.003	.013
Controls								
Economic growth	-.004	.004	-.004	.004	-.000	.005	.000	.005
Climate vulnerability	-.013	.004**	-.012	.004***	-.013	.005*	-.013	.005**
Decentralisation	-.005	.001***	-.005	.001***	-.006	.002***	-.006	.002***
n	110		110		89		89	
p	***		***		***		***	

Note: *p≤0.05, **p≤0.01, ***p≤0.001

Table A6.1. Explaining Green Bond underpricing (credit spread gap measure)

	Credit Spread Gap	
	(1)	**(2)**
Home Turf	-0.22**	-0.29***
	(0.08)	(0.07)
Syndicated	-0.03	-0.09
	(0.09)	(0.07)
HT×Syndycate	0.16+	0.25**
	(0.09)	(0.08)
Credit Rating	-0.00	-0.00
	(0.01)	(0.01)
Amount ($bn)	-0.00	0.00
	(0.00)	(0.00)
Maturity	-0.00	-0.00*
	(0.00)	(0.00)
Liquidity	0.07	0.06
	(0.05)	(0.05)
Big-hitter	0.09*	0.03
	(0.04)	(0.04)
Country FE	Yes	Yes
Year FE	Yes	Yes
Observations	261	261
R2	0.360	0.344

Note: Clustered standard errors in parentheses. +p<0.1, * p<0.05,** p<0.01, ***p<0.001

Table A6.2. Explaining Green Bond underpricing (primary-to-secondary market spread measure)

	Primary-to-secondary market spreads

	(3)	(4)
Home Turf	-0.19**	-0.29***
	(0.08)	(0.07)
Syndicated	-0.00	-0.09
	(0.09)	(0.08)
HT×Syndycate	0.17⁺	0.29**
	(0.09)	(0.09)
Credit Rating	-0.01⁺	-0.00
	(0.01)	(0.01)
Amount ($bn)	-0.00	0.00
	(0.00)	(0.00)
Maturity	-0.00	-0.00*
	(0.00)	(0.00)
Liquidity	0.06	0.05
	(0.05)	(0.05)
Big-hitter	0.11**	0.05
	(0.04)	(0.04)
Country FE	Yes	Yes
Year FE	Yes	Yes
Observations	261	261
R2	0.357	0.335

Note: Clustered standard errors in parentheses. ⁺p<0.1, * p<0.05,** p<0.01, ***p<0.001

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